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Organization of Islamic Cooperation (OIC)
Department of Civil and Environmental Engineering (CEE)

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FINAL REPORT

Part: Structural Design & Analysis

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2. Setback, FAR & MGC

Summary Table

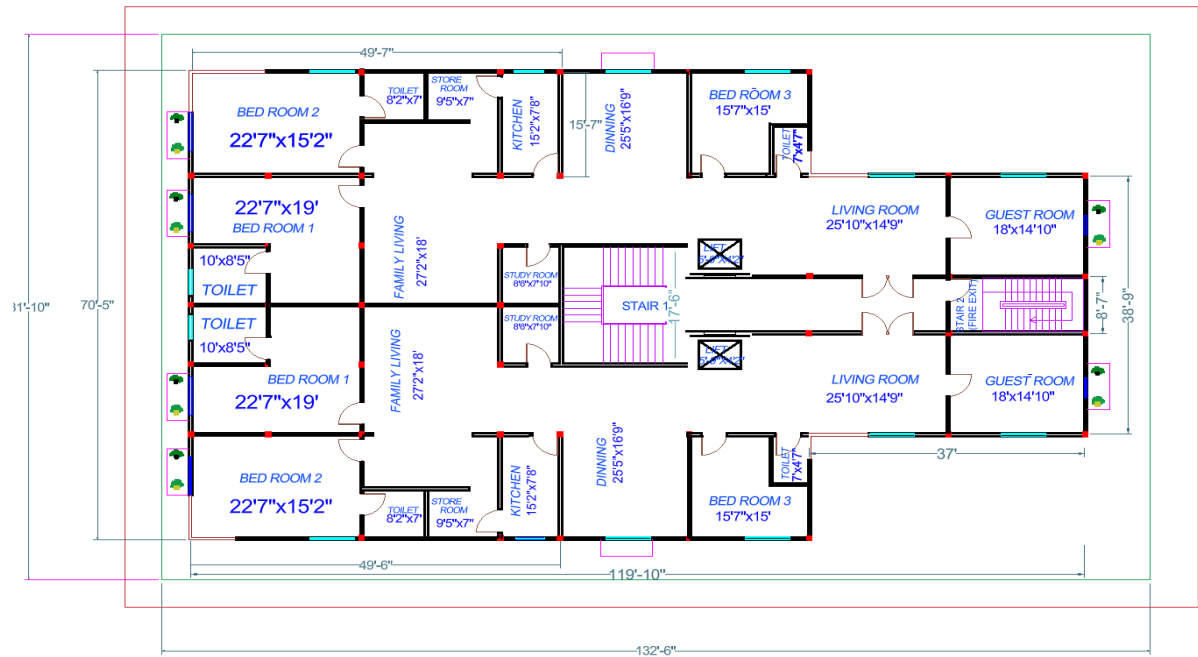
Table 1: Reference Information from DMINB used for the FAR and MGC Calculation

Setback				
For Building Height: Up to 33m or 10 floors	9 to 10	1.5	2	1.25
For Building Height: Up to 33m or 10 floors	Plot (Katha)	Front(m)	Back(m)	Each Side (m)
	Any Amount	1.5	3	3
FAR and MGC				
Plot Area: 14 Katha to 16 Katha	Road Width	FAR	MGC(%)	
	9	5	52.5	
Plot Area: 16 Katha to 18 Katha	Road Width	FAR	MGC(%)	
	9	5.25	52.5	

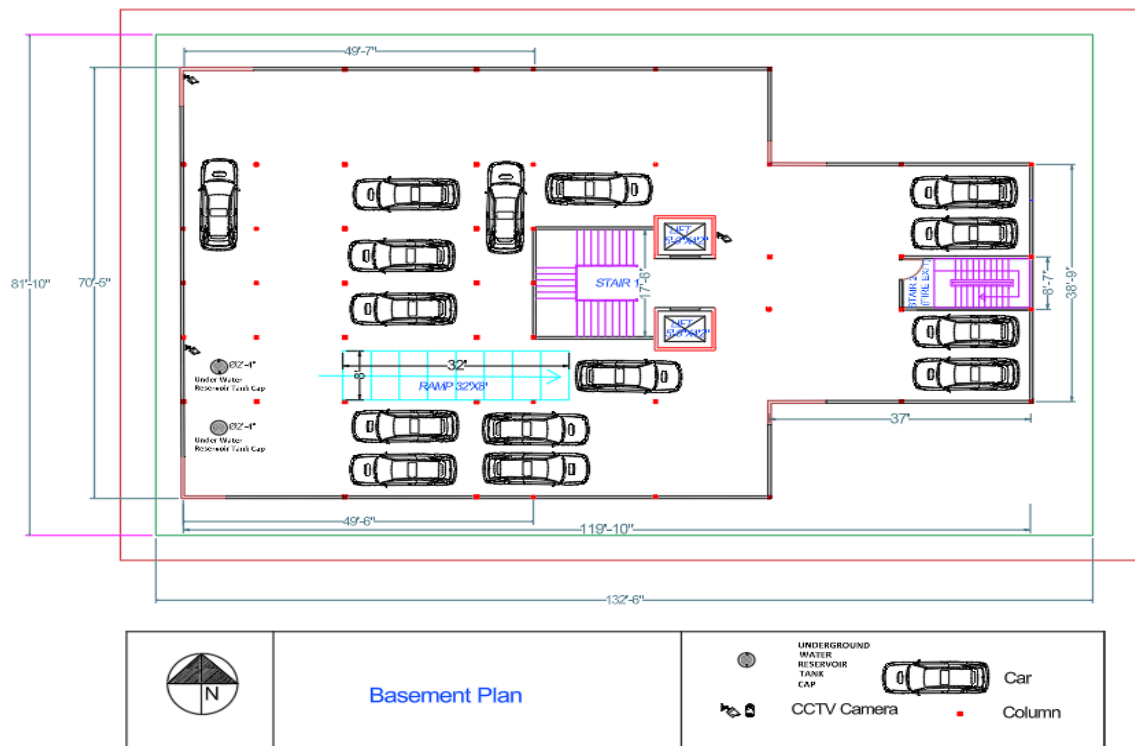
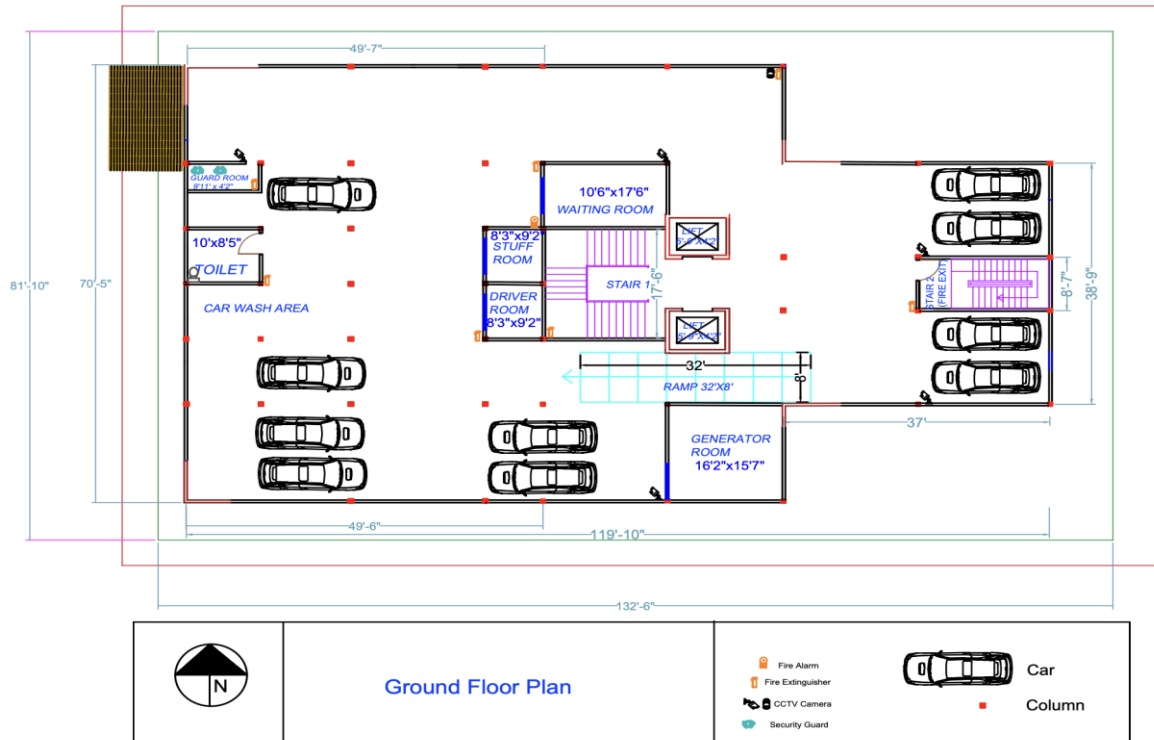
Table 2: Summary Table of Setback, FAR and MGC Calculations

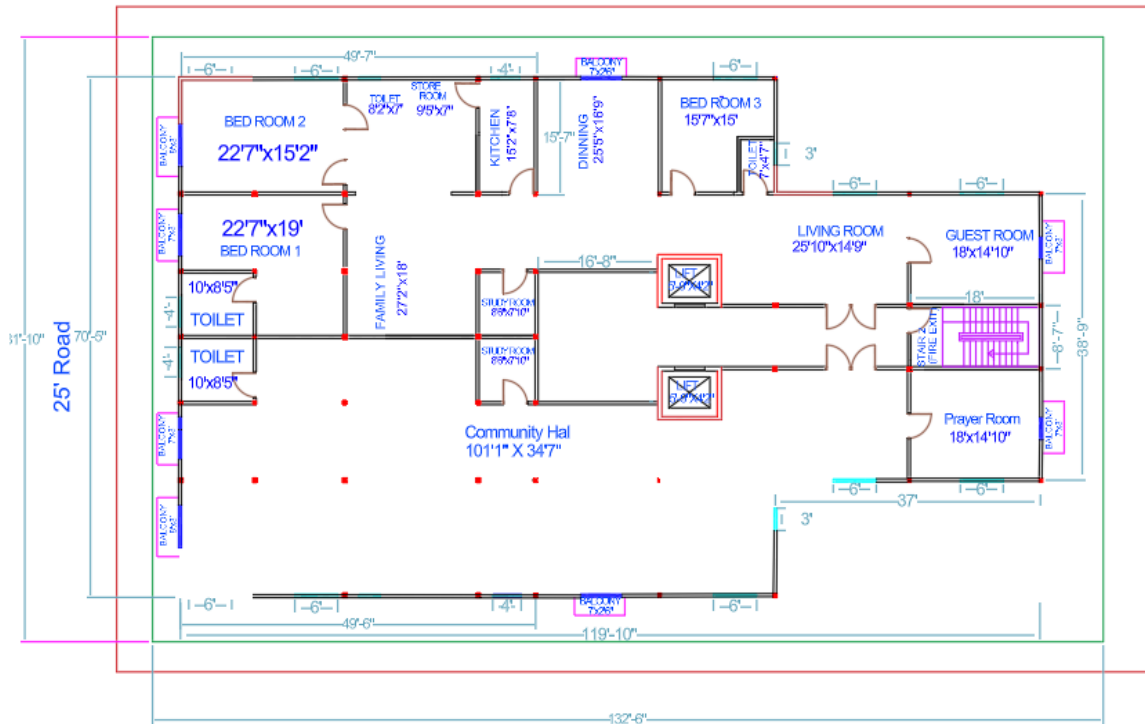
Total Plot Area	Katha	Ft^2
	18	12960
Road Width	Feet	Meter
	25	7.62195122
Considered Length for Total Plot Area (Ft)	144	
Considered Width for Total Plot Area (Ft)	90	
Length After Setback (Ft)	136	
Width After Setback (Ft)	78.5	
Effective Land Area Excluding Setback (ft^2)	10676	
Need to Satisfy the maximum FAR based on 100% MGC		
Built Area (ft^2)	6804	
Built Length (ft)	119.84	
Built Width (ft)	70.42	
No of Floors	10	

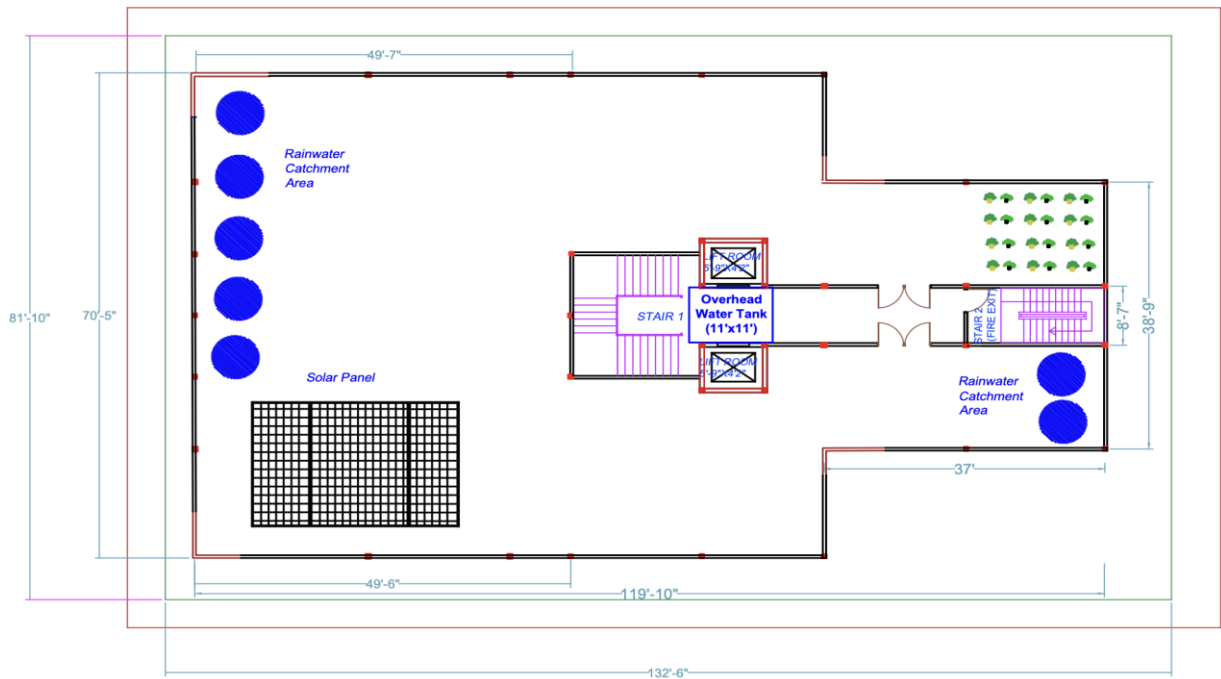
3. PLAN VIEWS OF LAYOUT

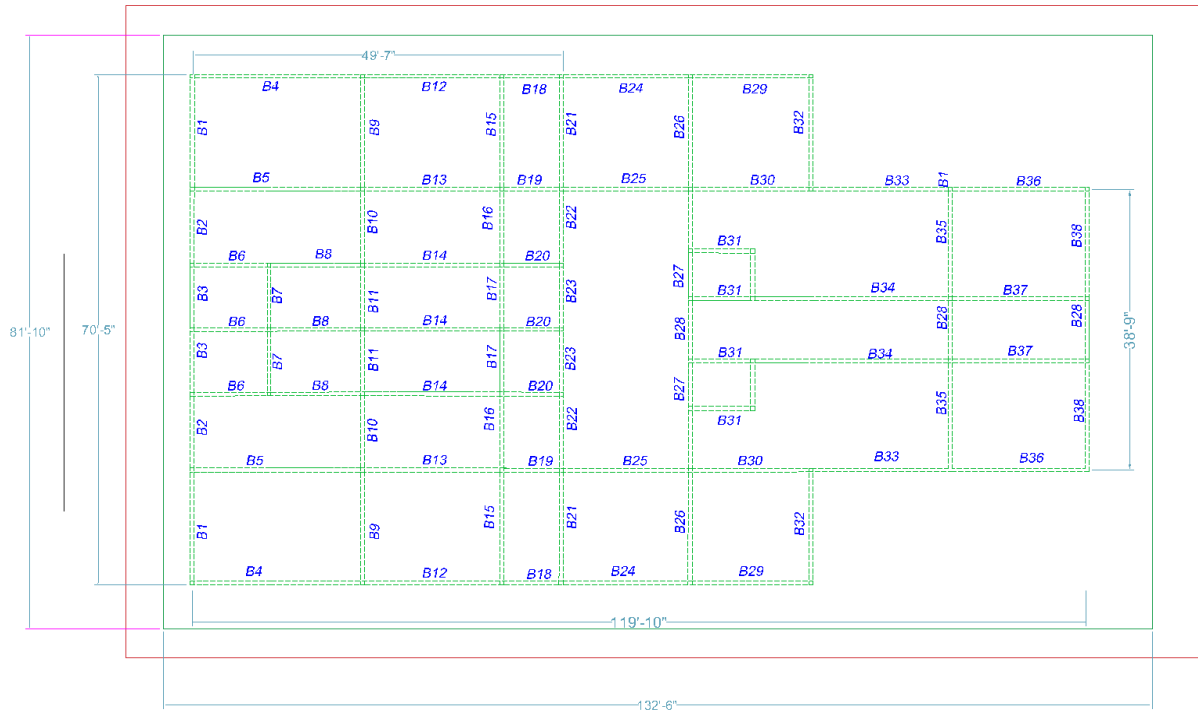


	PLAN VIEW OF A TYPICAL FLOOR	■ COLUMN — WINDOW — SLIDING DOOR  Fire Extinguisher
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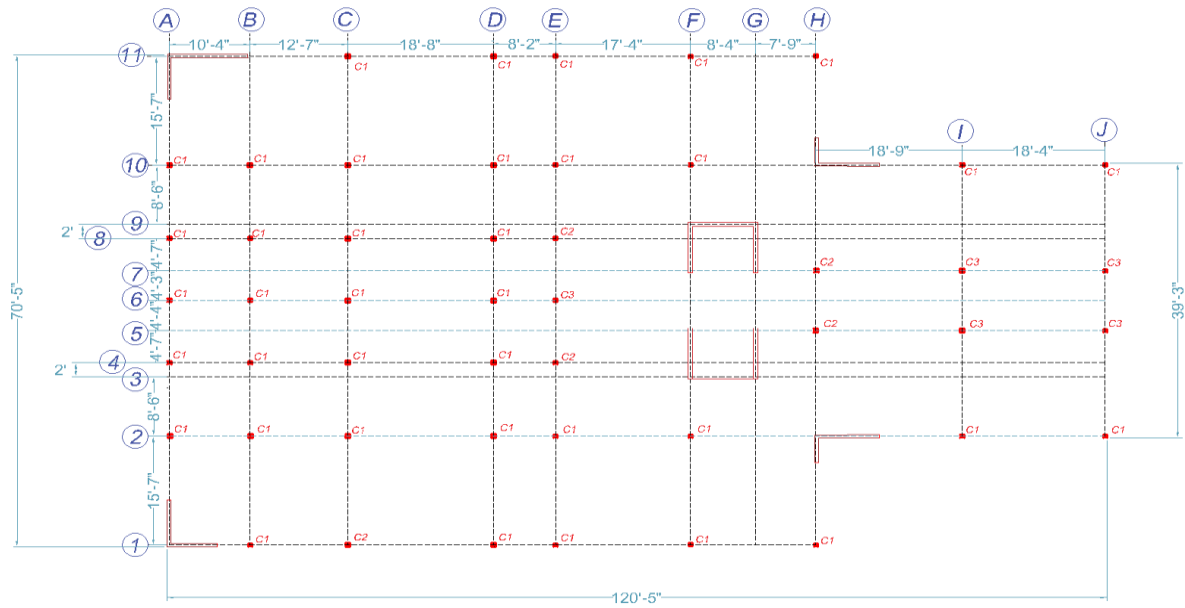




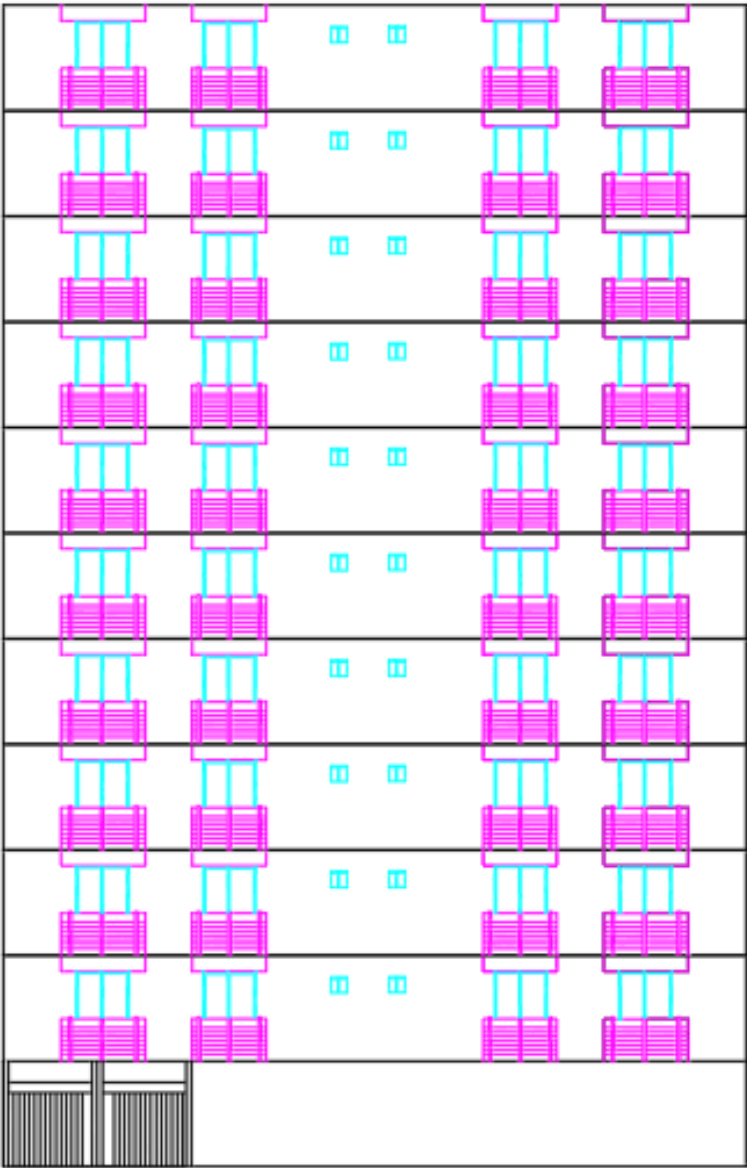


Beam Layout

- Floor Beam 15 x 10 in²
- Grade Beam 17 x 12 in²



	Column Layout	■ COLUMN
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Front Elevation

APPENDIX

3.1. Setback, MGC and FAR Calculations:

Land Informations,

- ❖ Land Area= 18 katha = 18×720 sq.ft=12,960 sq.ft=1204 sq.m
- ❖ Road Width=25 ft =7.62 m
- ❖ 52.5% MGC is given
- ❖ 5.25 FAR is given

Setback rule:

Setback Rule (m) [DMINB 2008]			
For less than 10 floors	Front	Rear	Both Side
	1.5	2	1.25

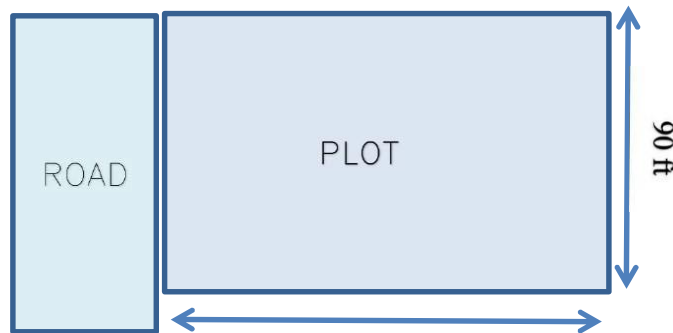
Setback area:

Length=144ft - $(1.25 \times 2) \times 3.28 = 135.8\text{ft} \sim 136\text{ft}$

Width=90ft- $(2+1.5) \times 3.28 = 78.5\text{ft}$

So, Effective Area= (78.5×136) sq.ft=10,676 sq.ft= 14.8 katha

144 ft



3.1.1. Floor Area Ratio (FAR) & Maximum Ground Coverage (MGC):

For Road=9m, FAR= 5.25 [DMINB2023]

MGC (Maximum Ground Coverage) = $(52.5\% \times 12960) = 6,804\text{sq.ft}$

No.of Floors = $(5.25 \times 12960) / 6804 = 10$ (G+10) floors

Each Floor Area = $(5.25 \times 12960) / 10 = 6,804$ sq. ft (2 units Apartment)

$A/L = 37/(119+10/12) = 0.31 > 0.15$ [So, re-entering corner ok]
Ramp size = Basement Height / max slope = $8'/(1/8) = 64\text{ft}$

3.2. Provisions for Setback, MGC and FAR

ইয়ার্ডের উচ্চতা ৩০ মিটার অথবা ১০ ডগার পর্যন্ত				
বর্গমিটার	প্রটের পরিমাণ	কাঠা	সমুদ্র (মিটার)	ন্যূনতম সেটিং ব্যাক
			পচাং (মিটার)	প্রতি পার্স (মিটার)
১৩৪ বর্গমিটার বা ইয়ার্ডের নীচে	২ কাঠা বা ইয়ার্ডের নীচে	১.৫০	১.০০	০.৮০
১৩৪ বর্গমিটার এর উপর হইতে ২০১ বর্গমিটার পর্যন্ত	২ কাঠার উপর হইতে ৩ কাঠা	১.৫০	১.০০	১.০০
২০১ বর্গমিটার এর উপর হইতে ২৬৮ বর্গমিটার পর্যন্ত	৩ কাঠার উপর হইতে ৪ কাঠা	১.৫০	১.৫০	১.০০
২৬৮ বর্গমিটার এর উপর হইতে ৩৩৫ বর্গমিটার পর্যন্ত	৪ কাঠার উপর হইতে ৫ কাঠা	১.৫০	২.০০	১.২৫
৩৩৫ বর্গমিটার এর উপর হইতে ৪০২ বর্গমিটার পর্যন্ত	৫ কাঠার উপর হইতে ৬ কাঠা	১.৫০	২.০০	১.২৫
৪০২ বর্গমিটার এর উপর হইতে ৪৬৯ বর্গমিটার পর্যন্ত	৬ কাঠার উপর হইতে ৭ কাঠা	১.৫০	২.০০	১.২৫
৪৬৯ বর্গমিটার এর উপর হইতে ৫৩৬ বর্গমিটার পর্যন্ত	৭ কাঠার উপর হইতে ৮ কাঠা	১.৫০	২.০০	১.২৫
৫৩৬ বর্গমিটার এর উপর হইতে ৬০৩ বর্গমিটার পর্যন্ত	৮ কাঠার উপর হইতে ৯ কাঠা	১.৫০	২.০০	১.২৫
৬০৩ বর্গমিটার এর উপর হইতে ৬৭০ বর্গমিটার পর্যন্ত	৯ কাঠার উপর হইতে ১০ কাঠা	১.৫০	২.০০	১.২৫
৬৭০ বর্গমিটার এর উপর হইতে ৭৩৭ বর্গমিটার পর্যন্ত	১০ কাঠার উপর হইতে ১১ কাঠা	১.৫০	২.০০	১.২৫
৭৩৭ বর্গমিটার এর উপর হইতে ৮০৪ বর্গমিটার পর্যন্ত	১১ কাঠার উপর হইতে ১২ কাঠা	১.৫০	২.০০	১.২৫
৮০৪ বর্গমিটার এর উপর হইতে ৮৭১ বর্গমিটার পর্যন্ত	১২ কাঠার উপর হইতে ১৩ কাঠা	১.৫০	২.০০	১.২৫
৮৭১ বর্গমিটার এর উপর হইতে ৯৩৮ বর্গমিটার পর্যন্ত	১৩ কাঠার উপর হইতে ১৪ কাঠা	১.৫০	২.০০	১.২৫
৯৩৮ বর্গমিটার এর উপর হইতে ১০০৫ বর্গমিটার পর্যন্ত	১৪ কাঠার উপর হইতে ১৫ কাঠা	১.৫০	২.০০	১.২৫
১০০৫ বর্গমিটার এর উপর হইতে ১০৭২ বর্গমিটার পর্যন্ত	১৫ কাঠার উপর হইতে ১৬ কাঠা	১.৫০	২.০০	১.২৫
১০৭২ বর্গমিটার এর উপর হইতে ১১৩৯ বর্গমিটার পর্যন্ত	১৬ কাঠার উপর হইতে ১৭ কাঠা	১.৫০	২.০০	১.২৫
১১৩৯ বর্গমিটার এর উপর হইতে ১২০৬ বর্গমিটার পর্যন্ত	১৭ কাঠার উপর হইতে ১৮ কাঠা	১.৫০	২.০০	১.২৫
১২০৬ বর্গমিটার এর উপর হইতে ১২৭৩ বর্গমিটার পর্যন্ত	১৮ কাঠার উপর হইতে ১৯ কাঠা	১.৫০	২.০০	১.২৫
১২৭৩ বর্গমিটার এর উপর হইতে ১৩৪০ বর্গমিটার পর্যন্ত	১৯ কাঠার উপর হইতে ২০ কাঠা	১.৫০	২.০০	১.২৫
১৩৪০ বর্গমিটার এর উপর	২০ কাঠার উপর	১.৫০	২.০০	১.৫০

(৩) রাজ্যের গ্রহস্থ যাহাই থাকুক না কেন, বিদ্যমান রাজ্যের কেন্দ্র হইতে ৪.৫ মিটার অথবা প্রান্তের সীমানা হইতে ১.৫ মিটার দূরত্বের মধ্যে যাহা অধিক (রাজ্যের ভবিষ্যৎ বর্ধিতকরণের জন্য প্রয়োজনীয় জমি বাদ দিয়া) তাহার চাইতে কম দূরত্বে কোন ইমারত নির্মাণ করা যাইবে না।

SETBACK MEASUREMENTS

1. Compliance with Standard Dimensions

Stair	Length	Width	
Provided	17'6"	16'8"	
Minimum Dimension		3.67'	
Maximum Dimension			
Reference	BNBC 2020 Chapter B.3 Section 11		

Lift	Area	Length	Width
Provided	23.95 ft ²	5'9"	4'2"
Minimum Dimension	27.8 ft ²	4' 11"	5' 8"
Reference	DINB 2023 Chapter 7 Section 6.१		

Dining	Area	Length	Width
Provided	425.73ft ²	25'5"	16'9"
Minimum Dimension	53.8 ft ²		6' 6"
Reference	DINB 2023 Chapter 5 Section 60.क.1		

Kitchen	Area	Length	Width
Provided	116.27 ft ²	15'2"	7'8"
Minimum Dimension	43 ft ²		6' 6"
Reference	DINB 2023 Chapter 5 Section 60.५.1		

Family Living	Area	Length	Width
Provided	489 ft ²	27'2"	18'
Minimum Dimension	53.8 ft ²		6' 6"
Reference	DINB 2023 Chapter 5 Section 60.क.1		

Study Room	Area	Length	Width
Provided	66.58ft ²	8'6"	7'10"
Minimum Dimension	53.8 ft ²		6' 6"
Reference	DINB 2023 Chapter 5 Section 60.क.1		

Bed 1	Area	Length	Width
Provided	429.08ft ²	22'7"	19'
Minimum Dimension	53.8 ft ²		6' 6"
Reference	DINB 2023 Chapter 5 Section 60.क.1		

Bed 2	Area	Length	Width
Provided	342.51ft ²	22'7"	15'2"
Minimum Dimension	53.8 ft ²		6' 6"
Reference	DINB 2023 Chapter 5 Section 60.क.1		

Bed 3	Area	Length	Width
Provided	233.75 ft ²	15'7"	15'
Minimum dimension	53.8 ft ²		6' 6"
Reference	DINB 2023 Chapter 5 Section 60.क.1		

Drawing Room	Area	Length	Width
Provided	381.04 ft ²	25'10"	14' 9"
Minimum dimension	53.8 ft ²		6' 6"
Reference	DINB 2023 Chapter 5 Section 60.क.1		

Guest Room	Area	Length	Width
Provided	312ft ²	18'	14'10"
Minimum dimension	53.8 ft ²		6'6"
Reference	DINB 2023 Chapter 5 Section 60.क.1		

Toilet 1	Area	Length	Width
Provided	84.167 ft ²	10'	8'5"
Minimum dimension	29.6 ft ²		3'4"
Reference	DINB 2023 Chapter 5 Section 60.ग.1		

Toilet 2	Area (ft²)	Length(ft)	Width (ft)
Provided	32.08ft ²	7'	4'7"
Minimum Dimension	29.6 ft ²		
Reference	DINB 2023 Chapter 5 Section 60.१.1		

Window (Bedroom)	Area	Height	Width
Provided	27ft ²	4' 6"	6'
Minimum Dimension	23.4 ft ²		
Reference	DINB 2023 Chapter 5 Section 65.२.2		

Window (Kitchen)	Area	Height	Width
Provided	18ft ²	4' 6"	4'
Minimum Dimension	10 ft ²		
Reference	DINB 2023 Chapter 5 Section 58.३.3		

Window (Washroom)	Area	Height	Width
Provided	18ft ²	4' 6"	4'
Minimum Dimension	3.98 ft ²		
Reference	DINB 2023 Chapter 5 Section 65.१.6		

Stair (Fire Exit)	Area	Length	Width
Provided	154.5 ft ²	18'	8' 7"
Minimum Dimension	3.28 ft ²		
Reference	BNBC20 Section 1.14.5 page 2650		

4. Structural Properties

Structural Properties					
Section		Dimension	Material	Material Properties	
				Strength (psi)	Unit Weight (lb/cft)
Frame Section	C1	15" x 15"	Concrete	4500	150
	C2	18" x 18"	Concrete	4500	150
	C3	20" x 20"	Concrete	4500	150
	GC1	17" x 17"	Concrete	4500	150
	GC2	20" x 20"	Concrete	4500	150
	GC3	22" x 22"	Concrete	4500	150
	Floor Beam	15" x 10"	Concrete	4500	150
	Grade Beam	17" x 12"	Concrete	4500	150
Slab Section	Landing Slab	Thickness = 8"	Concrete	3500	150
	Slab 4"	Thickness = 4"	Concrete	3500	150
	Slab 6"	Thickness = 6"	Concrete	3500	150
	Stair	Thickness = 10"	Concrete	3500	150
Wall Section	Shear Wall	Thickness = 10"	Concrete	3500	150
	Interior Wall	Thickness = 5"	Brick	-	120
	Exterior Wall	Thickness = 10"	Brick	-	120
	Parapet Wall	Thickness = 5"	Brick	-	120
		Height = 3'	Brick	-	120

5. Earthquake Load Parameters

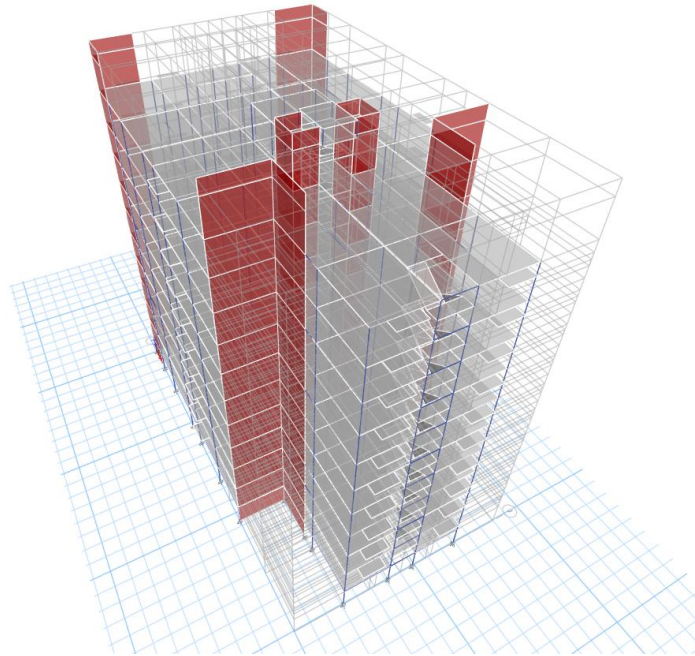
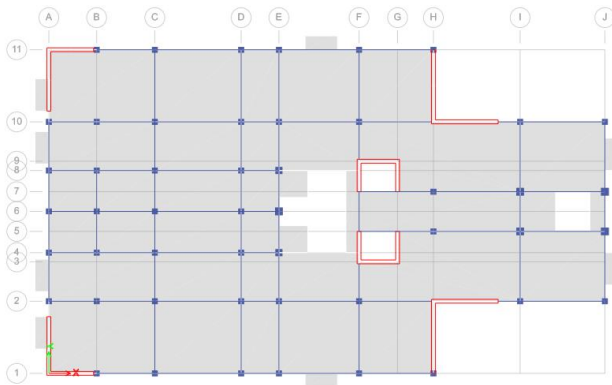
Earthquake Load	
Occupancy Category	II
Importance Factor	1
Site Class	SD
Seismic Zone Coefficient (Dhaka)	0.20
Seismic Design Category (SDC)	D
Structural System	IMRF
Response Modification, R	6.5
System Overstrength, Ω	2.5
Deflection Amplification, C_d	5
Time Period, T	1.21
Site Coefficient, F_a	1.35
Site Coefficient, F_v	2.70
Wind Speed (m/sec)	65.7

6. Wind Load as per BNBC2020

Wind Load Parameters	Value		BNBC 2020 References
Basic Wind Speed, V (mph)	147.168		Table 6.2.8
Exposure Type	A (ETABS B)		Article 2.4.6.3
Importance Factor, I	1		Table 6.2.9
Occupancy Category	II (Residential)		Table 6.1.1
Topographical Factor, K_{zt}	1		Article 2.4.7.2
Directionality Factor, K_d	0.85		Table 6.2.12
Gust Factor, G	X-direction	1.17	Article 2.4.8
	Y-direction	1.06	
Wind Pressure Co-efficients	C_{pw}	0.8	Figure 6.2.6
	C_{pl}	-0.35	

7.ETABS ANALYSIS

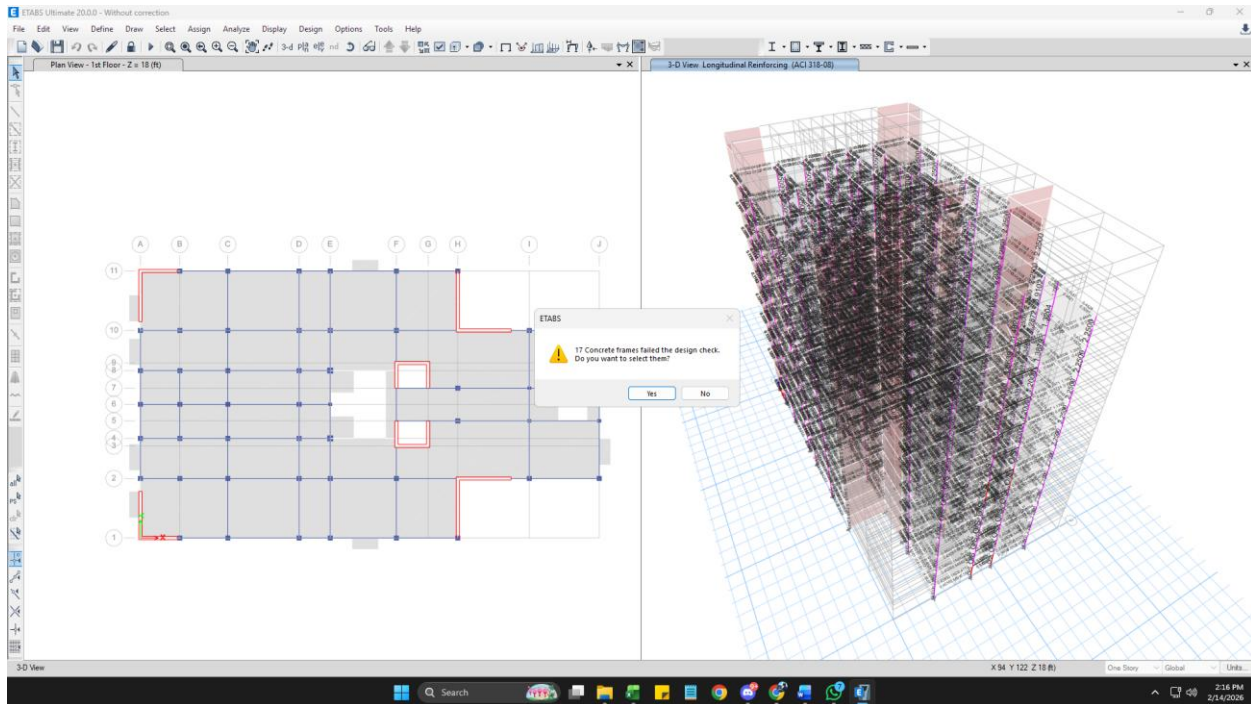
Structural Model



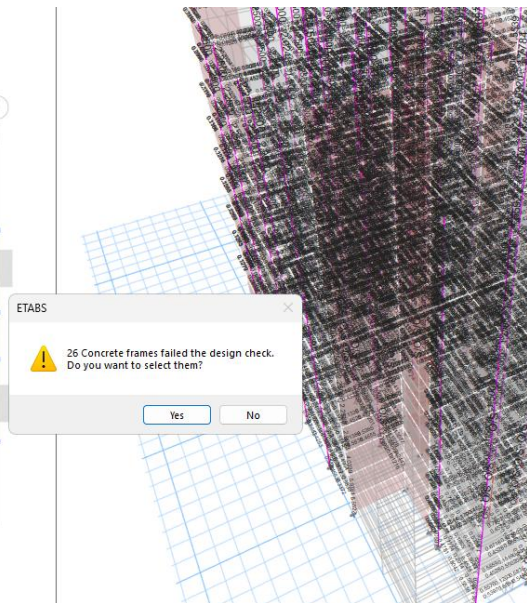
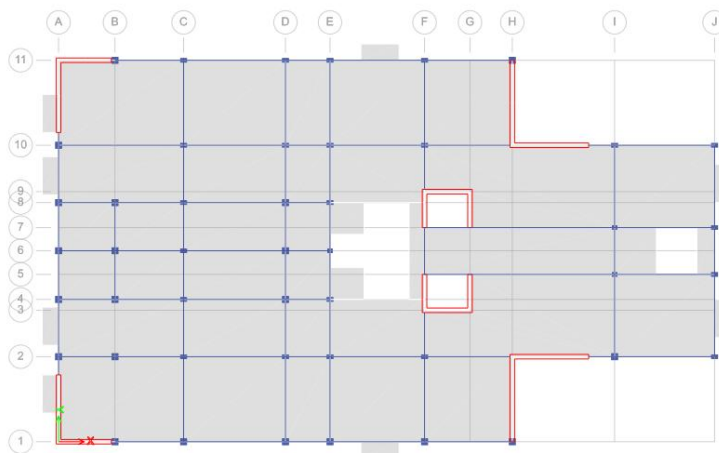
8. INITIAL MODEL CONFIGURATION REQUIRING REVISION (FAILED TRIALS)

These Trials did not pass in ETABS.

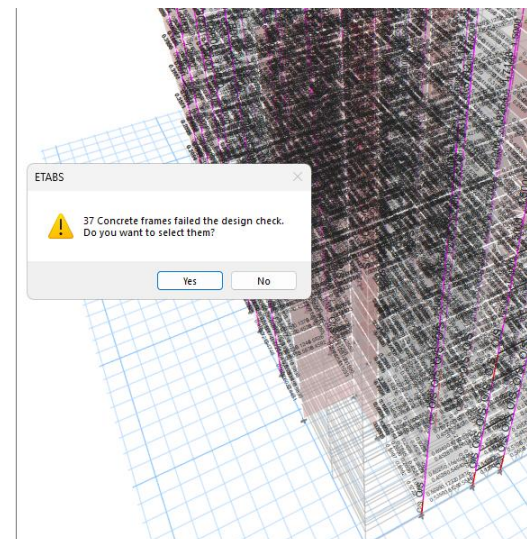
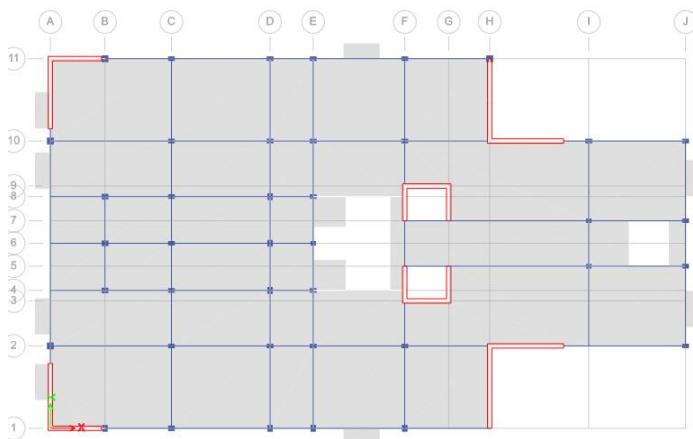
Using Column 10 x 12



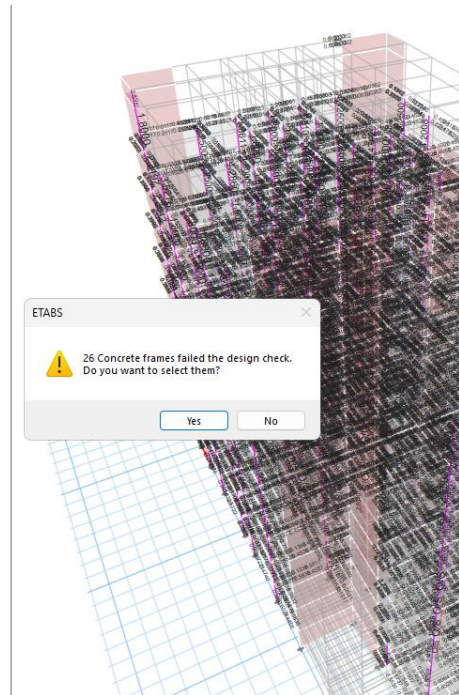
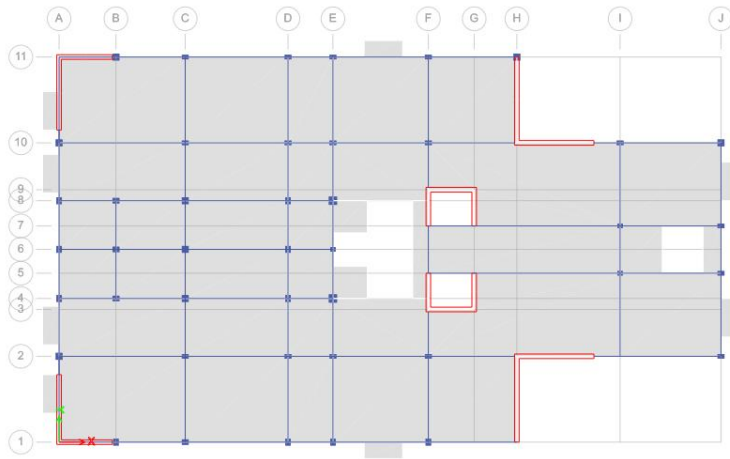
Using Column 10 x 15



Using Column 12 x 15

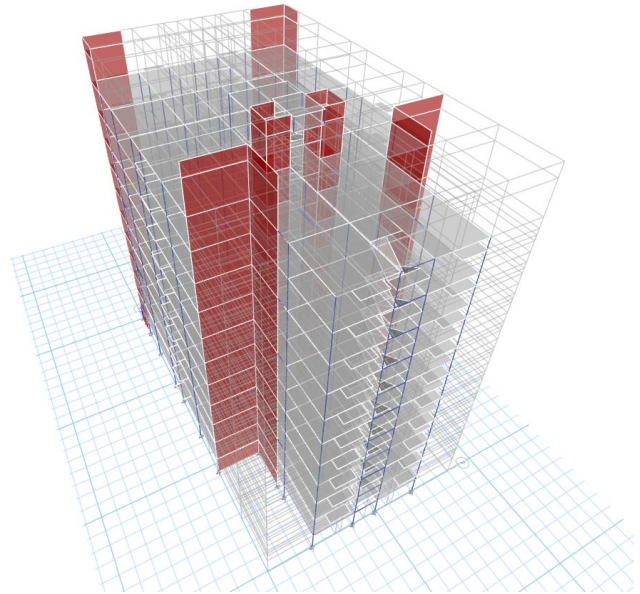
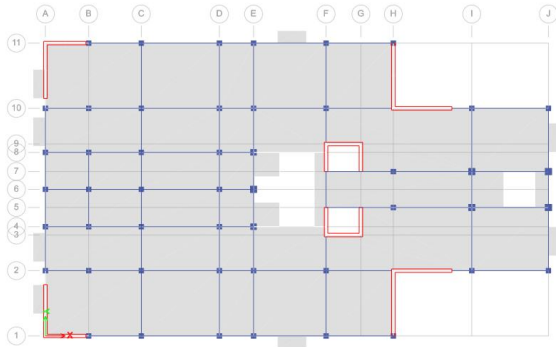


After Changing Orientation



9. FINAL ACCEPTED MODEL

All members have passed.



Used Columns	
C1	15" x 15"
C2	18" x 18"
C3	20" x 20"
GC1	17" x 17"
GC2	20" x 20"
GC3	22" x 22"

9.1. Material Properties

Material Property Data

General Data

Material Name

$f_c = 4500$ psi

Material Type

Concrete

Directional Symmetry Type

Isotropic

Material Display Color

Change...

Material Notes

Modify/Show Notes...

Material Weight and Mass

☒ Specify Weight Density

☐ Specify Mass Density

Weight per Unit Volume

150

lb/ft³

Mass per Unit Volume

4.662

lb-s²/ft⁴

Mechanical Property Data

Modulus of Elasticity, E

3823676

lb/in²

Poisson's Ratio, U

0.2

Coefficient of Thermal Expansion, A

0.0000055

1/F

Shear Modulus, G

1593198.33

lb/in²

Design Property Data

Modify/Show Material Property Design Data...

Advanced Material Property Data

Nonlinear Material Data...

Material Damping Properties...

Time Dependent Properties...

Modulus of Rupture for Cracked Deflections

☒ Program Default (Based on Concrete Slab Design Code)

☐ User Specified

OK

Cancel

E

Material Property Data

×

General Data

Material Name

A992Fy50

Material Type

Steel

▼

Directional Symmetry Type

Isotropic

▼

Material Display Color

Change...

Material Notes

Modify/Show Notes...

Material Weight and Mass

☒ Specify Weight Density
☐ Specify Mass Density

Weight per Unit Volume

490

lb/ft³

Mass per Unit Volume

15.23

lb-s²/ft⁴

Mechanical Property Data

Modulus of Elasticity, E

29000000

lb/in²

Poisson's Ratio, U

0.3

Coefficient of Thermal Expansion, A

0.0000065

1/F

Shear Modulus, G

11153846.15

lb/in²

Design Property Data

Modify/Show Material Property Design Data...

Advanced Material Property Data

Nonlinear Material Data...

Material Damping Properties...

Time Dependent Properties...

OK

Cancel

E

Material Property Data

×

General Data

Material Name

$f_c = 3500$ psi

Material Type

Concrete

▼

Directional Symmetry Type

Isotropic

▼

Material Display Color

Change...

Material Notes

Modify/Show Notes...

Material Weight and Mass

☒ Specify Weight Density
☐ Specify Mass Density

Weight per Unit Volume

150

lb/ft³

Mass per Unit Volume

4.662

lb-s²/ft⁴

Mechanical Property Data

Modulus of Elasticity, E

3372166

lb/in²

Poisson's Ratio, U

0.2

Coefficient of Thermal Expansion, A

0.0000055

1/F

Shear Modulus, G

1405069.17

lb/in²

Design Property Data

Modify/Show Material Property Design Data...

Advanced Material Property Data

Nonlinear Material Data...

Material Damping Properties...

Time Dependent Properties...

Modulus of Rupture for Cracked Deflections

☒ Program Default (Based on Concrete Slab Design Code)
☐ User Specified

OK

Cancel

E

Material Property Data

×

General Data

Material Name

F_y = 60 ksi (Rebar)

Material Type

Rebar

▼

Directional Symmetry Type

Uniaxial

Material Display Color

Change...

Material Notes

Modify/Show Notes...

Material Weight and Mass

☒ Specify Weight Density

☐ Specify Mass Density

Weight per Unit Volume

490

lb/ft³

Mass per Unit Volume

15.23

lb-s²/ft⁴

Mechanical Property Data

Modulus of Elasticity, E

29000000

lb/in²

Coefficient of Thermal Expansion, A

0.0000065

1/F

Design Property Data

Modify/Show Material Property Design Data...

Advanced Material Property Data

Nonlinear Material Data...

Material Damping Properties...

Time Dependent Properties...

OK

Cancel

Section Property

E Frame Section Property Data ✕

General Data

Property Name:

Material: ...

Notional Size Data:

Display Color:

Notes:

Shape

Section Shape:

Section Property Source

Source: User Defined

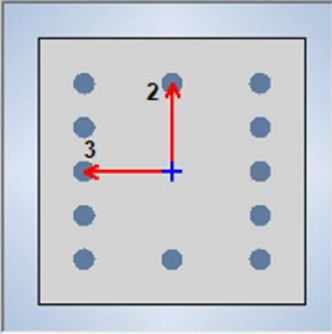
Section Dimensions

Depth: in

Width: in

☐ Include Automatic Rigid Zone Area Over Column

Reinforcement Diagram



Property Modifiers

Currently User Specified

Reinforcement

E

Frame Section Property Data

General Data

Property Name

C2 18 X 18

Material

f_c = 4500 psi

...

Notional Size Data

Modify/Show Notional Size...

Display Color

Change...

Notes

Modify/Show Notes...

Shape

Section Shape

Concrete Rectangular

Section Property Source

Source: User Defined

Section Dimensions

Depth

18

in

Width

18

in

Show Section Properties...

Reinforcement Diagram

Property Modifiers

Modify/Show Modifiers...

Currently User Specified

Reinforcement

Modify/Show Rebar...

OK

Cancel

☐ Include Automatic Rigid Zone Area Over Column

E

Frame Section Property Data

X

General Data

Property Name

C3 20 X 20

Material

f_c = 4500 psi

...

Notional Size Data

Modify/Show Notional Size...

Display Color

Change...

Notes

Modify/Show Notes...

Shape

Section Shape

Concrete Rectangular

Section Property Source

Source: User Defined

Section Dimensions

Depth

20

in

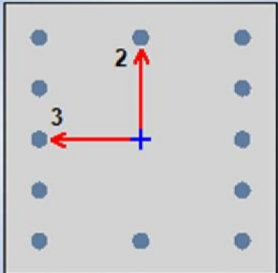
Width

20

in

Show Section Properties...

Reinforcement Diagram



Property Modifiers

Modify/Show Modifiers...

Currently User Specified

Reinforcement

Modify/Show Rebar...

OK

Cancel

☐ Include Automatic Rigid Zone Area Over Column

E

Frame Section Property Data

✕

General Data

Property Name

Floor Beam 15 X 10

Material

f_c = 4500 psi

▼

...

Notional Size Data

Modify/Show Notional Size...

Display Color

Change...

Notes

Modify/Show Notes...

Shape

Section Shape

Concrete Rectangular

▼

Section Property Source

Source: User Defined

Section Dimensions

Depth

15

in

Width

10

in

Show Section Properties...

☐ Include Automatic Rigid Zone Area Over Column

Property Modifiers

Modify/Show Modifiers...

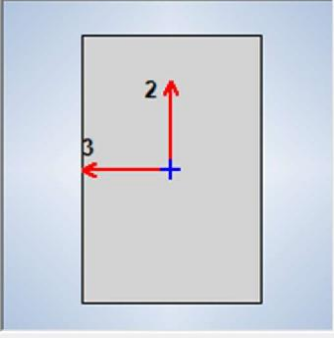
Currently User Specified

Reinforcement

Modify/Show Rebar...

OK

Cancel



Frame Section Property Data

General Data

Property Name

GC1 17 X 17

Material

Fc = 4500 psi

...

Notional Size Data

Modify/Show Notional Size...

Display Color

Change...

Notes

Modify/Show Notes...

Shape

Section Shape

Concrete Rectangular

Section Property Source

Source: User Defined

Section Dimensions

Depth

17

in

Width

17

in

Show Section Properties...

Include Automatic Rigid Zone Area Over Column

2

3

Property Modifiers

Modify/Show Modifiers...

Currently User Specified

Reinforcement

Modify/Show Rebar...

OK

Cancel

E

Frame Section Property Data

×

General Data

Property Name

GC2 20 X 20

Material

f_c = 4500 psi

...

Notional Size Data

Modify/Show Notional Size...

Display Color

Change...

Notes

Modify/Show Notes...

Shape

Section Shape

Concrete Rectangular

▼

Section Property Source

Source: User Defined

Section Dimensions

Depth

20

in

Width

20

in

Show Section Properties...

Property Modifiers

Modify/Show Modifiers...

Currently User Specified

Reinforcement

Modify/Show Rebar...

OK

Cancel

2

3

☐ Include Automatic Rigid Zone Area Over Column

34

E

Frame Section Property Data

×

General Data

Property Name

GC3 22 X 22

Material

fc = 4500 psi

...

Notional Size Data

Modify/Show Notional Size...

Display Color

Change...

Notes

Modify/Show Notes...

Shape

Section Shape

Concrete Rectangular

▼

Section Property Source

Source: User Defined

Section Dimensions

Depth

22

in

Width

22

in

Show Section Properties...

Property Modifiers

Modify/Show Modifiers...

Currently User Specified

Reinforcement

Modify/Show Rebar...

OK

Cancel

2

3

☐ Include Automatic Rigid Zone Area Over Column

35

E

Frame Section Property Data

×

General Data

Property Name

Grade Beam 17 X 12

Material

f_c = 4500 psi

...

Notional Size Data

Modify/Show Notional Size...

Display Color

Change...

Notes

Modify/Show Notes...

Shape

Section Shape

Concrete Rectangular

▼

Section Property Source

Source: User Defined

Section Dimensions

Depth

17

in

Width

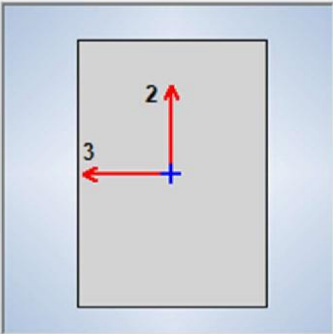
12

in

Show Section Properties...

Include Automatic Rigid Zone Area Over Column

☐



Property Modifiers

Modify/Show Modifiers...

Currently User Specified

Reinforcement

Modify/Show Rebar...

OK

Cancel

Slab Property Data

General Data

Property Name

Landing Slab

Slab Material

f_c = 3500 psi

...

Notional Size Data

Modify/Show Notional Size...

Modeling Type

Shell-Thin

Modifiers (Currently Default)

Modify/Show...

Display Color

Change...

Property Notes

Modify/Show...

Property Data

Type

Slab

Thickness

8

in

OK

Cancel

Slab Property Data

General Data

Property Name	Slab = 4"	
Slab Material	f _c = 3500 psi	...
Notional Size Data	Modify/Show Notional Size...	
Modeling Type	Shell-Thin	
Modifiers (Currently Default)	Modify/Show...	
Display Color		Change...
Property Notes	Modify/Show...	

Property Data

Type	Slab	
Thickness	4	in

OK Cancel

E

Wall Property Data

×

General Data

Property Name

Shear Wall 10 "

Property Type

Specified

▼

Wall Material

f_c = 3500 psi

▼

...

Notional Size Data

Modify/Show Notional Size...

Modeling Type

Shell-Thin

▼

Modifiers (Currently User Specified)

Modify/Show...

Display Color

Change...

Property Notes

Modify/Show...

Property Data

Thickness

10

in

☐ Include Automatic Rigid Zone Area Over Wall

OK

Cancel

9.2. Load Pattern

E Define Load Patterns

Click To:

Add New Load

Modify Load

Modify Lateral Load...

Delete Load

OK Cancel

Load	Type	Self Weight Multiplier	Auto Lateral Load
Dead	Dead	1	
Live	Live	0	
PW	Super Dead	0	
FF	Super Dead	0	
Roof Live	Roof Live	0	
EQ(x)	Seismic	0	ASCE 7-05
EQ(y)	Seismic	0	ASCE 7-05
WL(x)	Wind	0	ASCE 7-05
WL(y)	Wind	0	ASCE 7-05

E ASCE 7-05 Seismic Loading

Direction and Eccentricity

☒ X Dir ☐ Y Dir

☒ X Dir + Eccentricity ☐ Y Dir + Eccentricity

☒ X Dir - Eccentricity ☐ Y Dir - Eccentricity

Ecc. Ratio (All Diaph.) 0.05

Overwrite Eccentricities Overwrite...

Time Period

☐ Approximate ☐ Program Calculated ☒ User Defined

C_t (ft), x =

C_t (ft), x =

T = 1.21 sec

Story Range

Top Story for Seismic Loads Water Tank

Bottom Story for Seismic Loads Basement

Factors

Response Modification, R 6.5

System Overstrength, Omega 2.5

Deflection Amplification, Cd 5

Occupancy Importance, I 1

Seismic Coefficients

☐ Ss and S1 from USGS Database - by Latitude/Longitude

☐ Ss and S1 from USGS Database - by Zip Code

☒ Ss and S1 - User Defined

Site Latitude (degrees) ?

Site Longitude (degrees) ?

Site Zip Code (5-Digits) ?

0.2 Sec Spectral Accel, Ss 0.5

1 Sec Spectral Accel, S1 0.2

Long-Period Transition Period 2 sec

Site Class F

Site Coefficient, Fa 1.35

Site Coefficient, Fv 2.7

Calculated Coefficients

SDS = (2/3) * Fa * Ss 0.45

SD1 = (2/3) * Fv * S1 0.36

OK Cancel

E

Wind Load Pattern - ASCE 7-05

Exposure and Pressure Coefficients

☒ Exposure from Extents of Diaphragms

☐ Exposure from Frame and Shell Objects

- ☐ Include Shell Objects
- ☐ Include Frame Objects (Open Structure)

Wind Pressure Coefficients

☒ User Specified

☐ Program Determined

Windward Coefficient, Cpw

0.8

Leeward Coefficient, Cpl

0.35

Wind Exposure Parameters

Wind Direction and Exposure Width

Modify/Show...

Case (ASCE 7-05 Fig. 6-9)

Create All Sets

e1 Ratio (ASCE 7-05 Fig. 6-9)

0.15

e2 Ratio (ASCE 7-05 Fig. 6-9)

0.15

Wind Coefficients

Wind Speed (mph)

147.168

Exposure Type

B

Importance Factor

1

Topographical Factor, Kzt

1

Gust Factor

1.17

Directionality Factor, Kd

0.85

Solid / Gross Area Ratio

Exposure Height

Top Story

Water Tank

Bottom Story

Basement

☒ Include Parapet

Parapet Height

3 ft

OK

Cancel

E

Wind Load Pattern - ASCE 7-05

X

Exposure and Pressure Coefficients

☒ Exposure from Extents of Diaphragms

☐ Exposure from Frame and Shell Objects

☐ Include Shell Objects☐ Include Frame Objects (Open Structure)

Wind Pressure Coefficients

☒ User Specified☐ Program Determined

Windward Coefficient, Cpw0.8

Leeward Coefficient, Cpl0.35

Wind Exposure Parameters

Wind Direction and Exposure WidthModify/Show...

Case (ASCE 7-05 Fig. 6-9)Create All Sets ⓘ

e1 Ratio (ASCE 7-05 Fig. 6-9)0.15

e2 Ratio (ASCE 7-05 Fig. 6-9)0.15

Wind Coefficients

Wind Speed (mph)147.168

Exposure TypeB ▾

Importance Factor1

Topographical Factor, Kzt1

Gust Factor1.06

Directionality Factor, Kd0.85

Solid / Gross Area Ratio

Exposure Height

Top StoryWater Tank ▾

Bottom StoryBasement ▾

☒ Include Parapet
Parapet Height3 ft

OK

Cancel

E

Load Combinations

×

Combinations

0.9D - 1.6W_x

0.9D - 1.6W_y

0.9D + 1.6W_x

0.9D + 1.6W_y

0.854D + E_x

0.854D + E_y

0.854D - E_x

0.854D - E_y

1.2 D - 1.6W_x + L + 0.5L_r

1.2 D - 1.6W_y + L + 0.5 L_r

1.2D + 1.6L + 0.5L_r

1.2D + 1.6L_r - 0.8W_x

1.2D + 1.6L_r - 0.8W_y

1.2D + 1.6L_r + 0.8W_x

1.2D + 1.6L_r + 0.8W_y

Click to:

Add New Combo...

Add Copy of Combo...

Modify/Show Combo...

Delete Combo

Add Default Design Combos...

Convert Combos to Nonlinear Cases...

OK

Cancel

E

Load Combinations

×

Combinations

1.2D + 1.6L_r + 0.8W_y

1.2D + 1.6L_r + L

1.2 D + 1.6 W_x + L + 0.5 L_r

1.2 D + 1.6 W_y + L + 0.5 L_r

1.4D

1.246D - 1.0E_x + 1.0L

1.246D - 1.0E_y + 1.0L

1.246D + 1.0E_x + 1.0L

1.246D + 1.0E_y + 1.0L

D + 0.5L

D + 0.5L - 0.7W_x

D + 0.5L - 0.7 W_y

D + 0.5L + 0.7W_x

D + 0.5L + 0.7W_y

D + L

Click to:

Add New Combo...

Add Copy of Combo...

Modify/Show Combo...

Delete Combo

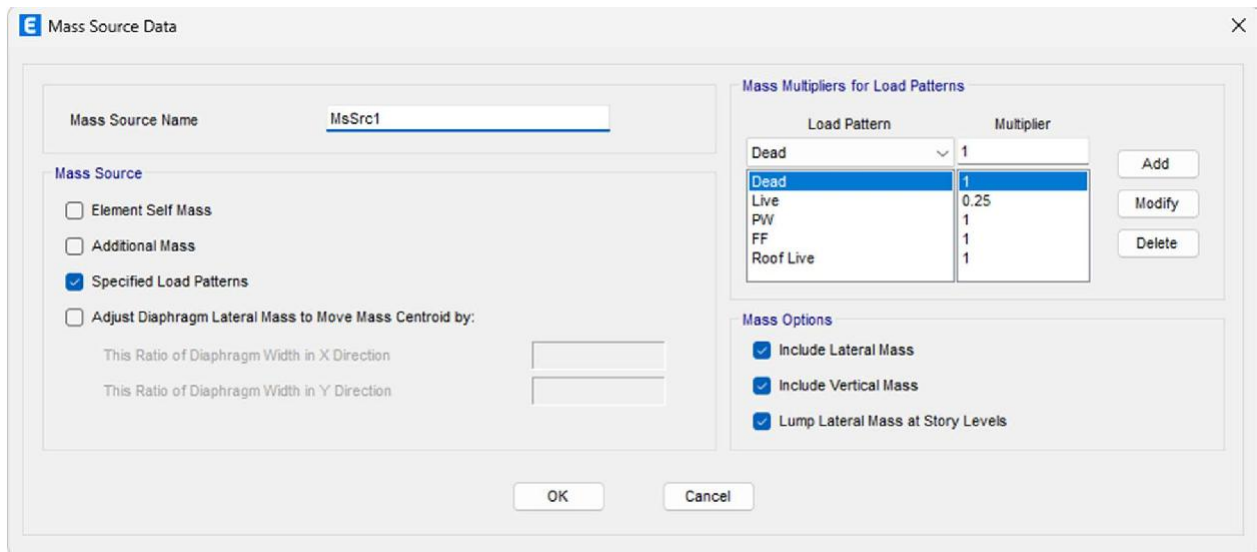
Add Default Design Combos...

Convert Combos to Nonlinear Cases...

OK

Cancel

9.3. Mass Source



The "Mass Source Data" dialog box is used to define mass sources for a structure. It includes fields for the mass source name, checkboxes for mass source types, and a table for load pattern multipliers.

Mass Source Name: MsSrc1

Mass Source:

- ☐ Element Self Mass
- ☐ Additional Mass
- ☒ Specified Load Patterns
- ☐ Adjust Diaphragm Lateral Mass to Move Mass Centroid by:
 - This Ratio of Diaphragm Width in X Direction:
 - This Ratio of Diaphragm Width in Y Direction:

Mass Multipliers for Load Patterns

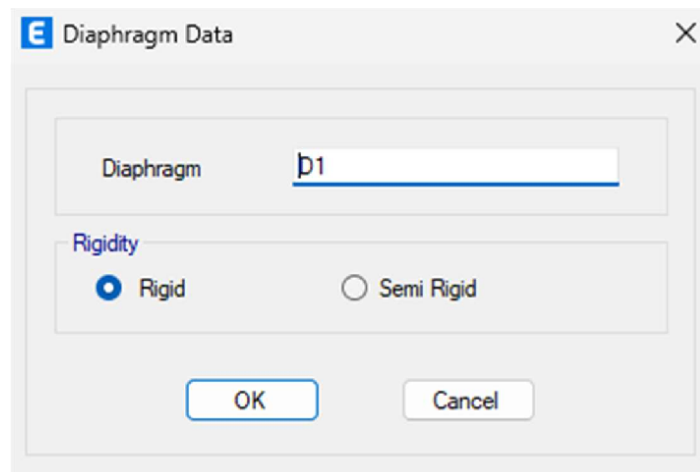
Load Pattern	Multiplier
Dead	1
Live	0.25
P/W	1
FF	1
Roof Live	1

Mass Options:

- ☒ Include Lateral Mass
- ☒ Include Vertical Mass
- ☒ Lump Lateral Mass at Story Levels

Buttons: OK, Cancel

Diaphragm



The "Diaphragm Data" dialog box is used to define diaphragm properties. It includes a field for the diaphragm name and a section for rigidity.

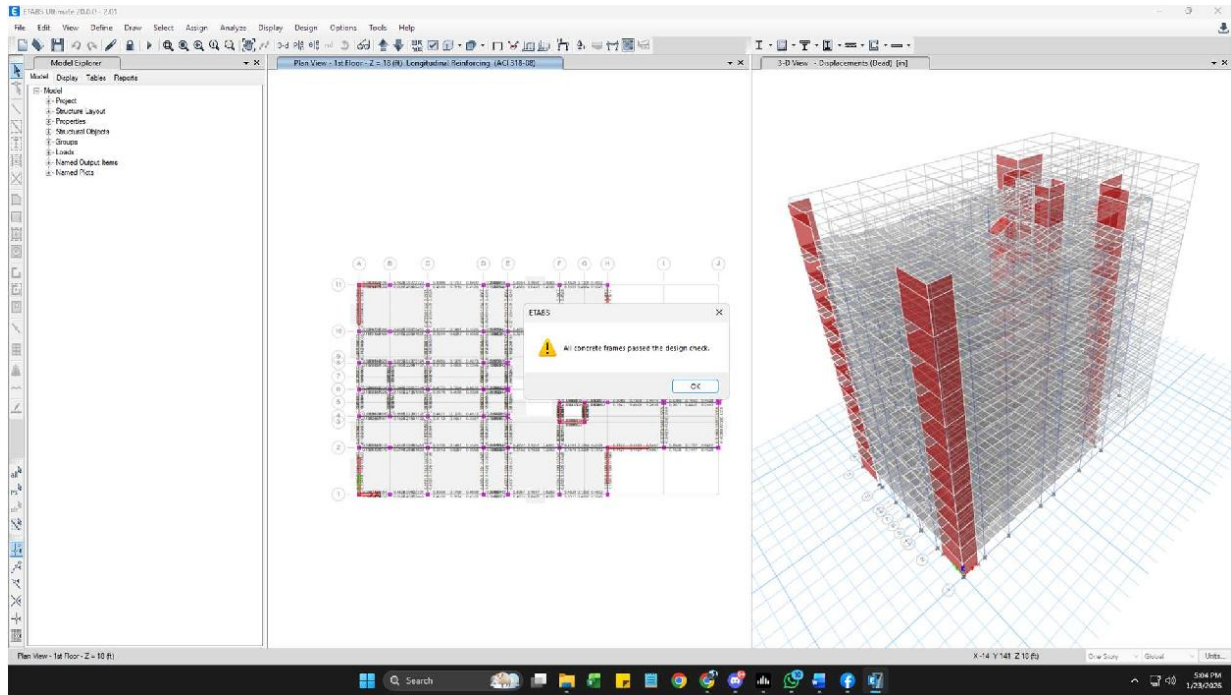
Diaphragm: p1

Rigidity:

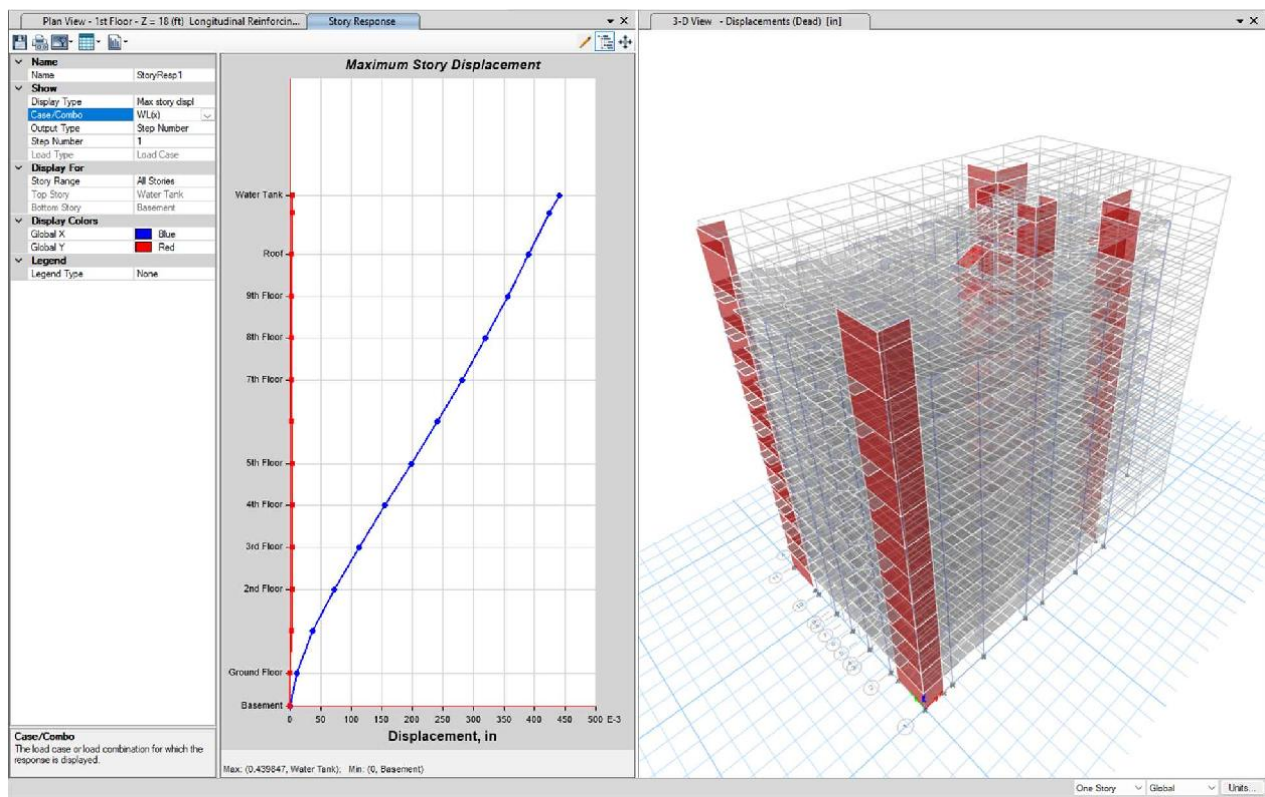
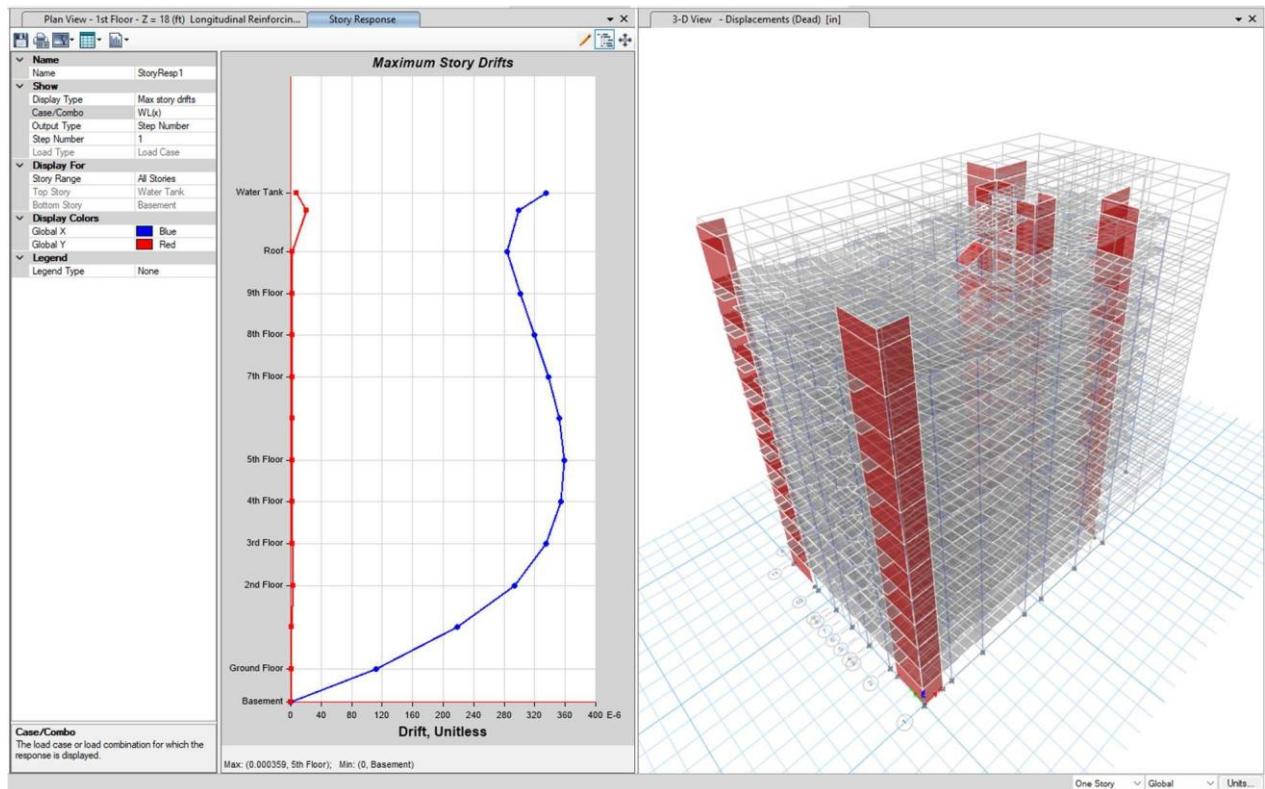
☒ Rigid ☐ Semi Rigid

Buttons: OK, Cancel

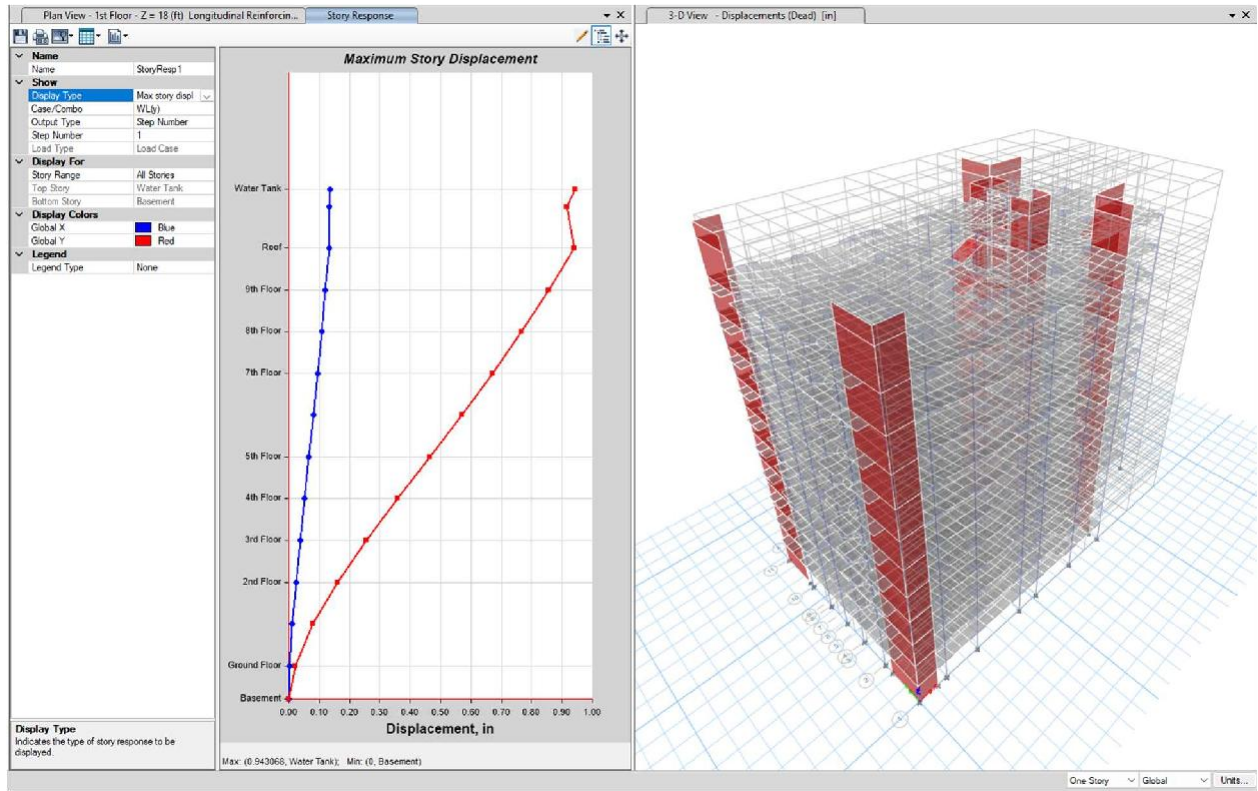
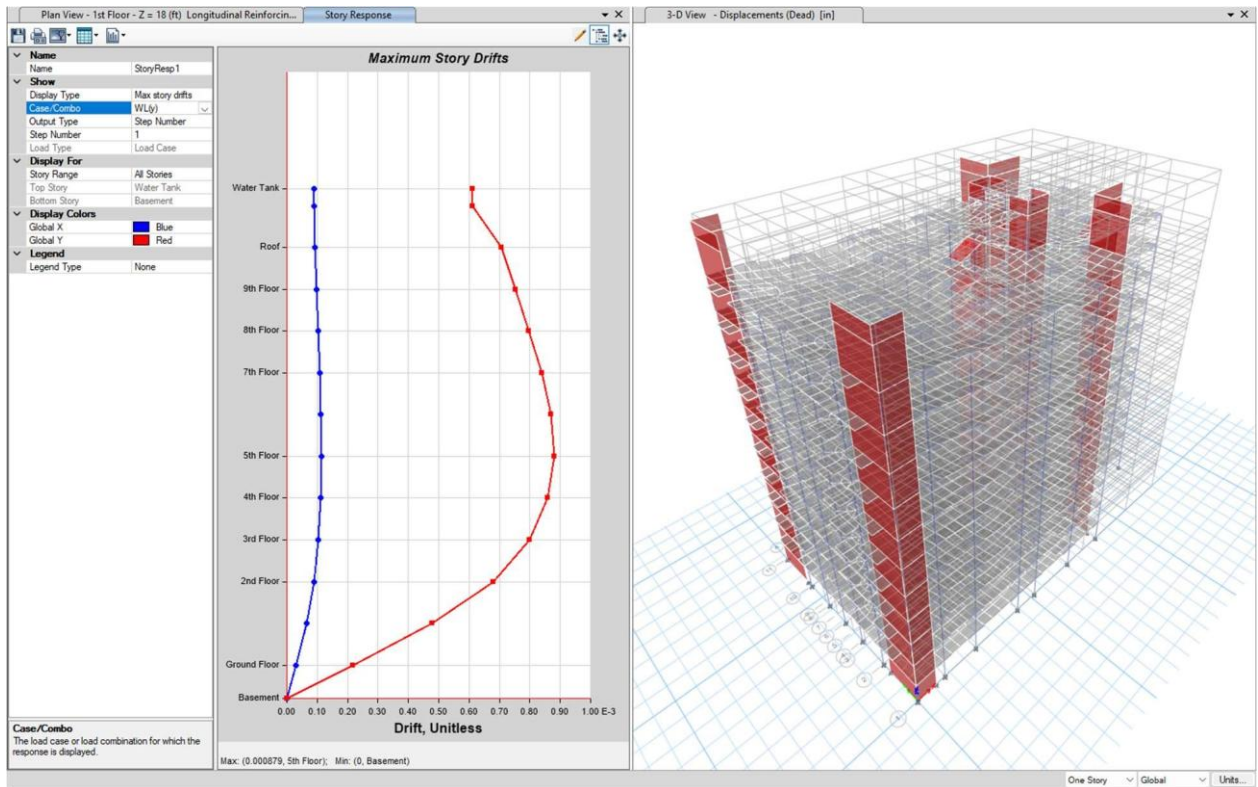
10. ETABS Run Analysis and Design Check



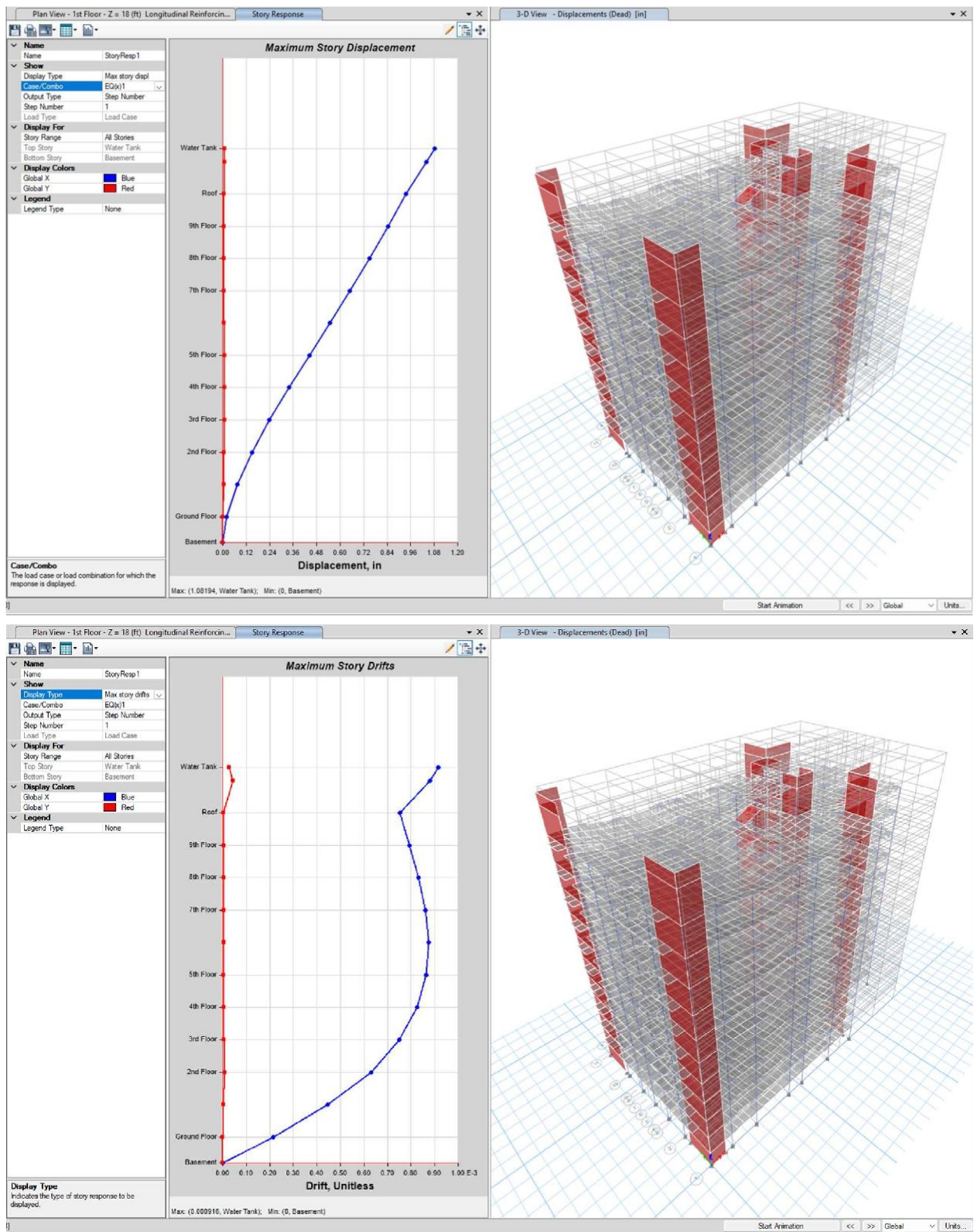
10.1. Wind Drift Check WL(x)



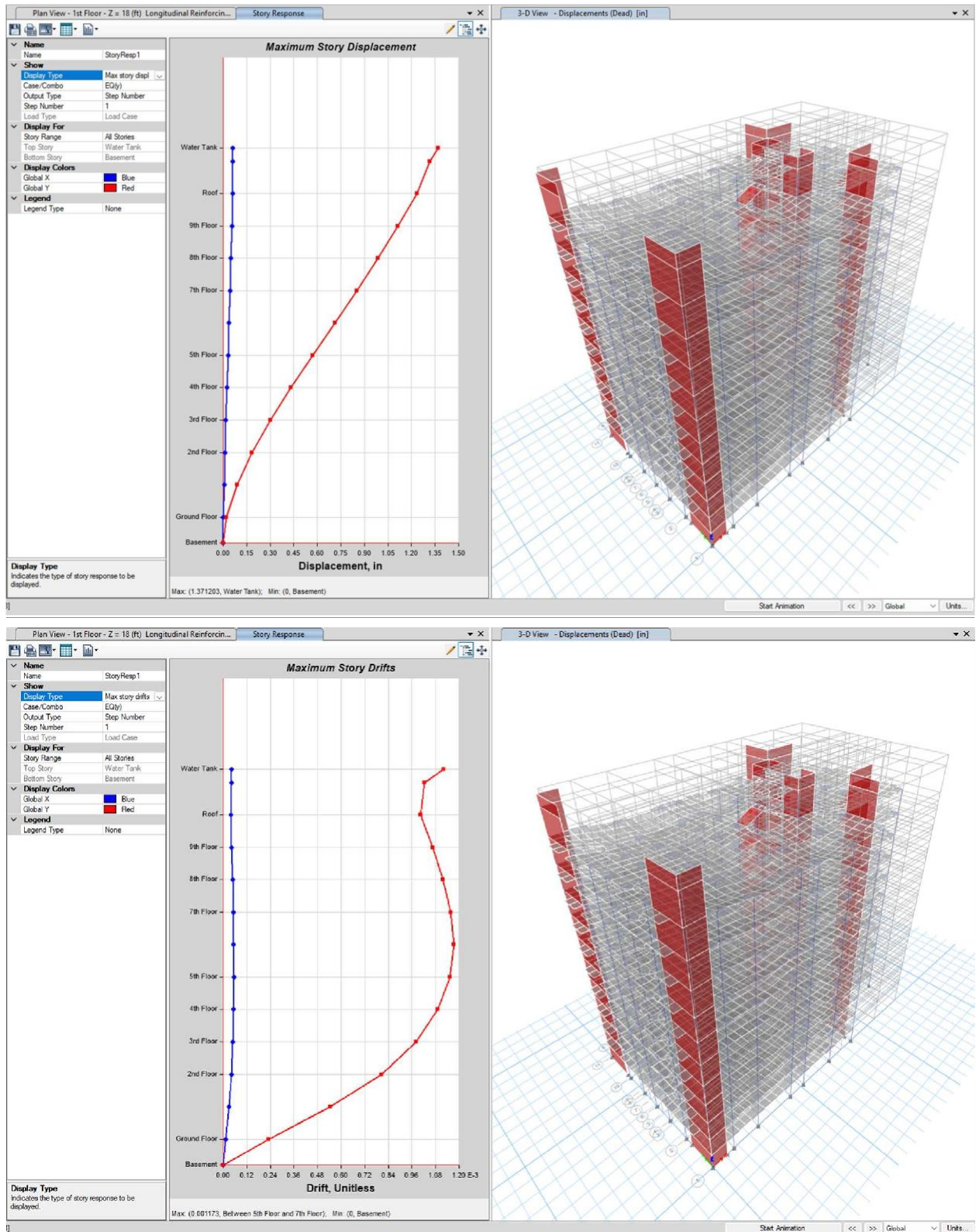
10.2. Wind Drift Check WL(y)



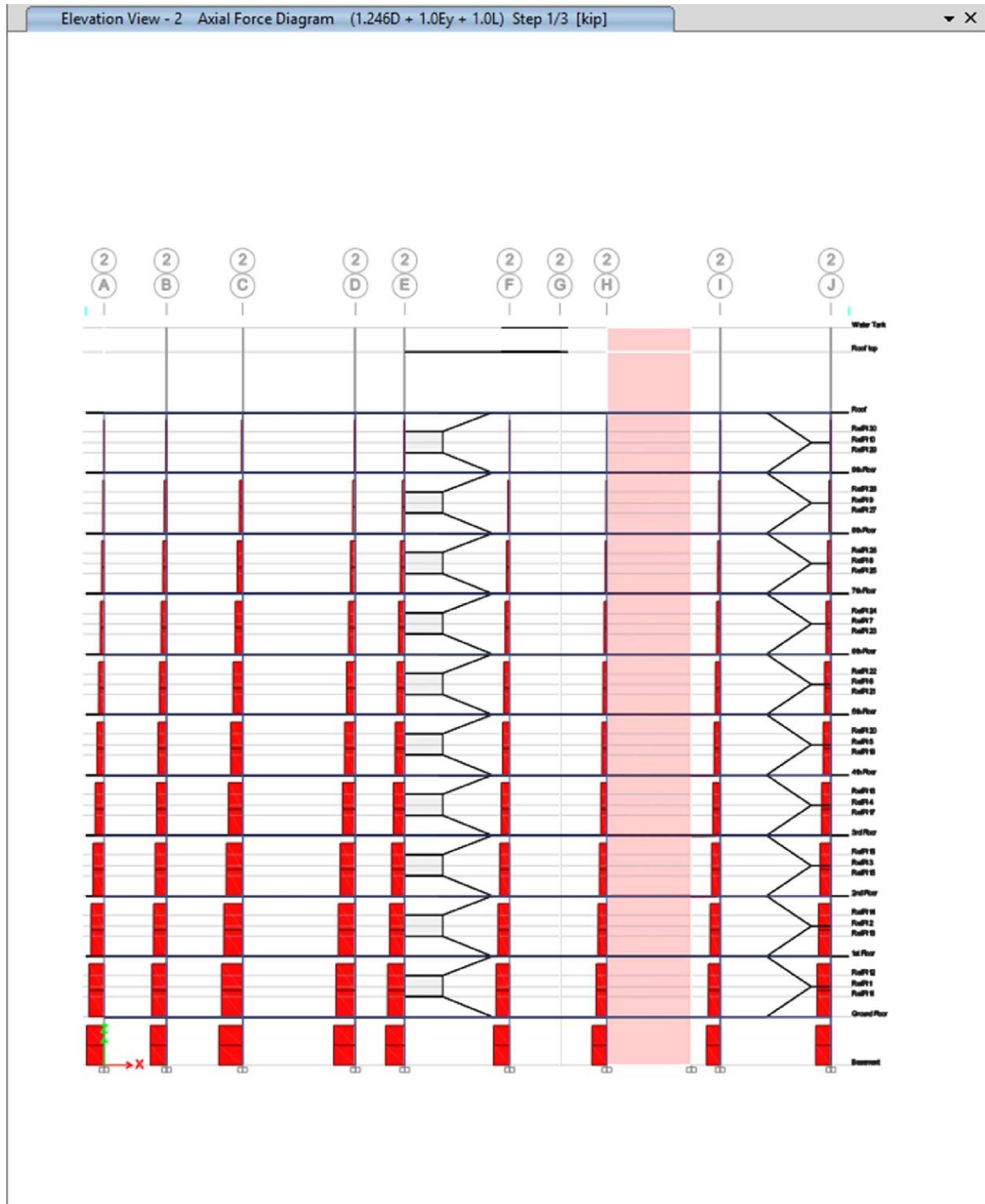
10.3. Earthquake Drift Check EQ(x)



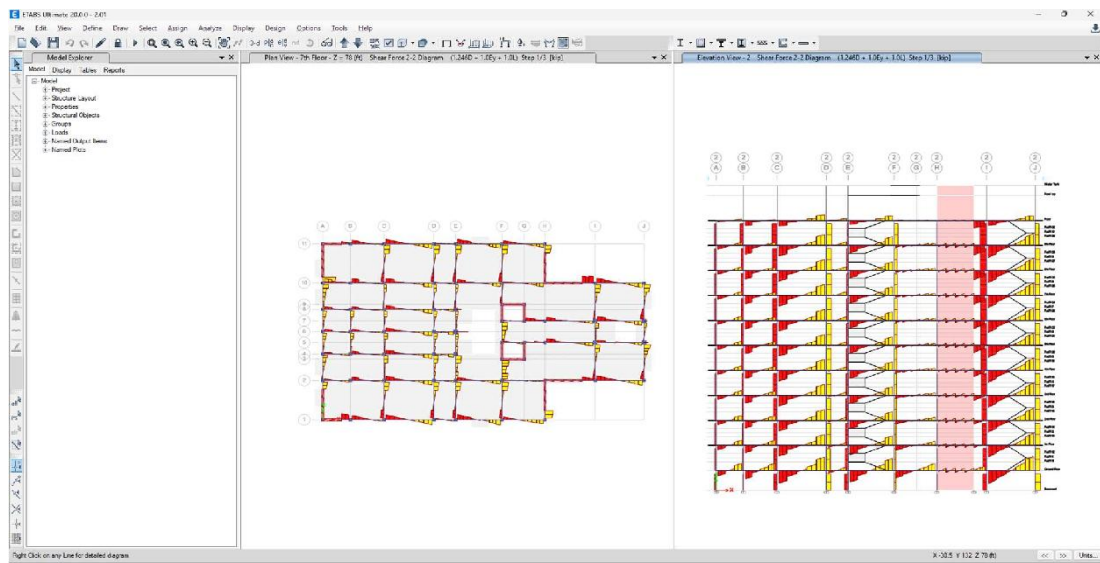
10.4. Earthquake Drift Check EQ(y)



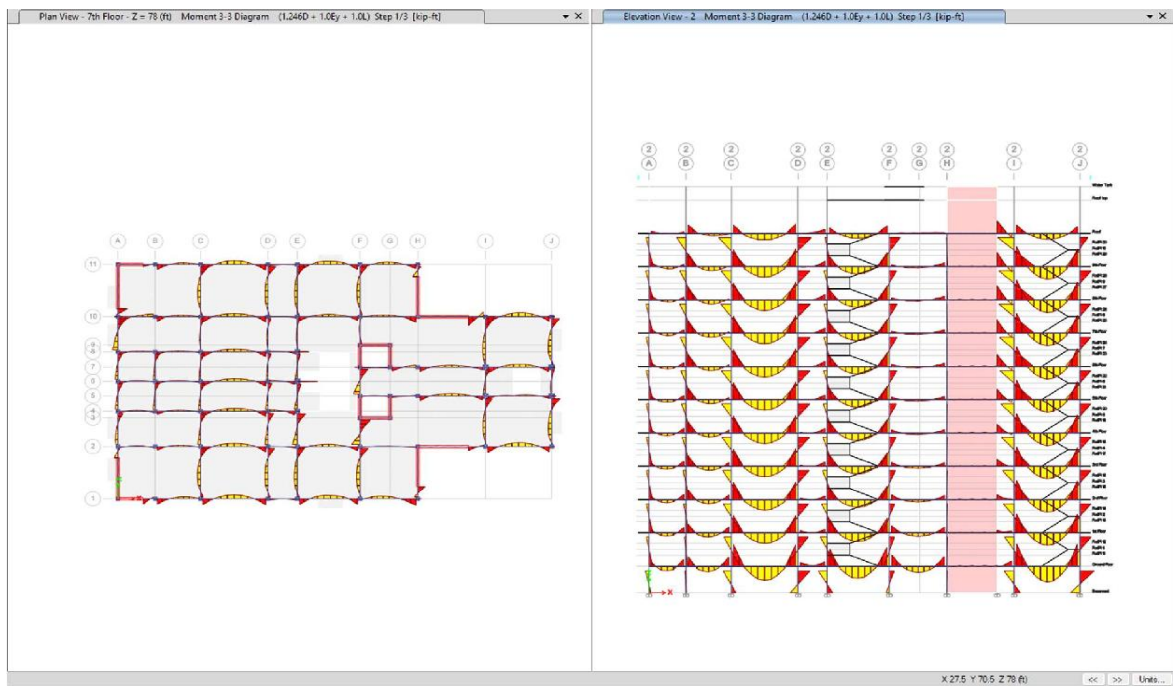
10.5. Axial Force Diagram



10.6. Shear Force Diagram



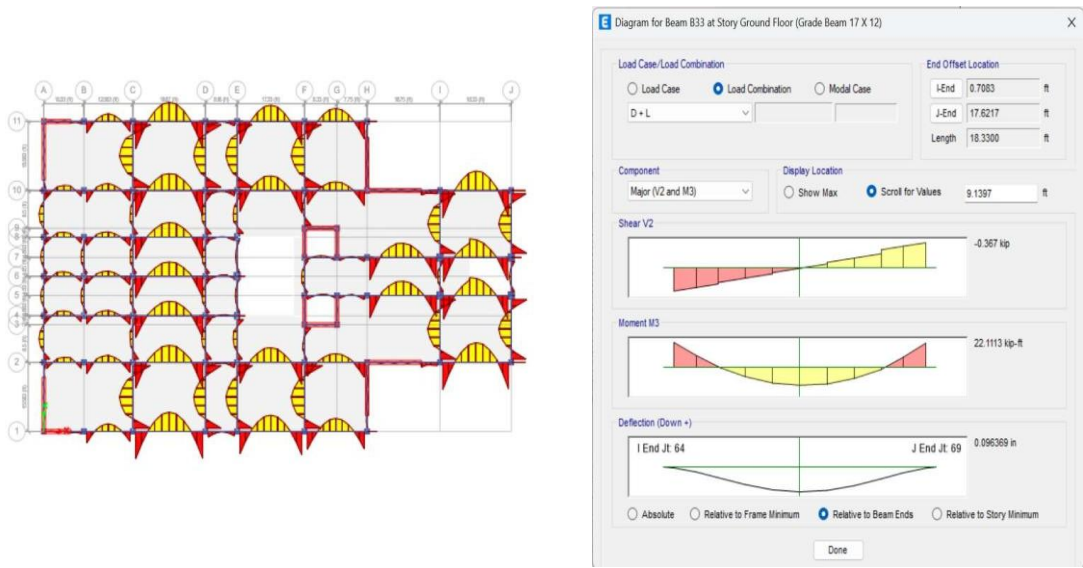
10.7. Bending Moment Diagram



11. SERVICEABILITY CHECK

Serviceability	Location	Value	Check
Beam Deflection	Ground Floor	0.9165	OK
Wind Sway X	Roof	0.793245	OK
Wind Sway Y	Roof	0.656617	OK
Wind Drift X	5 th Floor	0.042913	OK
Wind Drift Y	5 th Floor	0.103978	OK
Torsion X	6 th Floor	0.104848	OK
Torsion Y	6 th Floor	0.133873	OK
EQ Drift X	6 th Floor	0.522345	OK
EQ Drift Y	6 th Floor	0.69886	OK

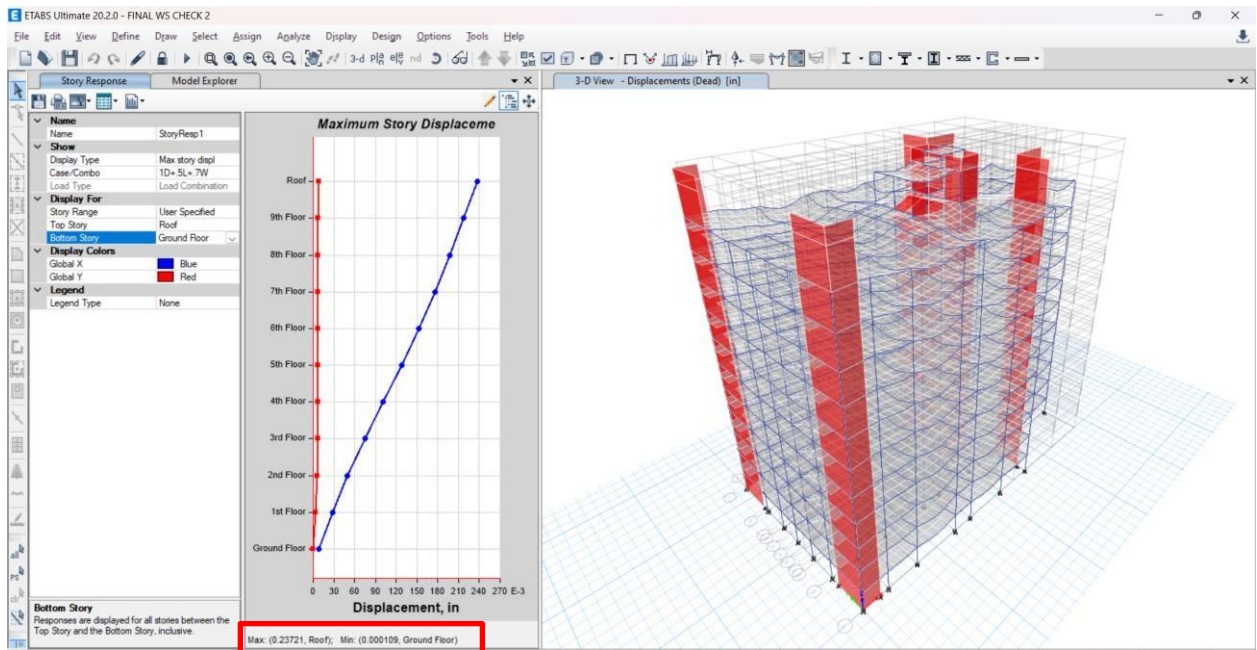
11.1. Beam Deflection



Maximum Beam Deflection (in)	Allowable Beam Deflection (1/240) in	Remarks
0.096369	0.9165	OK

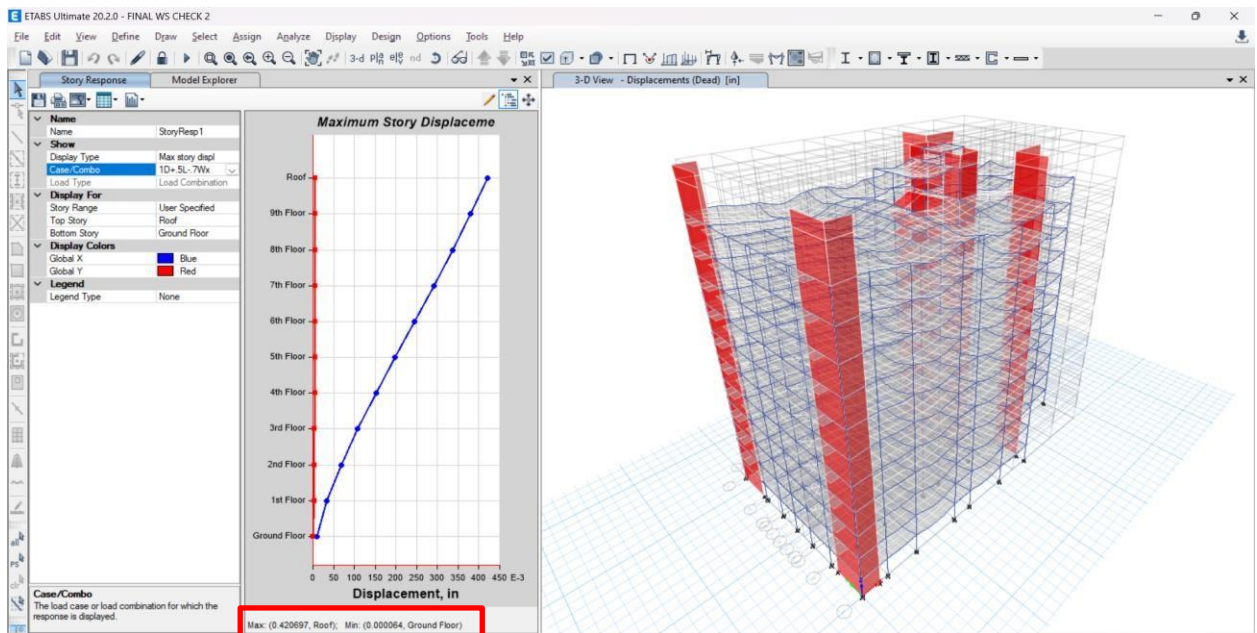
11.2. Wind Sway Check

$W_x +$



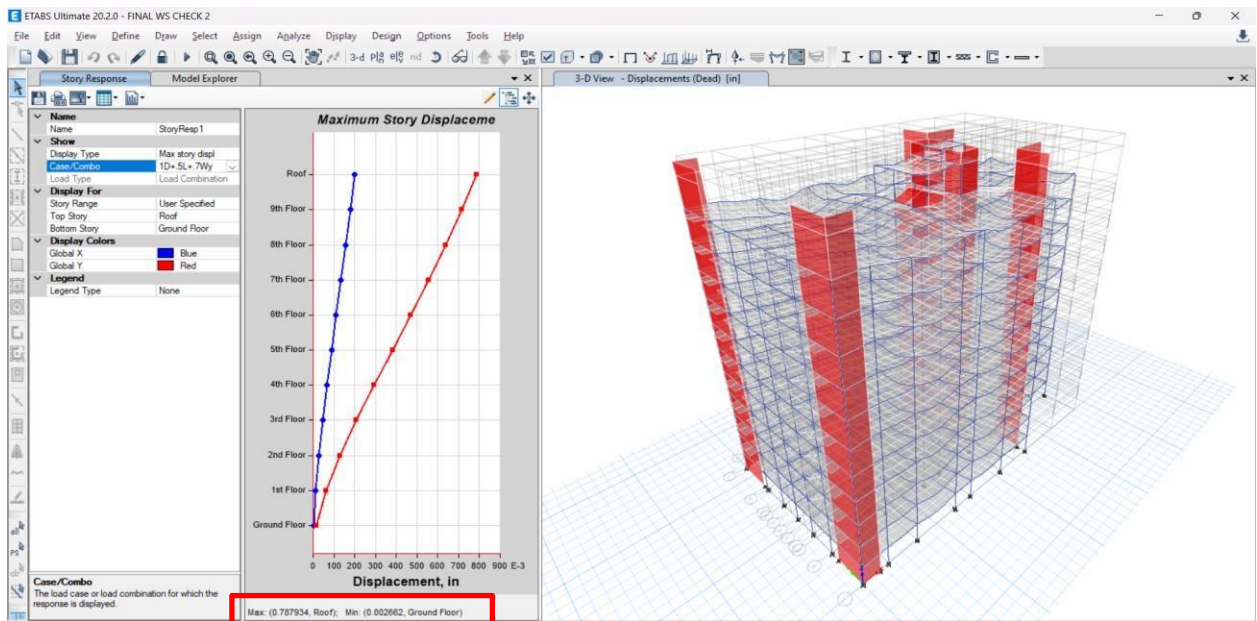
Max= 0.23721 < 2.4 (OK)

$W_x -$



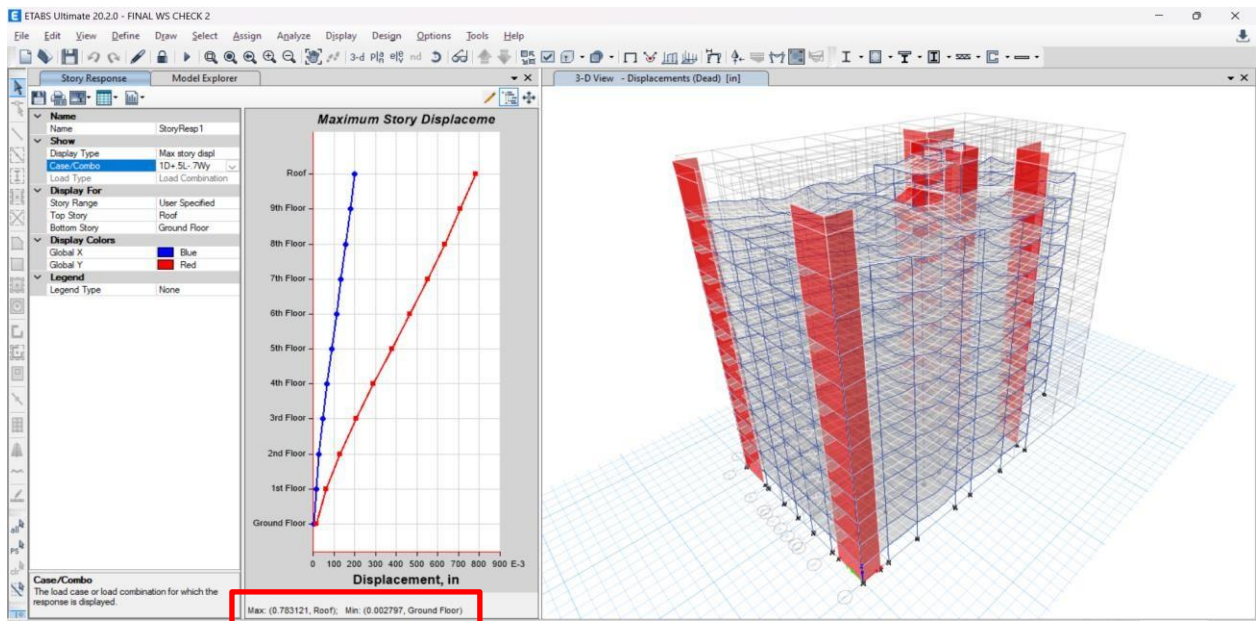
Max= 0.420697 < 2.4 (OK)

$W_y +$



Max= 0.787934 < 2.4 (OK)

$W_y -$



Max= 0.783121 < 2.4 (OK)

Load Combo	Max story displacement (in)	Story ht above ground (ft)	Story ht above ground (in)	Allowable story displacement(in)	Remarks
1D+.5L+.7W _x	0.23721	100	1200	(100/500)*12=2.4	OK
1D+.5L-.7W _x	0.420697				OK
1D+.5L+.7W _y	0.787934				OK
1D+.5L-.7W _y	0.783121				OK

11.3. Story Drift Check

W _x				
	Elastic Displacement (in)	Story Drift (in)	Allowable Drift (in)	Result
Roof top	0.422237	0.03469	0.48	Ok
Roof	0.387547	0.033765	0.48	Ok
9th Floor	0.353782	0.035904	0.48	Ok
8th Floor	0.317878	0.038158	0.48	Ok
7th Floor	0.27972	0.040372	0.48	Ok
6th Floor	0.239348	0.042089	0.48	Ok
5th Floor	0.197259	0.042913	0.48	Ok
4th Floor	0.154346	0.042397	0.48	Ok
3rd Floor	0.111949	0.040015	0.48	Ok
2nd Floor	0.071934	0.035013	0.48	Ok
1st Floor	0.036921	0.026191	0.48	Ok
Ground Floor	0.01073	0.01073	0.48	Ok

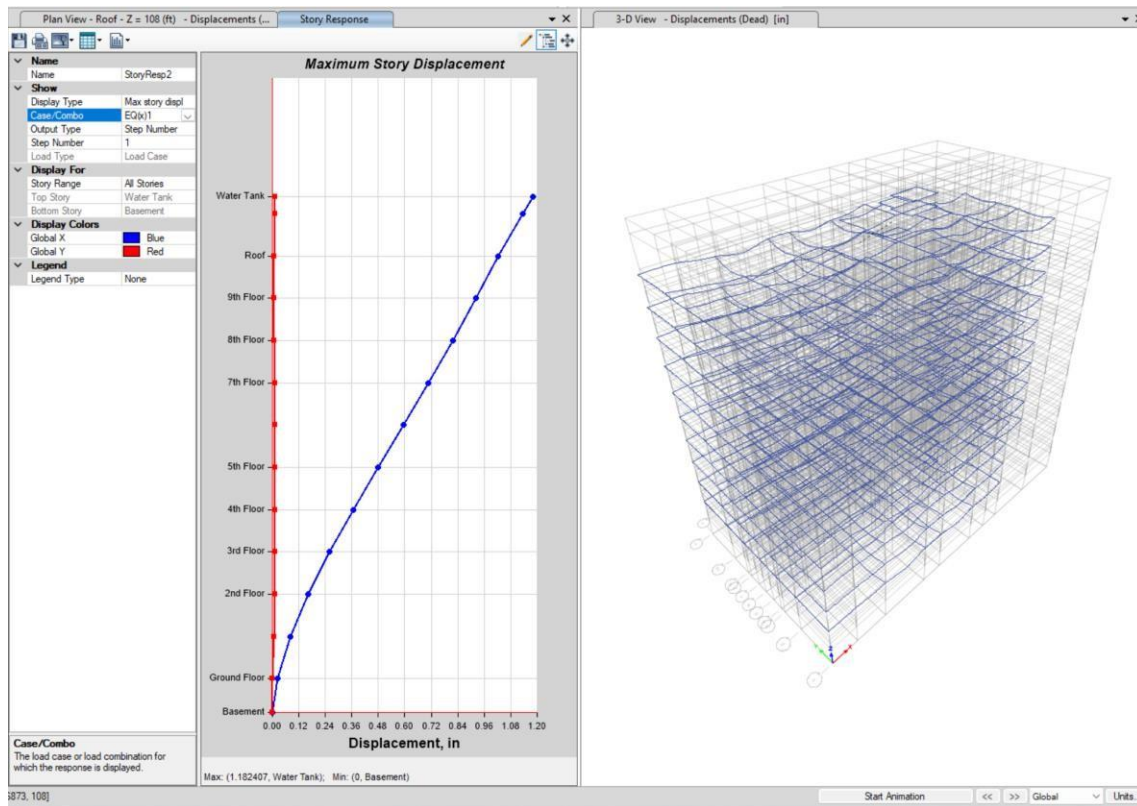
W _y				
	Elastic Displacement (in)	Story Drift (in)	Allowable Drift (in)	Result
Roof top	0.844838	-0.082314	0.48	Ok
Roof	0.927152	0.083535	0.48	Ok
9th Floor	0.843617	0.088871	0.48	Ok
8th Floor	0.754746	0.094276	0.48	Ok
7th Floor	0.66047	0.099273	0.48	Ok
6th Floor	0.561197	0.102828	0.48	Ok
5th Floor	0.458369	0.103978	0.48	Ok
4th Floor	0.354391	0.10161	0.48	Ok
3rd Floor	0.252781	0.094359	0.48	Ok
2nd Floor	0.158422	0.080396	0.48	Ok
1st Floor	0.078026	0.057123	0.48	Ok
Ground Floor	0.020903	0.020903	0.48	Ok

11.4. Torsional Irregularity

Torsional Irregularity								
EQ(x)								
Story	Output Case	Case Type	Step Type	Step Number	Direction	Max Drift in	Avg Drift in	Ratio
Water Tank	EQ(x)1	LinStatic	Step By Step	1	X	0.043985	0.03991	1.102
Water Tank	EQ(x)1	LinStatic	Step By Step	2	X	0.046655	0.040169	1.161
Water Tank	EQ(x)1	LinStatic	Step By Step	3	X	0.04581	0.039678	1.155
Roof top	EQ(x)1	LinStatic	Step By Step	1	X	0.1059	0.097862	1.082
Roof top	EQ(x)1	LinStatic	Step By Step	2	X	0.112228	0.098083	1.144
Roof top	EQ(x)1	LinStatic	Step By Step	3	X	0.111073	0.097336	1.141
Roof	EQ(x)1	LinStatic	Step By Step	1	X	0.090351	0.089943	1.005
Roof	EQ(x)1	LinStatic	Step By Step	2	X	0.096439	0.089733	1.075
Roof	EQ(x)1	LinStatic	Step By Step	3	X	0.096934	0.089783	1.08
9th Floor	EQ(x)1	LinStatic	Step By Step	1	X	0.095187	0.094936	1.003
9th Floor	EQ(x)1	LinStatic	Step By Step	2	X	0.101913	0.094932	1.074
9th Floor	EQ(x)1	LinStatic	Step By Step	3	X	0.102422	0.09498	1.078
8th Floor	EQ(x)1	LinStatic	Step By Step	1	X	0.099857	0.099622	1.002
8th Floor	EQ(x)1	LinStatic	Step By Step	2	X	0.106779	0.099598	1.072
8th Floor	EQ(x)1	LinStatic	Step By Step	3	X	0.107298	0.099646	1.077
7th Floor	EQ(x)1	LinStatic	Step By Step	1	X	0.103487	0.103245	1.002
7th Floor	EQ(x)1	LinStatic	Step By Step	2	X	0.110503	0.103218	1.071
7th Floor	EQ(x)1	LinStatic	Step By Step	3	X	0.11104	0.103272	1.075
6th Floor	EQ(x)1	LinStatic	Step By Step	1	X	0.105064	0.104818	1.002
6th Floor	EQ(x)1	LinStatic	Step By Step	2	X	0.112023	0.104789	1.069
6th Floor	EQ(x)1	LinStatic	Step By Step	3	X	0.112573	0.104848	1.074
5th Floor	EQ(x)1	LinStatic	Step By Step	1	X	0.103829	0.103577	1.002
5th Floor	EQ(x)1	LinStatic	Step By Step	2	X	0.110545	0.103554	1.068
5th Floor	EQ(x)1	LinStatic	Step By Step	3	X	0.111093	0.103619	1.072
4th Floor	EQ(x)1	LinStatic	Step By Step	1	X	0.099049	0.0988	1.003
4th Floor	EQ(x)1	LinStatic	Step By Step	2	X	0.105301	0.098787	1.066
4th Floor	EQ(x)1	LinStatic	Step By Step	3	X	0.105825	0.098855	1.071
3rd Floor	EQ(x)1	LinStatic	Step By Step	1	X	0.089932	0.089687	1.003
3rd Floor	EQ(x)1	LinStatic	Step By Step	2	X	0.095462	0.089699	1.064
3rd Floor	EQ(x)1	LinStatic	Step By Step	3	X	0.095929	0.089767	1.069
2nd Floor	EQ(x)1	LinStatic	Step By Step	1	X	0.075623	0.075283	1.005
2nd Floor	EQ(x)1	LinStatic	Step By Step	2	X	0.0798	0.075113	1.062
2nd Floor	EQ(x)1	LinStatic	Step By Step	3	X	0.080165	0.075174	1.066
1st Floor	EQ(x)1	LinStatic	Step By Step	1	X	0.053618	0.053529	1.002
1st Floor	EQ(x)1	LinStatic	Step By Step	2	X	0.056723	0.053507	1.06
1st Floor	EQ(x)1	LinStatic	Step By Step	3	X	0.056946	0.053551	1.063
Ground Floor	EQ(x)1	LinStatic	Step By Step	1	X	0.020514	0.020496	1.001
Ground Floor	EQ(x)1	LinStatic	Step By Step	2	X	0.021673	0.020492	1.058
Ground Floor	EQ(x)	LinStatic	Step by Step	3	X	0.021718	0.0205	1.059

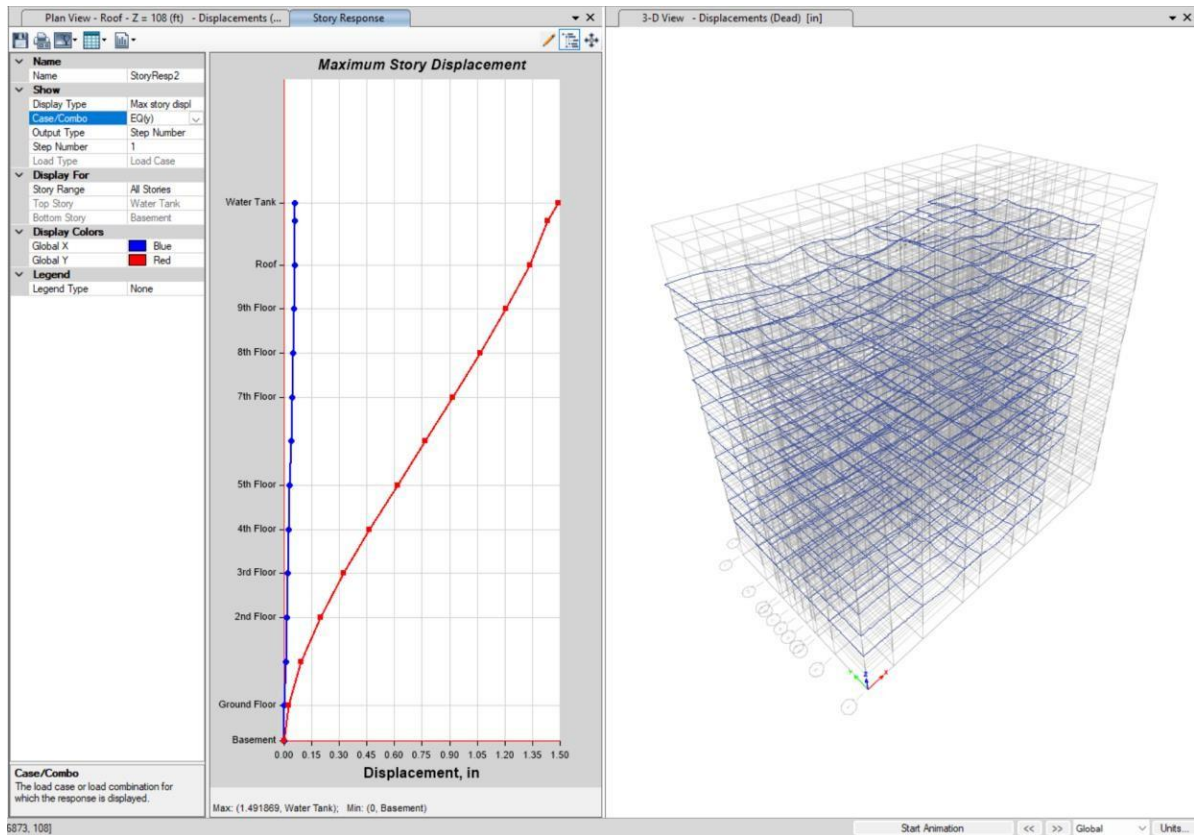
Torsional Irregularity								
EQ(Y)								
Story	Output Case	Case Type	Step Type	Step Number	Direction	Max Drift in	Avg Drift in	Ratio
Water Tank	EQ(y)	LinStatic	Step By Step	1	Y	0.053929	0.046969	1.148
Water Tank	EQ(y)	LinStatic	Step By Step	2	Y	0.058029	0.047055	1.233
Water Tank	EQ(y)	LinStatic	Step By Step	3	Y	0.051869	0.04509	1.15
Roof top	EQ(y)	LinStatic	Step By Step	1	Y	0.123222	0.112272	1.098
Roof top	EQ(y)	LinStatic	Step By Step	2	Y	0.13311	0.111668	1.192
Roof top	EQ(y)	LinStatic	Step By Step	3	Y	0.130537	0.116348	1.122
Roof	EQ(y)	LinStatic	Step By Step	1	Y	0.120866	0.113884	1.061
Roof	EQ(y)	LinStatic	Step By Step	2	Y	0.144418	0.117196	1.232
Roof	EQ(y)	LinStatic	Step By Step	3	Y	0.123829	0.110572	1.12
9th Floor	EQ(y)	LinStatic	Step By Step	1	Y	0.128278	0.11986	1.07
9th Floor	EQ(y)	LinStatic	Step By Step	2	Y	0.152942	0.123452	1.239
9th Floor	EQ(y)	LinStatic	Step By Step	3	Y	0.128921	0.116268	1.109
8th Floor	EQ(y)	LinStatic	Step By Step	1	Y	0.134662	0.125101	1.076
8th Floor	EQ(y)	LinStatic	Step By Step	2	Y	0.160147	0.12891	1.242
8th Floor	EQ(y)	LinStatic	Step By Step	3	Y	0.133405	0.121291	1.1
7th Floor	EQ(y)	LinStatic	Step By Step	1	Y	0.139327	0.128813	1.082
7th Floor	EQ(y)	LinStatic	Step By Step	2	Y	0.165293	0.132779	1.245
7th Floor	EQ(y)	LinStatic	Step By Step	3	Y	0.136332	0.124847	1.092
6th Floor	EQ(y)	LinStatic	Step By Step	1	Y	0.141029	0.129845	1.086
6th Floor	EQ(y)	LinStatic	Step By Step	2	Y	0.166925	0.133873	1.247
6th Floor	EQ(y)	LinStatic	Step By Step	3	Y	0.136499	0.125816	1.085
5th Floor	EQ(y)	LinStatic	Step By Step	1	Y	0.13876	0.127266	1.09
5th Floor	EQ(y)	LinStatic	Step By Step	2	Y	0.16388	0.131238	1.249
5th Floor	EQ(y)	LinStatic	Step By Step	3	Y	0.132948	0.123294	1.078
4th Floor	EQ(y)	LinStatic	Step By Step	1	Y	0.131488	0.120152	1.094
4th Floor	EQ(y)	LinStatic	Step By Step	2	Y	0.154974	0.123921	1.251
4th Floor	EQ(y)	LinStatic	Step By Step	3	Y	0.124761	0.116382	1.072
3rd Floor	EQ(y)	LinStatic	Step By Step	1	Y	0.118029	0.107468	1.098
3rd Floor	EQ(y)	LinStatic	Step By Step	2	Y	0.138857	0.110857	1.253
3rd Floor	EQ(y)	LinStatic	Step By Step	3	Y	0.110958	0.10408	1.066
2nd Floor	EQ(y)	LinStatic	Step By Step	1	Y	0.096811	0.087919	1.101
2nd Floor	EQ(y)	LinStatic	Step By Step	2	Y	0.113757	0.090711	1.254
2nd Floor	EQ(y)	LinStatic	Step By Step	3	Y	0.09039	0.085128	1.062
1st Floor	EQ(y)	LinStatic	Step By Step	1	Y	0.065413	0.059419	1.101
1st Floor	EQ(y)	LinStatic	Step By Step	2	Y	0.076524	0.061096	1.253
1st Floor	EQ(y)	LinStatic	Step By Step	3	Y	0.061183	0.057743	1.06
Ground Floor	EQ(y)	LinStatic	Step By Step	1	Y	0.022199	0.020415	1.087
Ground Floor	EQ(y)	LinStatic	Step By Step	2	Y	0.026194	0.020987	1.248
Ground Floor	EQ(y)	LinStatic	Step By Step	3	Y	0.021483	0.019843	1.083

11.5. Drift Check For Eqx



Ex					
	Elastic Displacement (in)	Amplified Displacement (in)	Story Drift (in)	Allowable Drift (in)	Result
Roof top	1.020071	5.100355	0.44742	2.4	Ok
Roof	0.930587	4.652935	0.44686	2.4	Ok
9th Floor	0.841215	4.206075	0.472885	2.4	Ok
8th Floor	0.746638	3.73319	0.49622	2.4	Ok
7th Floor	0.647394	3.23697	0.514385	2.4	Ok
6th Floor	0.544517	2.722585	0.522345	2.4	Ok
5th Floor	0.440048	2.20024	0.51632	2.4	Ok
4th Floor	0.336784	1.68392	0.492665	2.4	Ok
3rd Floor	0.238251	1.191255	0.447435	2.4	Ok
2nd Floor	0.148764	0.74382	0.37474	2.4	Ok
1st Floor	0.073816	0.36908	0.266955	2.4	Ok
Ground Floor	0.020425	0.102125	0.102125	2.4	Ok

11.6. Drift Check For E_y



E _y					
	Elastic Displacement (in)	Amplified Displacement (in)	Story Drift (in)	Allowable Drift (in)	Result
Roof top	1.267751	6.338755	0.207085	2.4	Ok
Roof	1.226334	6.13167	0.59907	2.4	Ok
9th Floor	1.10652	5.5326	0.63576	2.4	Ok
8th Floor	0.979368	4.89684	0.667365	2.4	Ok
7th Floor	0.845895	4.229475	0.69046	2.4	Ok
6th Floor	0.707803	3.539015	0.69886	2.4	Ok
5th Floor	0.568031	2.840155	0.6876	2.4	Ok
4th Floor	0.430511	2.152555	0.651565	2.4	Ok
3rd Floor	0.300198	1.50099	0.584895	2.4	Ok
2nd Floor	0.183219	0.916095	0.47981	2.4	Ok
1st Floor	0.087257	0.436285	0.325745	2.4	Ok
Ground Floor	0.022108	0.11054	0.11054	2.4	Ok

Ex					
	Elastic Displacement (in)	Amplified Displacement (in)	Story Drift (in)	Allowable Drift (in)	Result
Roof top	1.020071	5.100355	0.44742	2.4	Ok
Roof	0.930587	4.652935	0.44686	2.4	Ok
9th Floor	0.841215	4.206075	0.472885	2.4	Ok
8th Floor	0.746638	3.73319	0.49622	2.4	Ok
7th Floor	0.647394	3.23697	0.514385	2.4	Ok
6th Floor	0.544517	2.722585	0.522345	2.4	Ok
5th Floor	0.440048	2.20024	0.51632	2.4	Ok
4th Floor	0.336784	1.68392	0.492665	2.4	Ok
3rd Floor	0.238251	1.191255	0.447435	2.4	Ok
2nd Floor	0.148764	0.74382	0.37474	2.4	Ok
1st Floor	0.073816	0.36908	0.266955	2.4	Ok
Ground Floor	0.020425	0.102125	0.102125	2.4	Ok

Ey					
	Elastic Displacement (in)	Amplified Displacement (in)	Story Drift (in)	Allowable Drift (in)	Result
Roof top	1.267751	6.338755	0.207085	2.4	Ok
Roof	1.226334	6.13167	0.59907	2.4	Ok
9th Floor	1.10652	5.5326	0.63576	2.4	Ok
8th Floor	0.979368	4.89684	0.667365	2.4	Ok
7th Floor	0.845895	4.229475	0.69046	2.4	Ok
6th Floor	0.707803	3.539015	0.69886	2.4	Ok
5th Floor	0.568031	2.840155	0.6876	2.4	Ok
4th Floor	0.430511	2.152555	0.651565	2.4	Ok
3rd Floor	0.300198	1.50099	0.584895	2.4	Ok
2nd Floor	0.183219	0.916095	0.47981	2.4	Ok
1st Floor	0.087257	0.436285	0.325745	2.4	Ok
Ground Floor	0.022108	0.11054	0.11054	2.4	Ok

11.7. Soft Story Check

Story	Output Case	Case Type	Step Type	Step Number	Shear X kip	Drift X in	Stiff X kip/in	Ratio	Avarage	Shear Y kip	Drift Y in	Stiff Y kip/in
Water Tank	EQ(x)1	LinStatic	Step By Step	1	5.468	0.03991	137.0165	#VALUE!	1	0	0.000659	0
Roof top	EQ(x)1	LinStatic	Step By Step	1	30.092	0.097862	307.4961	0.445588	1	0	0.002644	0
Roof	EQ(x)1	LinStatic	Step By Step	1	221.25	0.0921	2402.269	0.128002	1	0	0.000532	0
9th Floor	EQ(x)1	LinStatic	Step By Step	1	353.175	0.096951	3642.83	0.659451	1	0	0.000494	0
8th Floor	EQ(x)1	LinStatic	Step By Step	1	467.197	0.101147	4619.007	0.788661	1	0	0.000381	0
7th Floor	EQ(x)1	LinStatic	Step By Step	1	564.026	0.103979	5424.405	0.851523	1	0	0.000354	0
6th Floor	EQ(x)1	LinStatic	Step By Step	1	644.427	0.104395	6172.966	0.878736	1	0	0.000269	0
5th Floor	EQ(x)1	LinStatic	Step By Step	1	709.24	0.102031	6951.227	0.88804	1	0	0.000249	0
4th Floor	EQ(x)1	LinStatic	Step By Step	1	759.393	0.095503	7951.507	0.874202	1	0	0.000374	0
3rd Floor	EQ(x)1	LinStatic	Step By Step	1	795.937	0.084119	9462.051	0.840358	1	0	0.000631	0
2nd Floor	EQ(x)1	LinStatic	Step By Step	1	820.098	0.066958	12248.02	0.772537	1	0	0.000855	0
1st Floor	EQ(x)1	LinStatic	Step By Step	1	834.035	0.053529	15580.99	0.786088	1	0	0.000106	0
Ground Floor	EQ(x)1	LinStatic	Step By Step	1	838.667	0.020496	40918.12	0.380784	1	0	5.80E-05	0
Water Tank	EQ(x)1	LinStatic	Step By Step	2	5.468	0.040169	136.1336	300.5733	1	0	0.003093	0
Roof top	EQ(x)1	LinStatic	Step By Step	2	30.092	0.098083	306.8023	0.443718	1	0	0.007932	0
Roof	EQ(x)1	LinStatic	Step By Step	2	221.25	0.089733	2465.638	0.124431	1	0	0.006958	0
9th Floor	EQ(x)1	LinStatic	Step By Step	2	353.175	0.094932	3720.274	0.662757	1	0	0.007338	0
8th Floor	EQ(x)1	LinStatic	Step By Step	2	467.197	0.099598	4690.849	0.793092	1	0	0.007518	0
7th Floor	EQ(x)1	LinStatic	Step By Step	2	564.026	0.103218	5464.394	0.858439	1	0	0.007657	0
6th Floor	EQ(x)1	LinStatic	Step By Step	2	644.427	0.104789	6149.778	0.888551	1	0	0.007605	0
5th Floor	EQ(x)1	LinStatic	Step By Step	2	709.24	0.103554	6848.976	0.897912	1	0	0.007398	0
4th Floor	EQ(x)1	LinStatic	Step By Step	2	759.393	0.098106	7740.505	0.884823	1	0	0.006959	0
3rd Floor	EQ(x)1	LinStatic	Step By Step	2	795.937	0.086465	9205.347	0.840871	1	0	0.006188	0
2nd Floor	EQ(x)1	LinStatic	Step By Step	2	820.098	0.068774	11924.45	0.771972	1	0	0.005149	0
1st Floor	EQ(x)1	LinStatic	Step By Step	2	834.035	0.053507	15587.39	0.765006	1	0	0.003353	0
Ground Floor	EQ(x)1	LinStatic	Step By Step	2	838.667	0.020492	40926.35	0.380865	1	0	0.001191	0
Water Tank	EQ(x)1	LinStatic	Step By Step	3	5.468	0.039678	137.8172	296.9611	1	0	0.003177	0
Roof top	EQ(x)1	LinStatic	Step By Step	3	30.092	0.097336	309.1564	0.445785	1	0	0.008806	0
Roof	EQ(x)1	LinStatic	Step By Step	3	221.25	0.089783	2464.277	0.125455	1	0	0.007717	0
9th Floor	EQ(x)1	LinStatic	Step By Step	3	353.175	0.09498	3718.43	0.66272	1	0	0.007823	0
8th Floor	EQ(x)1	LinStatic	Step By Step	3	467.197	0.099646	4688.589	0.793081	1	0	0.008044	0
7th Floor	EQ(x)1	LinStatic	Step By Step	3	564.026	0.103272	5461.567	0.85847	1	0	0.008165	0
6th Floor	EQ(x)1	LinStatic	Step By Step	3	644.427	0.104848	6146.293	0.888595	1	0	0.00812	0
5th Floor	EQ(x)1	LinStatic	Step By Step	3	709.24	0.103619	6844.682	0.897966	1	0	0.007819	0
4th Floor	EQ(x)1	LinStatic	Step By Step	3	759.393	0.098855	7681.865	0.891018	1	0	0.00727	0
3rd Floor	EQ(x)1	LinStatic	Step By Step	3	795.937	0.088031	9041.575	0.849616	1	0	0.006391	0
2nd Floor	EQ(x)1	LinStatic	Step By Step	3	820.098	0.069916	11729.81	0.77082	1	0	0.005246	0
1st Floor	EQ(x)1	LinStatic	Step By Step	3	834.035	0.053551	15574.59	0.753138	1	0	0.003423	0
Ground Floor	EQ(x)1	LinStatic	Step By Step	3	838.667	0.0205	40909.91	0.380705	1	0	0.001268	0

SOFT STORY CHECK

Story	Output Case	Case Type	Step Type	Step Number	Shear X kip	Drift X in	Stiff X kip/in	Ratio	Avarage	Shear Y kip	Drift Y in	Stiff Y kip/in
Water Tank	EQ(y)	LinStatic	Step By Step	1	0	0.001036	0	#DIV/0!	#VALUE!	5.468	0.046969	116.4233
Roof top	EQ(y)	LinStatic	Step By Step	1	0	0.002678	0	#DIV/0!	#DIV/0!	30.092	0.112272	268.0286
Roof	EQ(y)	LinStatic	Step By Step	1	0	0.002653	0	#DIV/0!	#DIV/0!	221.25	0.116	1907.32
9th Floor	EQ(y)	LinStatic	Step By Step	1	0	0.002837	0	#DIV/0!	#DIV/0!	353.175	0.121006	2918.644
8th Floor	EQ(y)	LinStatic	Step By Step	1	0	0.003181	0	#DIV/0!	#DIV/0!	467.197	0.125978	3708.563
7th Floor	EQ(y)	LinStatic	Step By Step	1	0	0.003406	0	#DIV/0!	#DIV/0!	564.026	0.128946	4374.136
6th Floor	EQ(y)	LinStatic	Step By Step	1	0	0.003522	0	#DIV/0!	#DIV/0!	644.427	0.129572	4973.505
5th Floor	EQ(y)	LinStatic	Step By Step	1	0	0.003578	0	#DIV/0!	#DIV/0!	709.24	0.126986	5585.187
4th Floor	EQ(y)	LinStatic	Step By Step	1	0	0.003506	0	#DIV/0!	#DIV/0!	759.393	0.119875	6334.862
3rd Floor	EQ(y)	LinStatic	Step By Step	1	0	0.003275	0	#DIV/0!	#DIV/0!	795.937	0.106062	7504.462
2nd Floor	EQ(y)	LinStatic	Step By Step	1	0	0.002797	0	#DIV/0!	#DIV/0!	820.098	0.083069	9872.55
1st Floor	EQ(y)	LinStatic	Step By Step	1	0	0.001875	0	#DIV/0!	#DIV/0!	834.035	0.059419	14036.41
Ground Floor	EQ(y)	LinStatic	Step By Step	1	0	0.000582	0	#DIV/0!	#DIV/0!	838.667	0.020415	41080.69
Water Tank	EQ(y)	LinStatic	Step By Step	2	0	0.003629	0	#DIV/0!	#DIV/0!	5.468	0.047055	116.2097
Roof top	EQ(y)	LinStatic	Step By Step	2	0	0.008701	0	#DIV/0!	#DIV/0!	30.092	0.111668	269.4787
Roof	EQ(y)	LinStatic	Step By Step	2	0	0.009005	0	#DIV/0!	#DIV/0!	221.25	0.118742	1863.279
9th Floor	EQ(y)	LinStatic	Step By Step	2	0	0.009325	0	#DIV/0!	#DIV/0!	353.175	0.123762	2853.662
8th Floor	EQ(y)	LinStatic	Step By Step	2	0	0.009831	0	#DIV/0!	#DIV/0!	467.197	0.128882	3624.993
7th Floor	EQ(y)	LinStatic	Step By Step	2	0	0.010132	0	#DIV/0!	#DIV/0!	564.026	0.131987	4273.35
6th Floor	EQ(y)	LinStatic	Step By Step	2	0	0.010237	0	#DIV/0!	#DIV/0!	644.427	0.133068	4842.848
5th Floor	EQ(y)	LinStatic	Step By Step	2	0	0.010063	0	#DIV/0!	#DIV/0!	709.24	0.130443	5437.175
4th Floor	EQ(y)	LinStatic	Step By Step	2	0	0.009597	0	#DIV/0!	#DIV/0!	759.393	0.123164	6165.688
3rd Floor	EQ(y)	LinStatic	Step By Step	2	0	0.008698	0	#DIV/0!	#DIV/0!	795.937	0.110174	7224.364
2nd Floor	EQ(y)	LinStatic	Step By Step	2	0	0.007202	0	#DIV/0!	#DIV/0!	820.098	0.09015	9097.075
1st Floor	EQ(y)	LinStatic	Step By Step	2	0	0.004651	0	#DIV/0!	#DIV/0!	834.035	0.061096	13651.22
Ground Floor	EQ(y)	LinStatic	Step By Step	2	0	0.001635	0	#DIV/0!	#DIV/0!	838.667	0.020987	39960.89
Water Tank	EQ(y)	LinStatic	Step By Step	3	0	0.00231	0	#DIV/0!	#DIV/0!	5.468	0.04509	121.2761
Roof top	EQ(y)	LinStatic	Step By Step	3	0	0.008326	0	#DIV/0!	#DIV/0!	30.092	0.116348	258.6396
Roof	EQ(y)	LinStatic	Step By Step	3	0	0.004538	0	#DIV/0!	#DIV/0!	221.25	0.110895	1995.135
9th Floor	EQ(y)	LinStatic	Step By Step	3	0	0.004209	0	#DIV/0!	#DIV/0!	353.175	0.116576	3029.562
8th Floor	EQ(y)	LinStatic	Step By Step	3	0	0.00403	0	#DIV/0!	#DIV/0!	467.197	0.121586	3842.516
7th Floor	EQ(y)	LinStatic	Step By Step	3	0	0.003722	0	#DIV/0!	#DIV/0!	564.026	0.125127	4507.641
6th Floor	EQ(y)	LinStatic	Step By Step	3	0	0.003426	0	#DIV/0!	#DIV/0!	644.427	0.126076	5111.407
5th Floor	EQ(y)	LinStatic	Step By Step	3	0	0.003033	0	#DIV/0!	#DIV/0!	709.24	0.12263	5783.593
4th Floor	EQ(y)	LinStatic	Step By Step	3	0	0.002628	0	#DIV/0!	#DIV/0!	759.393	0.114423	6636.737
3rd Floor	EQ(y)	LinStatic	Step By Step	3	0	0.002147	0	#DIV/0!	#DIV/0!	795.937	0.100511	7918.869
2nd Floor	EQ(y)	LinStatic	Step By Step	3	0	0.001974	0	#DIV/0!	#DIV/0!	820.098	0.079355	10334.61
1st Floor	EQ(y)	LinStatic	Step By Step	3	0	0.001087	0	#DIV/0!	#DIV/0!	834.035	0.057743	14443.97
Ground Floor	EQ(y)	LinStatic	Step By Step	3	0	0.000535	0	#DIV/0!		838.667	0.019843	42265.06

11.8. Dual System Check

EQx			
Base Reaction V=	-838.667		
Column taking C=	223.259		
C*100/V =	-26.6207	>25%	OK

EQy			
Base Reaction, V	-838.667		
Column Taking C	227.386		
C*100/v	-27.1125	>25%	ok

V2 kin	V3 kin
1.503	1.24
1.503	1.24
1.503	1.24
1.598	1.021
1.598	1.021
1.598	1.021
1.408	1.458
1.408	1.458
1.408	1.458
1.855	4.709
1.855	4.709
1.855	4.709
1.886	3.97
1.886	3.97
1.886	3.97
1.824	5.447
1.824	5.447
1.824	5.447
1.954	2.406
1.954	2.406
1.954	2.406
1.979	2.027
1.979	2.027
1.979	2.027
1.928	2.785
1.928	2.785
1.928	2.785
1.5	1.208
1.5	1.208
1.5	1.208
1.405	0.993
1.405	0.993
1.405	0.993
1.594	1.423
1.594	1.423
1.594	1.423
1.884	4.402
1.884	4.402
1.884	4.402
1.844	3.711
1.844	3.711
1.844	3.711
1.924	5.093
1.924	5.093
1.924	5.093
1.941	2.344
1.941	2.344
1.941	2.344
1.915	1.974
1.915	1.974
1.915	1.974
1.966	2.713
1.966	2.713
1.966	2.713
1.941	2.36
1.941	2.36
1.941	2.36
1.941	1.988
1.941	1.988
1.941	1.988
1.941	2.732
1.941	2.732
1.941	2.732
4.372	1.001
4.372	1.001
4.372	1.001
4.69	0.872
4.69	0.872
4.69	0.872
4.055	1.13
4.055	1.13
4.055	1.13
2.622	1.314
2.622	1.314
2.622	1.314
2.67	1.152
2.67	1.152
2.67	1.152
2.574	1.477
2.574	1.477
2.574	1.477
223.259	227.384

11.9. Base Shear Check

Total Seismic Weight of The Building	
Load Type	Load (kip)
Dead	11826.148
Live	10508.193
PW	1601.012
FF	1744.787
Roof Live	669.231
EQ(x)	0
EQ(x)	0
EQ(x)	0
EQ(y)	0
EQ(y)	0
EQ(y)	0
Total Seismic Weight	26349.371

Verification of Base Shear	
Total Weight, W(kip)	26349.371
Base Shear, V (From ETABS)	838.667
Spectral Acceleration, Sa (ETABS)	0.0318
Spectral Acceleration, Sa (Hand Calculation)	0.04
Error (%)	20.5

Error Analysis:

A 20.5% error is observed between the ETABS results and hand calculations, which can be explained by the following factors:

- **Modeling Assumptions:** ETABS employs advanced modeling techniques that consider actual member continuity, complex geometries, and realistic load distribution. In contrast, hand calculations use simplified assumptions, such as idealized supports and uniform load distribution.
- **Load Combinations and Factors:** ETABS automatically applies code-based load factors and considers all load combinations accurately, whereas hand calculations may simplify or round loads, creating minor discrepancies.
- **Response Spectrum Evaluation:** ETABS interpolates structural response using detailed numerical techniques, whereas hand calculations use simplified spectrum estimates, causing minor variations.
- **Numerical Precision:** ETABS uses iterative solution algorithms with high precision, whereas hand calculations involve rounding and simplifications, resulting in small differences.
- **Boundary and Local Effects:** ETABS can model foundation flexibility and local restraints, which are usually ignored in manual calculations.
- **Diaphragm and Continuity Effects:** ETABS models rigid/semi-rigid diaphragms and continuous beams/columns. Whereas hand calculations often assume idealized lateral load distribution, which can create significant differences in forces.
- **Torsional Effects:** ETABS models torsion accurately, particularly in asymmetric structures, while manual calculations may neglect or simplify torsion.
- **Building Irregularities:** ETABS handles irregularities, while manual methods use approximations.

12. Dead Load & Live Load Calculation

12.1. Dead Load Calculations:

Ceiling Plaster (13 mm thickness)

$$= (0.287 * 20.89) = 6 \text{ psf} \quad \text{---(Section 2.2, Table 6.2.2)}$$

Floor Finish (FF)

$$= 25 \text{ psf}$$

Cement plaster (10 mm thickness)

$$= (0.23 * 20.89) = 4.8 \text{ psf} \quad \text{---(Section 2.2, Table 6.2.2)}$$

External Wall

Weight of 5'' wall

= Unit weight of brick * thickness of the wall * height of the wall excluding beam

$$= (18.9 * 6.4) \text{ lb/ft}^3 * \left(\frac{5}{12}\right) \text{ ft} * \left(10 - \frac{16}{12}\right)$$

$$= 436.8 \text{ lb/ft} = 0.4 \text{ kip/ft}$$

Parapet Wall

$$= (18.9 * 6.4) \text{ lb/ft}^3 * \left(\frac{5}{12}\right) \text{ ft} * 3 \text{ ft} \quad \text{[Unit weight of basic materials: Brick = 18.9 K/m}^3\text{]}$$

$$= 151.2 \text{ lb/ft} = 0.1512 \text{ kip/ft}$$

Partition Wall loads on Exterior Beam

$$10' \times 10'' / 12 \times 120 \text{ pcf} = 1000 \text{ lb/ft} = 1 \text{ kip/ft}$$

Partition Wall loads on Interior Beam

$$10' \times 5'' / 12 \times 120 \text{ pcf} = 500 \text{ lb/ft} = 0.5 \text{ kip/ft}$$

12.1.1. Provisions for Dead Load (BNBC2020)

2.2.4 Weight of Materials and Constructions

In estimating dead loads, the actual weights of materials and constructions shall be used, provided that in the absence of definite information, the weights given in Tables 6.2.1 and 6.2.2 shall be assumed for the purposes of design.

Table 6.2.1: Unit Weight of Basic Materials

Material	Unit Weight (kN/m ³)	Material	Unit Weight (kN/m ³)
Aluminium	27.0	Granite, Basalt	26.4
Asphalt	21.2	Iron - cast	70.7
Brass	83.6	- wrought	75.4
Bronze	87.7	Lead	111.0
Brick	18.9	Limestone	24.5
Cement	14.7	Marble	26.4
Coal, loose	8.8	Sand, dry	15.7

Table 6.2.2: Weight of Construction Materials.

Material/Component/Member	Weight per Unit Area (kN/m ²)	Material/Component/Member	Weight per Unit Area (kN/m ²)
Floor		Walls and Partitions	
Asphalt, 25 mm thick	0.526	Acrylic resin sheet, flat, per mm thickness	0.012
Clay tiling, 13 mm thick	0.268	Asbestos cement sheeting:	
Concrete slab (stone aggregate)*:		4.5 mm thick	0.072
solid, 100 mm thick	2.360	6.0 mm thick	0.106
solid, 150 mm thick	3.540	Brick masonry work, excl. plaster:	
Galvanized steel floor deck (excl. topping)	0.147- 0.383	burnt clay, per 100 mm thickness	1.910
Magnesium oxychloride:		sand-lime, per 100 mm thickness	1.980
normal (sawdust filler), 25 mm thick	0.345	Concrete (stone aggregate)*:	
heavy duty (mineral filler), 25 mm thick	0.527	100 mm thick	2.360
Terrazzo paving 16 mm thick	0.431		
Ceiling			
Fibrous plaster, 10 mm thick	0.081		
Cement plaster, 13 mm thick	0.287		
Suspended metal lath and plaster (two faced incl. studding)	0.480		

12.2. Live Load Calculations:

All others except stairs and balconies

$$= 2 \text{ KN/m}^2$$

$$= (2 * 20.89) \text{ psf}$$

$$= 41.78 \text{ psf}$$

Stairs and exit ways

$$= 4.80 \text{ KN/m}^2$$

$$= (4.80 * 20.89) \text{ psf}$$

$$= 100.27 \text{ psf}$$

Community hall

[Roofs used for gardens and assembly purposes]

$$= 4.8 \text{ KN/m}^2$$

$$= (4.8 * 20.89) \text{ psf}$$

$$= 100.27 \text{ psf}$$

Balconies (Exterior)

$$= 4.80 \text{ KN/m}$$

$$= (4.80 * 20.89)$$

$$= 100.27$$

12.2.1. Provisions for Live Load (BNBC2020)

Table 6.2.3: Minimum Uniformly Distributed and Concentrated Live Loads^a

Occupancy or Use	Uniform kN/m ²	Concentrated kN
Assembly areas and theaters		
Fixed seats (fastened to floor)	2.90	--
Lobbies	4.80	--
Movable seats	4.80	--
Platforms (assembly)	4.80	--
Stage floors	7.20	--
Balconies (exterior)	4.80	--
On one- and two-family residences only, and not exceeding 19.3 m ²	2.90	--
Stairs and exit ways	4.80	See Note ⁱ
One- and two-family residences only	2.00	below --
Roofs		
Ordinary flat roof	1.00 ^h	--
Pitched and curved roofs	See Table 6.2.4	
Roofs used for promenade purposes	2.90	--
Roofs used for roof gardens or assembly purposes	4.80	--
Roofs used for other special purposes	See Note ⁱ below	
Residential		
Dwellings (one- and two-family)		
Uninhabitable attics without storage	0.50	--
Uninhabitable attics with storage	1.00	--
Habitable attics and sleeping areas	1.50	--
All other areas except stairs and balconies	2.00	--

12.3. Water Tank Load Calculation

Length=11'

Width=11'

Height=4'

Let wall thickness = 10"

Effective height = $4' - 10''/12 \times 2 = 2.33\text{ft}$

Effective Width = $11' - 10''/12 \times 2 = 9.33\text{ ft}$

Effective Length = $11' - 10''/12 \times 2 = 9.33\text{ ft}$

Effective Slab Area, $A = 9.33 \times 9.33 = 87.04\text{ ft}^2$

According to BNBC $Q = 200\text{ lpcd}$

No of stories = 10 (excluding ground floor)

Each story has 2 flats

(Only 2nd floor has one unit)

So total number of flats = $9 \times 2 + 1 = 19$

Considering 4-people family for each flat, total population = $19 \times 4 = 76\text{ people}$

Now, $\text{lpcd} = \text{Annual water consumption/population served}$

So, water load = $76 \times 200 = 15200\text{ L}$

The water tank is replenished by pumping twice a day

So, water volume = $15200/2 = 7600\text{ L}$

8000L = 8000 kg of water

= $7600 \times 2.20462\text{ lbs of water}$

= $16755.112\text{ lbs of water}$

Unit load due to water = $16755.112/87.04 = 192.49\text{ psf}$

Let the tank top slab = 4"

Top slab weight of water tank = $(4''/12) \times 150 \text{ pcf} = 50 \text{ psf}$

Total weight on bottom slab of water tank = $192.49 + 50 = 242.49 \text{ psf}$

Total wall load = $(10''/12) \times 4' \times 150 \text{ pcf} = 500 \text{ lb/ft}$
 $= 0.5 \text{ kip/ft}$

12.3.1. Provisions for Water Tank Load (BNBC2020)

Table 8.5.1(a): Water Consumption for Domestic Purposes in Residential Buildings (Cities/Big District Towns)

Category	Socio-economic group, Type of Building, Source and Other Facilities	Water Consumption	
		Full Facility (lpcd)	Restricted Facility (lpcd)
A	Metropolitan Cities/City Corporation Area/District Towns		
x	High income group:		
A1	Single family dwelling (with garden and car washing)	260	200
A3	Big multi-family apartment/flat (> 2500 sft)	200	150
y	Middle income group:		
y1	Officer's qtr./Colony and moderate apartment (< 2000 sft)	180	135
y2	Small building/staff qtr. and small apartment (< 1500 sft)	---	120
z	Low income group:		
z1	Junior staff qtr./flat (< 1000 sft) and temporary shade	---	80
z2	Stand post connection in the fringe area	---	65
z3	Common yard (stand post) connection in the fringe area	---	50
z4	Slum dwellers collection from road side public stand post	---	40

5.9.2.2 Roof storage tank

The roof storage tank shall be constructed with pre-stressed or reinforced concrete or ferro-cement or galvanized steel or of the material that will resist any action by the plain or chlorinated water. The tank shall be made of water tight without the use of putty.

Tanks made of non-galvanized metal sheets shall be coated internally with a nontoxic material which does not impart a taste or odor. The metal storage tank shall be coated externally with a good quality anticorrosive weather resistant paint. The outlet of storage tank to the distribution system shall be at least 50 mm above the tank bottom.

To provide sufficient pressure, the bottom of the tank must be elevated sufficiently above the highest floor water fixtures.

Vent pipe should be provided to avoid any air lock and should be placed where the horizontal branch pipes connect the vertical down feed pipes (not adjacent to storage tank and at interconnection place between storage tank and distribution pipes).

5.9.2.3 Ground or underground storage tank

13. Wind Load Hand Calculation

Location: Dhaka

Basic Wind Speed, $V = 65.7 \text{ m/s} = 65.7 * 2.24 = 147.16 \text{ mph}$

Building Height, $h_n = 122 \text{ ft} = (122/3.28) = 37.2 \text{ m}$

$C_t = 0.0466$, $m = 009$ [BNBC-Table 6.2.20]

Time period, $T = C_t(h_n)^m = 0.0466 * (37.2)^{0.9} = 1.21 \text{ s} > 1 \text{ s}$ [Flexible Structure]

$$\bar{z} = 0.6h$$

$$= (0.6 * 122)$$

$$= 73.2 \text{ ft}$$

$$= 22.3 \text{ m} > 9.14 \text{ m} \text{ [BNBC2020- Table 6.2.10]}$$

$$I_{\bar{z}} = c (10/\bar{z})^{1/6}$$

$$= 0.30 * (10/22.3)^{1/6}$$

$$= 0.26$$

$$g_Q = g_v = 3.14$$

For Exposure Type A, $l = 97.54 \text{ m}$, $\epsilon = 1/3 = 0.33$

$$L_{\bar{z}} = l (\bar{z}/10)^{1/\epsilon}$$

$$= 97.54 * (22.3/10)^{1/3}$$

$$= 127.4$$

For X-direction, (W_x)

$$B = 78.5 \text{ ft} = 23.9 \text{ m}$$

$$H = 122 \text{ ft} = 37.2 \text{ m}$$

$$L = 136 \text{ ft} = 41.5 \text{ m}$$

$$Q = \sqrt{\frac{1}{1 + 0.63 \left(\frac{B+H}{L_{\bar{z}}} \right)^{.63}}}$$

$$= \sqrt{Q} = \sqrt{1 / (1 + 0.63 (23.9 + 37.2) / 127.4)^{.63}} = 0.85$$

$$n_1 = \frac{1}{T} = \frac{1}{1.21} = 0.83$$

$$g_R = \sqrt{2 \ln(3600 n_1)} + \frac{0.577}{\sqrt{2 \ln(3600 n_1)}} = 4.14$$

$$R_n = \frac{7.47 N_1}{(1 + 10.3 N_1)^{5/3}} = \frac{7.47 * 2.9}{(1 + 10.3 * 2.9)^{5/3}} = 0.07$$

$$N_1 = \frac{n_1 * L_{\bar{z}}}{V_{\bar{z}}} = \frac{0.83 * 127.4}{36.1} = 2.9$$

$$V_{\bar{z}} = \bar{b} \left(\frac{\bar{z}}{10} \right)^{\alpha} V = 0.45 * \left(\frac{22.3}{10} \right)^{1/4} * 65.7 = 36.1$$

$$\text{For } R_B, n = 4.6 n_1 \frac{B}{V_{\bar{z}}} = \frac{4.6 * 0.83 * 23.9}{36.1} = 2.5$$

$$\text{For } R_L, n=15.4n_1 \frac{L}{V\bar{Z}} = \frac{15.4*0.83*41.5}{36.1} = 14.7$$

$$\text{For } R_h, n=4.6n_1 \frac{H}{V\bar{Z}} = \frac{4.6*0.83*37.2}{36.1} = 3.9$$

Here, $\beta=5\%$

$$R = \sqrt{1/\beta \quad R_n \quad R_h \quad R_B(0.53 + 0.47RL)} = 0.989$$

$$\therefore \text{Gust Factor, } G_f = 0.925 * \left(\frac{1+1.7*I\bar{Z}*\sqrt{(gQ^2Q^2 + gR^2R^2)}}{1+1.7*gv*Iz} \right) = 1.17 \text{ [In X-direction]}$$

For Y-direction, (W_Y)

$$B = 78.5\text{ft} = 23.9 \text{ m}$$

$$H = 122 \text{ ft} = 37.2 \text{ m}$$

$$L = 136 \text{ ft} = 41.5 \text{ m}$$

$$\begin{aligned} \therefore Q_y &= \sqrt{\frac{1}{1+0.63\left(\frac{B+h}{Lz}\right).63}} \\ &= \sqrt{Q} = \sqrt{1/(1 + 0.63 (41.5 + 37.2)/127.4).63)} = 0.83 \end{aligned}$$

$$n_1 = \frac{1}{T} = \frac{1}{1.21} = 0.83$$

$$g_R = \sqrt{2\ln(3600n_1)} + \frac{0.577}{\sqrt{2\ln(3600n_1)}} = 4.14$$

$$R_n = \frac{7.47N_1}{(1+10.3N_1)^{5/3}} = \frac{7.47*2.9}{(1+10.3*2.9)^{5/3}} = 0.07$$

$$N_1 = \frac{n_1*L\bar{Z}}{V\bar{Z}} = \frac{0.83*127.4}{36.1} = 2.9$$

$$V_{\bar{Z}} = \bar{b} \left(\frac{\bar{z}}{10} \right)^{\bar{\alpha}} V = 0.45 * \left(\frac{22.3}{10} \right)^{1/4} * 65.7 = 36.1$$

$$\text{For } R_B, n=4.6n_1 \frac{B}{V\bar{Z}} = \frac{4.6*0.83*41.5}{36.1} = 4.4$$

$$\text{For } R_L, n=15.4n_1 \frac{L}{V\bar{Z}} = \frac{15.4*0.83*23.9}{36.1} = 8.5$$

$$\text{For } R_h, n=4.6n_1 \frac{H}{V\bar{Z}} = \frac{4.6*0.83*37.2}{36.1} = 3.9$$

Here, $\beta=5\%$

$$R = \sqrt{1/\beta \quad R_n \quad R_h \quad R_B(0.53 + 0.47RL)} = 0.79$$

$$\therefore \text{Gust Factor, } G_f = 0.925 * \left(\frac{1+1.7*I\bar{Z}*\sqrt{(gQ^2Q^2 + gR^2R^2)}}{1+1.7*gv*Iz} \right) = 1.06 \text{ [In Y-direction]}$$

13.1. Provisions for Wind Load (BNBC2020)

a) BNBC 2020

Our building height is 122 ft = 37.2 m (approximately) < 792 m

Thus, falls in exposure A category which is considered as B in ETABS

WIND LOAD CALCULATION-BNBC 2020



Exposure A

- upwind direction prevail distance at least 792 m
- 20 times the height of the building





Exposure B



- distance $\geq 1,524$ m
- 20 times building height

Exposure C



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2.4.6.3 Exposure categories

Exposure A: Exposure A shall apply where the ground surface roughness condition, as defined by Surface Roughness A, prevails in the upwind direction for a distance of at least 792 m or 20 times the height of the building, whichever is greater.

Exception: For buildings whose mean roof height is less than or equal to 9.1 m, the upwind distance may be reduced to 457 m.

Table 6.2.8: Basic Wind Speeds, V , for Selected Locations in Bangladesh

Location	Basic Wind Speed (m/s)	Location	Basic Wind Speed (m/s)
Angarpota	47.8	Lalmohirhat	63.7
Bagerhat	77.5	Madaripur	68.1
Bandarban	62.5	Magura	65.0
Barguna	80.0	Manikganj	58.2
Barisal	78.7	Meherpur	58.2
Bhola	69.5	Maheshkhali	80.0
Bogra	61.9	Moulvibazar	53.0
Brahmanbaria	56.7	Munshiganj	57.1
Chandpur	50.6	Mymensingh	67.4
Chapai Nawabganj	41.4	Naogaon	55.2
Chittagong	80.0	Narail	68.6
Chuadanga	61.9	Narayanganj	61.1
Comilla	61.4	Narsinghdi	59.7
Cox's Bazar	80.0	Natore	61.9
Dahagram	47.8	Netrokona	65.6
Dhaka	65.7	Nilphamari	44.7
Dinajpur	41.4	Noakhali	57.1
Faridpur	63.1	Pabna	63.1
Feni	64.1	Panchagarh	41.4
Gaibandha	65.6	Patuakhali	80.0
Gazipur	66.5	Pirojpur	80.0
Gopalganj	74.5	Rajbari	59.1
Habiganj	54.2	Rajshahi	49.2
Hatiya	80.0	Rangamati	56.7
Ishurdi	69.5	Rangpur	65.3
Joypurhat	56.7	Satkhira	57.6
Jamalpur	56.7	Shariatpur	61.9
Jessore	64.1	Sherpur	62.5
Jhalakati	80.0	Sirajganj	50.6
Jhenaidah	65.0	Srimangal	50.6
Khagrachhari	56.7	St. Martin's Island	80.0
Khulna	73.3	Sunamganj	61.1
Kutubdia	80.0	Sylhet	61.1
Kishoreganj	64.7	Sandwip	80.0
Kurigram	65.6	Tangail	50.6
Kushtia	66.9	Teknaf	80.0
Lakshmipur	51.2	Thakurgaon	41.4

Table 6.2.9: Importance Factor, I (Wind Loads)

Occupancy Category ¹ or Importance Class	Non-Cyclone Prone Regions and Cyclone Prone Regions with $V = 38-44$ m/s	Cyclone Prone Regions with $V > 44$ m/s
I	0.87	0.77
II	1.0	1.00
III	1.15	1.15
IV	1.15	1.15

¹ The building and structure classification categories are listed in Table 6.1.1

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Table 6.2.12: Wind Directionality Factor, K_d

Structure Type	Directionality Factor K_d *	Structure Type	Directionality Factor K_d *
Buildings		Solid Signs	0.85
Main Wind Force Resisting System	0.85	Open Signs and Lattice Framework	0.85
Components and Cladding	0.85	Trussed Towers	
Arched Roofs	0.85	Triangular, square, rectangular	0.85
Chimneys, Tanks, and Similar Structures		All other cross section	0.95
Square	0.90		
Hexagonal	0.95		
Round	0.95		

* Directionality Factor K_d has been calibrated with combinations of loads specified in Sec 2.7. This factor shall only be applied when used in conjunction with load combinations specified in Sections 2.7.2 and 2.7.3.

										BORELOG				
										Borehole Top RL, m: + 5.271				
										Date Started: 12-Aug-25				
										Date Completed: 13-Aug-25				
Method of Boring:		Rotary Method								TBM: 20, RL + 9.296m				
Depth of Boring (m):		33.0	Boring Dia (mm): 150											
Soil Classification:		ASTM D-2487 & D-2488												
Depth Below EGL (m)	Sample ID	Sample Type	RL (m)	Thickness (m)	Description of Soil Strata	Symbols	SPT Interval	Fines (%)	Liquid Limit (LL)	Plasticity Index (PI)	q _c (kPa)	N.M.C (%)	SPT N-Value / Penetration, cm	Graphical Representation of field SPT N-Value
1.5	D-1		3.77	2.25	Brown, soft, medium plasticity, Lean CLAY with Sand, CL		1.5	76					3	
3.0	D-2		2.27				3.0	97	43	22			6	
4.5	D-3		0.77	4.50	Grey, very soft to medium stiff, medium plasticity, Lean CLAY, CL		4.5						1	
6.0	D-4		(0.73)				6.0	99	46	22			1	
7.5	D-5		(2.23)		Grey, loose to medium dense, Silty SAND, SM , trace of mica		7.5	35					12	
9.0	D-6		(3.73)	3.00			9.0						9	
10.5	D-7		(5.23)		Grey, medium stiff, high plasticity, Sandy Fat CLAY, CH		10.5						5	
12.0	D-8		(6.73)	3.00			12.0	66					7	
13.5	D-9		(8.23)	1.50	Grey, loose, non-plastic, Sandy SILT, ML , trace of mica		13.5	63					7	
15.0	D-10		(9.73)	1.50	Grey, medium stiff, medium plasticity, Lean CLAY with Sand, CL		15.0	82	34	13			5	
16.5	D-11		(11.23)	1.50	Grey, medium stiff, medium plasticity, Lean CLAY, CL		16.5	86	40	18			8	
18.0	D-12		(12.73)				18.0	67					10	
19.5	D-13		(14.23)	4.50	Grey, medium stiff to stiff, medium plasticity, Sandy Lean CLAY, CL		19.5						7	
21.0	D-14		(15.73)				21.0	50					13	
22.5	D-15		(17.23)				22.5	33					16	
24.0	D-16		(18.73)				24.0	38					31	
25.5	D-17		(20.23)	11.25	Grey, medium dense to dense, Silty SAND, SM , trace of mica		25.5						26	
27.0	D-18		(21.73)				27.0	13					50/26	
28.5	D-19		(23.23)				28.5						50/8	
30.0	D-20		(24.73)				30.0						50/8	

Total depth= 1.5x20=30 m(thickness of each layer =1.5 m)

$$\text{Sum}(1.5/N_i)=6.027$$

$$N= 30/6.027= 4.98 < 15 \quad \text{So, site class SD}$$

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Table 6.2.13: Site Classification Based on Soil Properties

Site Class	Description of soil profile up to 30 meters depth	Average Soil Properties in top 30 meters		
		Shear wave velocity, $\bar{V}_s$ (m/s)	SPT Value, $\bar{N}$ (blows/30cm)	Undrained shear strength, $\bar{S}_u$ (kPa)
SA	Rock or other rock-like geological formation, including at most 5 m of weaker material at the surface.	> 800	--	--
SB	Deposits of very dense sand, gravel, or very stiff clay, at least several tens of metres in thickness, characterised by a gradual increase of mechanical properties with depth.	360 – 800	> 50	> 250

Site Class	Description of soil profile up to 30 meters depth	Average Soil Properties in top 30 meters		
		Shear wave velocity, $\bar{V}_s$ (m/s)	SPT Value, $\bar{N}$ (blows/30cm)	Undrained shear strength, $\bar{S}_u$ (kPa)
SC	Deep deposits of dense or medium dense sand, gravel or stiff clay with thickness from several tens to many hundreds of metres.	180 - 360	15 - 50	70 - 250
SD	Deposits of loose-to-medium cohesionless soil (with or without some soft cohesive layers), or of predominantly soft-to-firm cohesive soil.	< 180	< 15	< 70
SE	A soil profile consisting of a surface alluvium layer with V_s values of type SC or SD and thickness varying between about 5 m and 20 m, underlain by stiffer material with $V_s > 800$ m/s.	--	--	--
S ₁	Deposits consisting, or containing a layer at least 10 m thick, of soft clays/silts with a high plasticity index ($PI > 40$) and high water content	< 100 (indicative)	--	10 - 20
S ₂	Deposits of liquefiable soils, of sensitive clays, or any other soil profile not included in types SA to SE or S ₁	--	--	--

14. Earthquake load calculation

$$C_t = 0.0466 \quad [\text{Table 6.2.20}]$$

$$m = 0.9$$

$$h_n = 122 \text{ ft} = 37.19 \text{ m}$$

$$\text{For Site Class SD:} \quad [\text{Table 6.2.16}]$$

$$S = 1.35$$

$$T_B = 0.20 \text{ sec.}$$

$$T_C = 0.80 \text{ sec.}$$

$$T_D = 2 \text{ sec.}$$

$$\begin{aligned} T &= C_t * (h_n)^m \\ &= 0.0466 * (37.19)^{0.9} \\ &= 1.21 \text{ sec.} \end{aligned}$$

$$\text{For Special reinforced concrete shear walls:} \quad [\text{Table 6.2.19}]$$

$$R = 6.5$$

$$Z = 0.20$$

$$\eta = \sqrt{\left\{ \frac{10}{(5 + \epsilon)} \right\}} = 1$$

$$\begin{aligned} C_s &= 2.5 S \eta \left(\frac{T_C}{T} \right); T_C \leq T \leq T_D \quad [6.2.35c] \\ &= 2.5 * 1.35 * 1 * (0.8 / 1.21) = 2.23 \end{aligned}$$

$$S_a = \frac{2}{3} \frac{Z I C_s}{R} = (2/3) * (0.20 * 1 * 2.23) / 6.5 = 0.04$$

$$\text{Total Base Shear, } V = S_a * W = 0.04 * W$$

14.1. Provisions for EQ Load (BNBC2020)

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In our building, total residents = 76 < 300 So,
occupancy category of the building is I.

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Table 6.1.1: Occupancy Category of Buildings and other Structures for Flood, Surge, Wind and Earthquake Loads.

Nature of Occupancy	Occupancy Category
Buildings and other structures that represent a low hazard to human life in the event of failure, including, but not limited to: • Agricultural facilities • Certain temporary facilities • Minor storage facilities	I
All buildings and other structures except those listed in Occupancy Categories I, III and IV	II
Buildings and other structures that represent a substantial hazard to human life in the event of failure, including, but not limited to: • Buildings and other structures where more than 300 people congregate in one area	III

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Table 6.2.17: Importance Factors for Buildings and Structures for Earthquake design

Occupancy Category	Importance factor I
I, II	1.00
III	1.25
IV	1.50

2.5.7.2 Building period

The fundamental period T of the building in the horizontal direction under consideration shall be determined using the following guidelines:

- (a) Structural dynamics procedures (such as Rayleigh method or modal eigenvalue analysis), using structural properties and deformation characteristics of resisting elements, may be used to determine the fundamental period T of the building in the direction under consideration. This period shall not exceed the approximate fundamental period determined by Eq. 6.2.38 by more than 40 percent.
- (b) The building period T (in sec) may be approximated by the following formula:

$$T = C_t(h_n)^m \quad (6.2.38)$$

Where,

h_n = Height of building in metres from foundation or from top of rigid basement. This excludes the basement storeys, where basement walls are connected with the ground floor deck or fitted between the building columns. But it includes the basement storeys, when they are not so connected. C_t and m are obtained from Table 6.2.20

Table 6.2.20: Values for Coefficients to Estimate Approximate Period

Structure type	C_t	m	
Concrete moment-resisting frames	0.0466	0.9	Note: Consider moment resisting frames as frames which resist 100% of seismic force and are not enclosed or adjoined by components that are more rigid and will prevent the frames from deflecting under seismic forces.
Steel moment-resisting frames	0.0724	0.8	
Eccentrically braced steel frame	0.0731	0.75	
All other structural systems	0.0488	0.75	

Table 6.2.14: Description of Seismic Zones

Seismic Zone	Location	Seismic Intensity	Seismic Zone Coefficient, Z
1	Southwestern part including Barisal, Khulna, Jessore, Rajshahi	Low	0.12
2	Lower Central and Northwestern part including Noakhali, Dhaka, Pabna, Dinajpur, as well as Southwestern corner including Sundarbans	Moderate	0.20
3	Upper Central and Northwestern part including Brahmanbaria, Sirajganj, Rangpur	Severe	0.28
4	Northeastern part including Sylhet, Mymensingh, Kurigram	Very Severe	0.36

Table 6.2.15: Seismic Zone Coefficient Z for Some Important Towns of Bangladesh

Town	Z	Town	Z	Town	Z	Town	Z
Bagerhat	0.12	Gaibandha	0.28	Magura	0.12	Patuakhali	0.12
Bandarban	0.28	Gazipur	0.20	Manikganj	0.20	Pirojpur	0.12
Barguna	0.12	Gopalganj	0.12	Maulvibazar	0.36	Rajbari	0.20
Barisal	0.12	Habiganj	0.36	Meherpur	0.12	Rajshahi	0.12
Bhola	0.12	Jaipurhat	0.20	Mongla	0.12	Rangamati	0.28
Bogra	0.28	Jamalpur	0.36	Munshiganj	0.20	Rangpur	0.28
Brahmanbaria	0.28	Jessore	0.12	Mymensingh	0.36	Satkhira	0.12
Chandpur	0.20	Jhalokati	0.12	Narail	0.12	Shariatpur	0.20
Chapainababganj	0.12	Jhenaidah	0.12	Narayanganj	0.20	Sherpur	0.36
Chittagong	0.28	Khagrachari	0.28	Narsingdi	0.28	Sirajganj	0.28
Chuadanga	0.12	Khulna	0.12	Natore	0.20	Srimangal	0.36
Comilla	0.20	Kishoreganj	0.36	Naogaon	0.20	Sunamganj	0.36
Cox's Bazar	0.28	Kurigram	0.36	Netrakona	0.36	Sylhet	0.36
Dhaka	0.20	Kushtia	0.20	Nilphamari	0.12	Tangail	0.28
Dinajpur	0.20	Lakshmipur	0.20	Noakhali	0.20	Thakurgaon	0.20
Faridpur	0.20	Lalmanirhat	0.28	Pabna	0.20		
Feni	0.20	Madaripur	0.20	Panchagarh	0.20		

As our land location is in Dhaka, from the table we can say that it is in seismic Zone 2.

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Table 6.2.18: Seismic Design Category of Buildings

Site Class	Occupancy Category I, II and III				Occupancy Category IV			
	Zone 1	Zone 2	Zone 3	Zone 4	Zone 1	Zone 2	Zone 3	Zone 4
SA	B	C	C	D	C	D	D	D
SB	B	C	D	D	C	D	D	D
SC	B	C	D	D	C	D	D	D
SD	C	D	D	D	D	D	D	D
SE, S ₁ , S ₂	D	D	D	D	D	D	D	D

For occupancy category I, zone 2 and site class D, our building's seismic design category is D (SD).

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Table 6.2.16: Site Dependent Soil Factor and Other Parameters Defining Elastic Response Spectrum

Soil type	S	$T_B(s)$	$T_C(s)$	$T_D(s)$
SA	1.0	0.15	0.40	2.0
SB	1.2	0.15	0.50	2.0
SC	1.15	0.20	0.60	2.0
SD	1.35	0.20	0.80	2.0
SE	1.4	0.15	0.50	2.0

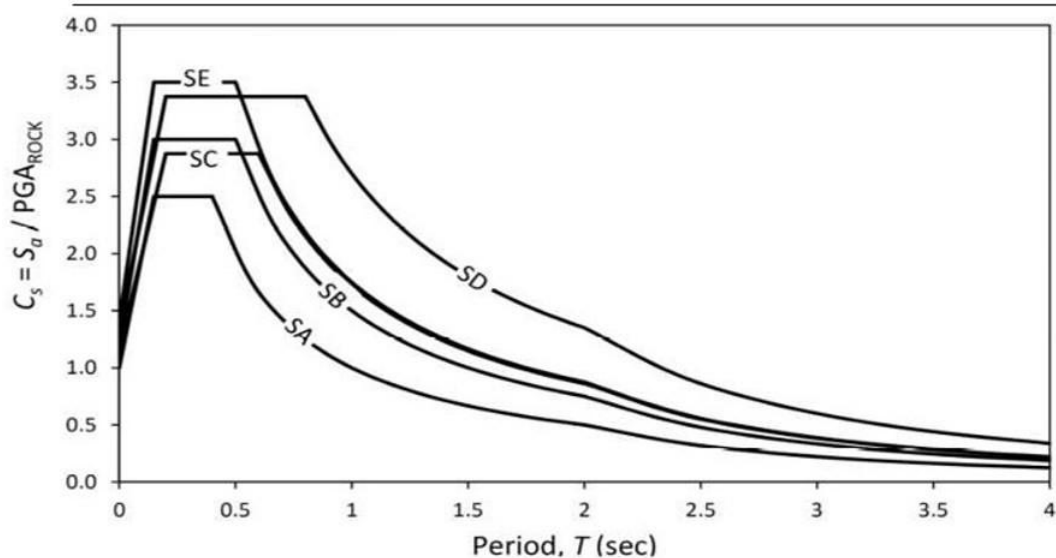


Figure 6.2.26 Normalized design acceleration response spectrum for different site classes.

selected at a ground shaking level that is 2/3 of the maximum considered earthquake (MCE) ground motion. The effect of local soil conditions on the response spectrum is incorporated in the normalized acceleration response spectrum C_s . The spectral acceleration for the design earthquake is given by the following equation:

$$S_a = \frac{2}{3} \frac{ZI}{R} C_s \quad (6.2.34)$$

Where,

- S_a = Design spectral acceleration (in units of g) which shall not be less than $0.67\beta ZIS$
- β = Coefficient used to calculate lower bound for S_a . Recommended value for β is 0.11
- Z = Seismic zone coefficient, as defined in Sec 2.5.4.2
- I = Structure importance factor, as defined in Sec 2.5.5.1
- R = Response reduction factor which depends on the type of structural system given in Table 6.2.19. The ratio $\frac{I}{R}$ cannot be greater than one.
- C_s = Normalized acceleration response spectrum, which is a function of structure (building) period and soil type (site class) as defined by Equations 6.2.35a to 6.2.35d.

$$C_s = S \left(1 + \frac{T}{T_B} (2.5\eta - 1) \right) \quad \text{for } 0 \leq T \leq T_B \quad (6.2.35a)$$

$$C_s = 2.5 S \eta \quad \text{for } T_B \leq T \leq T_C \quad (6.2.35b)$$

$$C_s = 2.5 S \eta \left(\frac{T_C}{T} \right) \quad \text{for } T_C \leq T \leq T_D \quad (6.2.35c)$$

$$C_s = 2.5 S \eta \left(\frac{T_C T_D}{T^2} \right) \quad \text{for } T_D \leq T \leq 4 \text{ sec} \quad (6.2.35d)$$

C_s depends on S and values of T_B , T_C and T_D , (Figure 6.2.25) which are all functions of the site class. Constant C_s value between periods T_B and T_C represents constant spectral acceleration.

S = Soil factor which depends on site class and is given in Table 6.2.16

T = Structure (building) period as defined in Sec 2.5.7.2

T_B = Lower limit of the period of the constant spectral acceleration branch given in Table 6.2.16 as a function of site class.

T_C = Upper limit of the period of the constant spectral acceleration branch given in Table 6.2.16 as a function of site class

T_D = Lower limit of the period of the constant spectral displacement branch given in Table 6.2.16 as a function of site class

η = Damping correction factor as a function of damping with a reference value of $\eta=1$ for 5% viscous damping. It is given by the following expression:

$$\eta = \sqrt{10 / (5 + \xi)} \geq 0.55 \quad (6.2.36)$$

Where, ξ is the viscous damping ratio of the structure, expressed as a percentage of critical damping. The value of η cannot be smaller than 0.55.

The anticipated (design basis earthquake) peak ground acceleration (PGA) for rock or very stiff soil (site class SA) is $\frac{2}{3}Z$. However, for design, the ground motion is modified through the use of response reduction factor R and importance factor I , resulting in $PGA_{rock} = \frac{2}{3} \left(\frac{ZI}{R} \right)$. Figure 6.2.26 shows the normalized acceleration response spectrum C_s for 5% damping, which may be defined as the 5% damped spectral acceleration (obtained by Eq. 6.2.34) normalized with respect to PGA_{rock} . This Figure demonstrates the significant influence of site class on the response spectrum.

E. DUAL SYSTEMS:

INTERMEDIATE MOMENT
FRAMES CAPABLE OF
RESISTING AT LEAST 25%
OF PRESCRIBED SEISMIC
FORCES

(with bracing or shear wall)

1. Special steel concentrically braced frames	6	2.5	5	NL	NL	11
2. Special reinforced concrete shear walls	6.5	2.5	5	NL	NL	50
3. Ordinary reinforced masonry shear walls	3	3	3	NL	50	NP

15. Justification of Design Procedure

(Based on BNBC 2020 and Dhaka Mohanagar Imarat Nirman Bidhimala)

Code Reference and Project Location

The structural modelling, analysis, and member design are performed following the Bangladesh National Building Code 2020 (BNBC 2020), Part 6 (Structural Design). For Dhaka city planning considerations (such as Floor Area Ratio and related development controls), the Dhaka Mohanagar Imarat Nirman Bidhimala (DMINB) is referenced.

Project Location: Dhaka, Bangladesh.

Basic Building Information

The building is a reinforced concrete (RC) multi-storey structure with a T-shaped plan. The ETABS model includes storeys up to the 9th floor, a roof and rooftop level, and an overhead water tank level.

Item	Adopted Value
Occupancy Category	Category II
Importance Factor, I	1.0
Total Seismic Height, h_n	122 ft (≈ 37.2 m)
Typical Floor-to-Floor Height	10 ft
Structural System	Dual System (IMRF + Shear Wall)
Seismic Zone Coefficient, Z (Dhaka)	0.20
Site Class	SD
Seismic Design Category (SDC)	D

Structural System and Seismic Parameters

A dual lateral resistance system is adopted, combining Intermediate Moment Resisting Frames (IMRF) with reinforced concrete shear walls. This improves lateral stiffness and helps control torsion for the T-shaped plan while maintaining ductility through the frame system.

Parameter	Value
Response Modification Factor, R	6.5
Overstrength Factor, Ω	2.5
Deflection Amplification Factor, C_d	5.0
Site Coefficient, F_a	1.35

Site Coefficient, F_v	2.70
Fundamental Time Period used, T	1.21 s
Damping Ratio	5%

Dual System Check: The moment-resisting frame must carry at least 25% of the design seismic base shear. From ETABS reactions, the frame contribution is 26.62% in EQX and 27.11% in EQY, which satisfies the $\geq 25\%$ criterion.

Justification: This system was chosen to improve **lateral stiffness** and control **torsion** for the building's T-shaped plan, while the frame system ensures necessary **ductility**. Per code requirements, the IMRF must carry at least 25% of the design seismic base shear; the ETABS model verified contributions of **26.62% in the X-direction** and **27.11% in the Y-direction**.

Seismic Base Shear Verification

Base shear from ETABS is verified against BNBC hand calculation using the effective seismic weight and spectral acceleration.

Quantity	Value
Total Seismic Weight, W	26349.371 kip
ETABS Base Shear, V	838.667 kip
Spectral Acceleration S_a (ETABS)	0.0318
Spectral Acceleration S_a (Hand Calculation)	0.04
Difference (Error)	$\approx 20.5\%$

The difference is acceptable for preliminary verification because ETABS performs modal participation and mass distribution more rigorously than simplified hand assumptions.

Material Properties

Material properties are selected consistently with common RC practice and project model definitions.

Material	Adopted Property
Concrete	$f'_c = 3500$ psi and 4500 psi (unit weight = 150 pcf)
Reinforcing Steel	$f_y = 60$ ksi (unit weight = 490 pcf)
Brick Masonry (for wall loads)	Unit weight = 120 pcf

Material Properties Justification

The selection of these material strengths and properties is fundamentally driven by structural safety, durability and code compliance for the Dhaka region. By using concrete strengths of 3500–4500 psi and 60 ksi reinforcing steel, the design ensures the building has the necessary stiffness and ductility to function as a **Dual System**, which is essential for managing the seismic demands of **Category D**. These materials

are integrated into an exhaustive list of load combinations to account for statistical probability and structural safety, ensuring the building can withstand various "worst-case" scenarios. Furthermore, the assumed unit weights (150 pcf for concrete and 490 pcf for steel) allow for an accurate calculation of the **Total Seismic Weight** (26349.371 kip), which is critical for verifying base shear. Finally, the use of approved cement and aggregates acts as a protective measure against corrosion in Dhaka's humid monsoon climate, satisfying BNBC 2020 durability requirements for inland urban environments.

Load Patterns and Gravity Loads

ETABS self-weight multiplier is kept ON for Dead Load (DL). Superimposed dead loads (SDL), live loads (LL), and other loads are assigned as area/line loads.

Pattern	Assigned Value / Note
DL	Self-weight (ETABS)
SDL	0.125 kip/ft ²
LL	0.040 kip/ft ²
Roof Live (RL)	0.033 kip/ft ²
EQX, EQY	BNBC 2020 seismic load cases
WX, WY	BNBC 2020 wind load cases
WT	Overhead water tank load

Wall line load calculation is based on 8 in brick wall height of 10 ft with 16 in beam depth and 3 ft parapet, resulting in ≈ 0.605 kip/ft (≈ 9.4 kN/m).

Load Combinations

Following factored load combinations are used in ETABS.

- $0.9D - 1.6W_x$
- $0.9D - 1.6W_y$
- $0.9D + 1.6W_x$
- $0.9D + 1.6W_y$
- $0.854D + E_x$
- $0.854D + E_y$
- $0.854D - E_x$
- $0.854D - E_y$
- $1.2D - 1.6W_x + L + 0.5L_r$
- $1.2D - 1.6W_y + L + 0.5L_r$
- $1.2D + 1.6L_r - 0.8W_x$
- $1.2D + 1.6L_r - 0.8W_y$
- $1.2D + 1.6L_r + 0.8W_x$
- $1.2D + 1.6L_r + 0.8W_y$

- $1.2D + 1.6L_r + L$
- $1.4D$
- $1.246D - 1.0E_x + 1.0L$
- $1.246D - 1.0E_y + 1.0L$
- $1.246D + 1.0E_x + 1.0L$
- $1.246D + 1.0E_y + 1.0L$
- $D + 0.5L$
- $D + L$
- $D + 0.5L - 0.7W_x$
- $D + 0.5L - 0.7W_y$
- $D + 0.5L + 0.7W_x$
- $D + 0.5L + 0.7W_y$

The reason we use such an exhaustive list of load combinations is to account for **statistical probability** and **structural safety**. Buildings rarely experience every type of maximum load at the exact same moment, but they also cannot be designed based on just one "worst-case" scenario.

Atmospheric Conditions (Dhaka)

Accordingly, the design adopts good practice measures such as adequate concrete cover, proper compaction and curing, and the use of approved cement, aggregates, and reinforcement to minimize corrosion risk under Dhaka's humid monsoon climate.

No project specific information indicates exposure to highly aggressive industrial chemicals. Therefore, a normal atmospheric exposure condition is assumed for durability design, and standard Bangladesh National Building Code (BNBC) 2020 provisions for concrete durability and corrosion protection are considered sufficient.

The project site is in Dhaka city, which is an inland urban environment. The site is not located in a coastal or saline zone and is not directly exposed to marine spray.

Wind Load Parameters (Dhaka)

Wind parameters for Dhaka are taken as: $V = 65.7 \text{ m/s}$ (147.16 mph), $K_d = 0.85$, $K_{zt} = 1.0$, $I = 1.0$, and $K_z = 0.80$ at the reference height. Velocity pressure is computed using q_z

$$= 0.00256 \cdot K_z \cdot K_{zt} \cdot K_d \cdot V^2 \cdot I.$$

The structure is treated as flexible because the computed fundamental period is $T = 1.21 \text{ s}$ ($> 1.0 \text{ s}$).

Serviceability and Irregularity Checks

Key ETABS checks and results are summarized below.

Check	Result
Beam deflection	Max = 0.096 in, Allowable (l/240) = 0.917 in, OK
EQ drift (X)	Max drift = 0.522 in at 6th floor, OK
EQ drift (Y)	Max drift = 0.699 in at 6th floor, OK
Wind sway	W-x max = 0.421 in < 2.4 in, W+y max = 0.788 in < 2.4 in, OK
Wind drift (storey drift)	Max $\approx$ 0.928 in (Roof level), Allowable = 2.4 in, OK
Torsional irregularity	Max torsional ratio $\approx$ 1.25 (>1.20), torsional irregularity identified; accidental torsion effects are considered

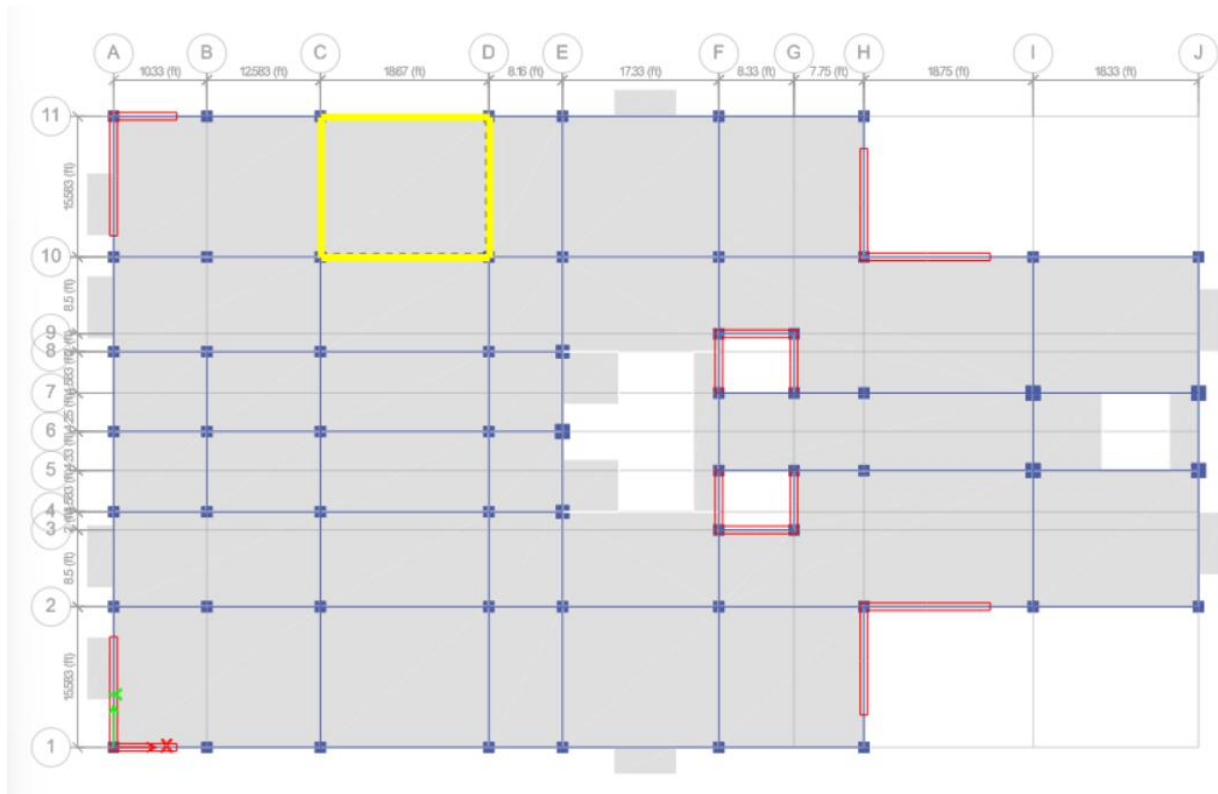
Planning Note from DMINB (FAR Concept)

DMINB defines Floor Area Ratio (FAR) as the ratio of total built-up floor area to total plot area. In general, FAR control is linked to maximum ground coverage, and reducing maximum ground coverage can allow higher total floor area under the same FAR rule. This report references DMINB for these planning constraints while structural design follows BNBC 2020.

Conclusion

Based on the adopted BNBC 2020 parameters for Dhaka, the ETABS model, load assignments, load combinations, and serviceability checks demonstrate code-consistent performance. Dual system compliance, base shear verification, and drift limits are satisfied, and the identified torsional irregularity is addressed through the analysis and design checks.

16. SLAB DESIGN



16.1. Slab Properties:

Floor	Label	Selected Slab (Unique Name)	Thickness	Type
3rd Floor	F4	49	6	Exterior

c/c long span (ft)	c/c short span(ft)
18.67	15.583

$$\frac{18.67}{15.583} = 1.198 < 2$$

∴ The slab is a two-way slab.

$$\text{Clear span length (long)} = 18.67 - \frac{15}{2 \times 12} \times 2 = 17.42 \text{ ft (l}_b\text{)}$$

$$\text{Clear span length (short)} = 15.583 - \frac{15}{2 \times 12} \times 2 = 14.33 \text{ ft (l}_a\text{)}$$

$$\text{Perimeter} = 2(17.42 + 14.33) = 63.5 \text{ ft}$$

$$\text{Floor area} = 249.63 \text{ ft}^2$$

$$\text{Slab thickness, } t = \frac{\text{Perimeter}}{180} = \frac{63.5 \times 12}{180} = 4.23 \text{ in} < 6 \text{ in}$$

$$\text{Effective depth (short)} = 6 - 1 = 5 \text{ in}$$

Effective depth (long) = $6 - 1.5 = 4.5$ in

16.2. Slab Loads:

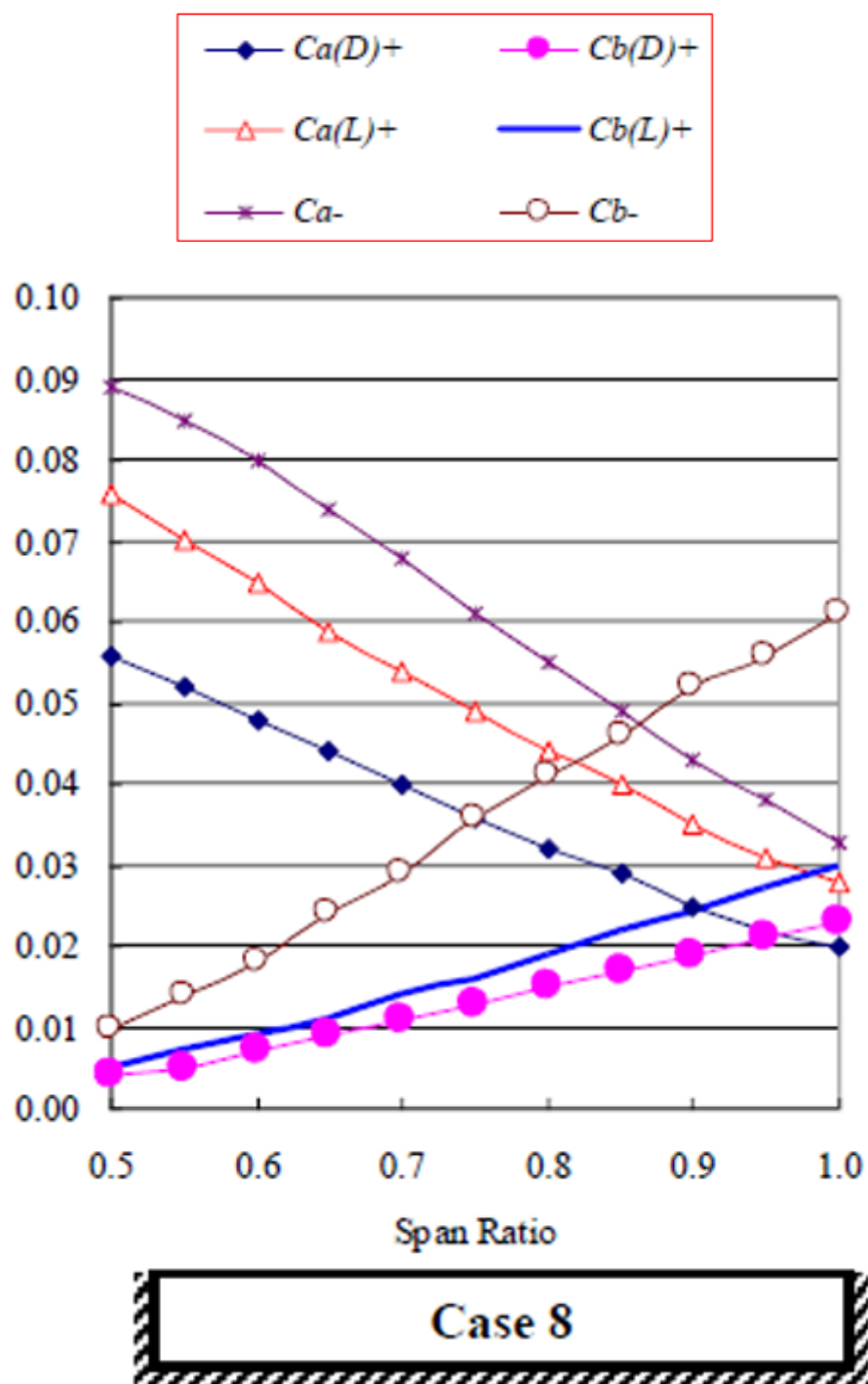
Dead Load Type	Magnitude (lb/ft²)		Live Load (lb/ft²)	
Floor Finish	25		41.77	
PW	25			
Self-weight	$\frac{6}{12} \times 150 = 75$			
Σ Dead Load	125		41.77	
Factored (lb/ft²)	W_{DL}	$1.4 \times 125 = 175$	W_{LL}	$1.7 \times 41.77 = 71.01$
Total Factored Load, w (lb/ft²)	246.01			

Table 3. Calculation of Loads on Slab

Clear span ratio, $m = \frac{l_a}{l_b} = 0.823$

Ca(D)+	0.031	Cb(D)+	0.016
Ca(L)+	0.043	Cb(L)+	0.02
Ca-	0.052	Cb-	0.043

Table 4. Moment Coefficients obtained from Graph



16.3. Calculation of moments and reinforcements

$$\begin{aligned} M_a(+) &= [C_a(DL)(+)w_{DL}l_a^2 + [C_a(LL)(+)w_{LL}l_a^2 \\ &= 1.74 \text{ k-ft} \end{aligned}$$

$$\begin{aligned} M_b(+) &= [C_b(DL)(+)W_{DL}l_b^2 + [C_b(LL)(+)W_{LL}l_b^2 \\ &= 1.28 \text{ k-ft} \end{aligned}$$

$$\begin{aligned} M_a(-) &= (C_a-)w l_a^2 = 2.628 \text{ k-ft} \\ M_b(-) &= (C_b-)w l_b^2 = 3.21 \text{ k-ft} \end{aligned}$$

Temperature reinforcement, i.e. minimum reinforcement that should be provided,
 $A_s = 0.0018bt = 0.1296 \text{ in}^2$ (for unit strip)

$$\begin{aligned} \text{Main reinforcement, } A_s &= \frac{0.85f_{cd}bd}{f_y} \left(1 - \sqrt{1 - \frac{2Mu}{\phi(0.85)f_{cd}bd^2}}\right) \\ &= \frac{0.85 \times 3 \times 12 \times d}{60} \left(1 - \sqrt{1 - \frac{2Mu}{0.9 \times 0.85 \times 3 \times 12 \times d^2}}\right) \text{-----(5)} \end{aligned}$$

Using equation (5), we get,

Moment (kip-ft)		Effective depth (in)	Reinforcement, A_s (in ² /ft)		Ast (in ² /ft)	Bar no.	Spacing = $\frac{12 \times 0.11}{A_s}$
$M_a(+)$	1.74	5	$A_{sa}(+)$	0.078	0.1296	1 #3	10" c/c
$M_b(+)$	1.28	4.5	$A_{sb}(+)$	0.057		1 #3	10" c/c
$M_a(-)$	2.628	5	$A_{sa}(-)$	0.1188		1 #3	10" c/c
$M_b(-)$	3.21	4.5	$A_{sb}(-)$	0.1457		1 #3	10" c/c

Table 5. Moments acting on Slab and main reinforcement required for resistance

Check for effective depth

$$\rho_{\max} = 0.75\rho_b = 0.75 \times 0.85^2 \times \frac{3.5}{60} \times \frac{87}{87+60} = 0.01886$$

Maximum moment obtained = $M_b(-) = 3.21 \text{ kip-ft}$

$$d_{\text{req}} = \sqrt{\frac{Mu}{\phi \rho f_y b (1 - 0.59 \rho \frac{f_y}{f'_c})}}$$

$$= 1.83 < 4.5 \text{ in}$$

∴ Design is adequate.

Spacing :

Short Direction:

- i. $S = \frac{0.11 \times 12}{.1296} = 10.18 \approx 10$
- ii. $S_{max} = 2 \times t = 2 \times 6 = 12''$
- iii. $S_{max} = 18''$

Long Direction:

- i. $S = \frac{0.11 \times 12}{.1296} = 10.18'' \approx 10''$
- ii. $S_{max} = 2 \times t = 2 \times 6 = 12''$
- iii. $S_{max} = 18''$

16.4. Provisions for Slab Reinforcement Calculation

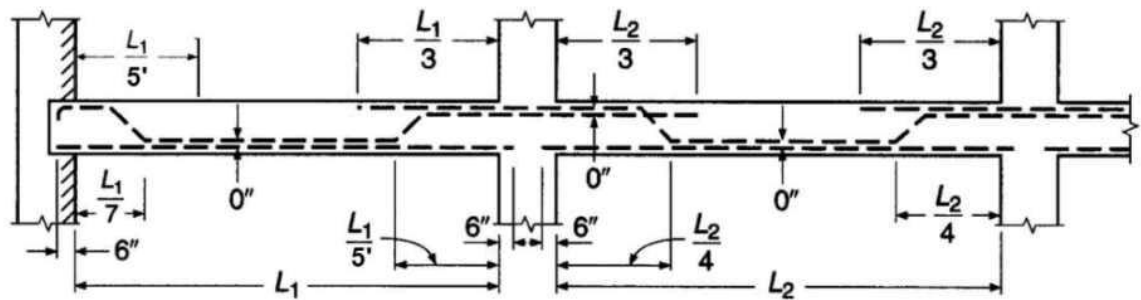


Fig: Slab Reinforcements Criteria

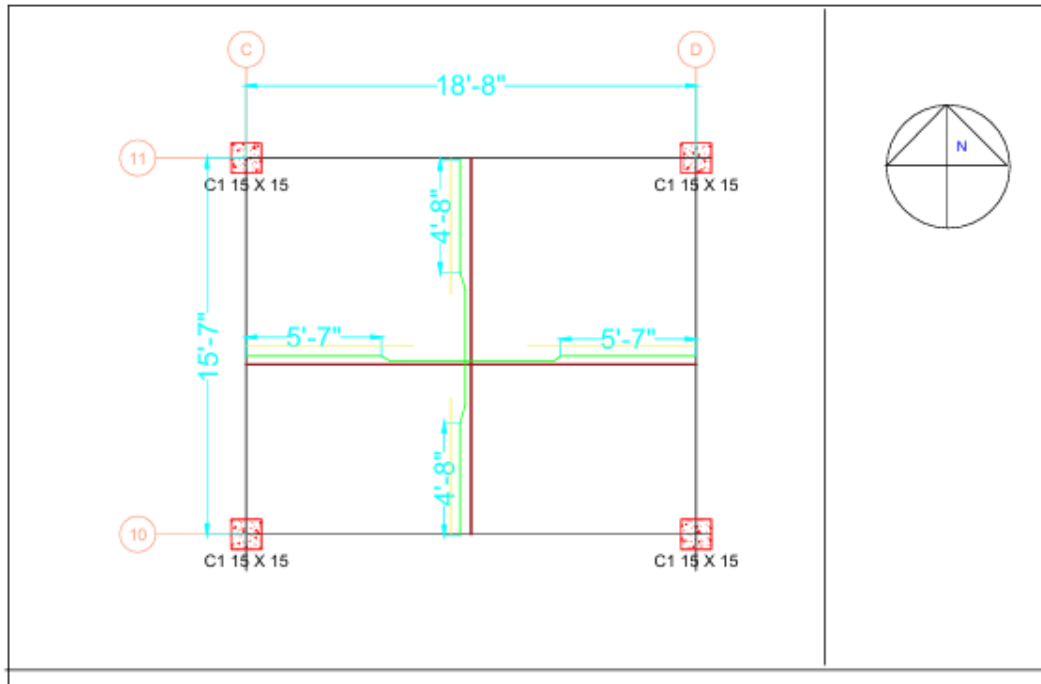
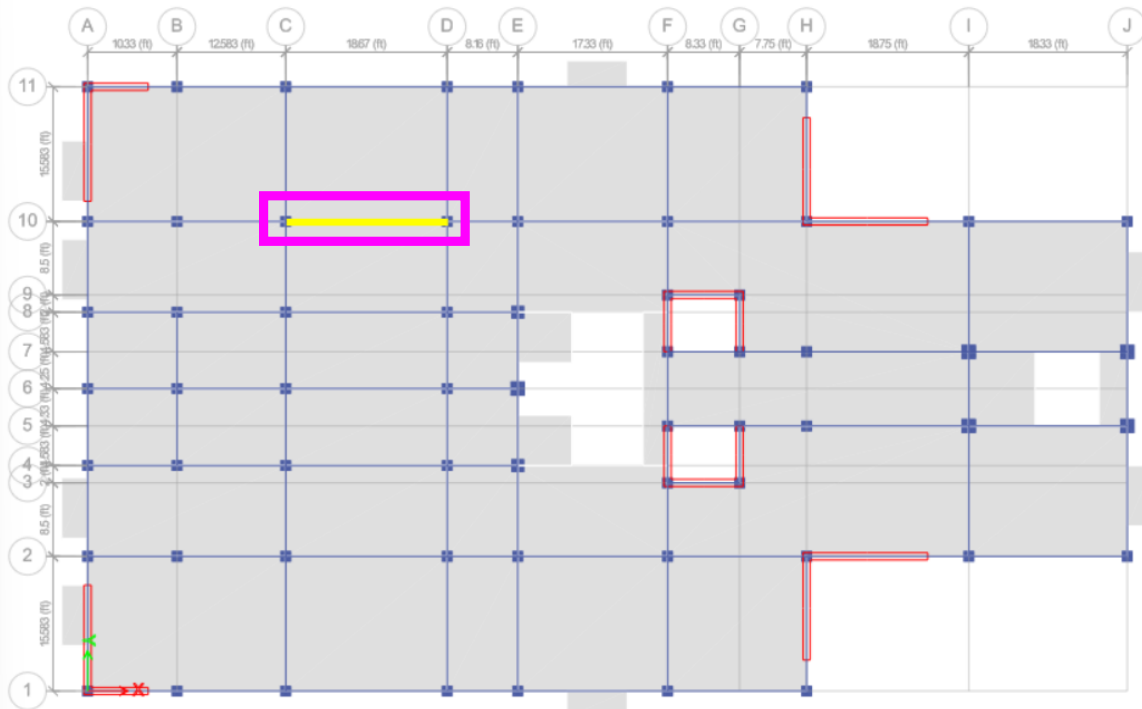


Fig : Reinforcement Detailing of slab

17. BEAM DESIGN



17.1. Beam Properties

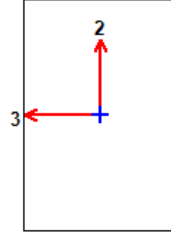
Beam Element Details				
Element	Unique Name	Level	Section Type	Critical Load Combination
B18	1603	3 rd	Interior Beam	1.246D - 1.0Ex+1.0L

Section Properties			
Length (ft)	Width, b (in)	Depth, h (in)	Distance to Rebar Center, d _c (in)
18.67	10	15	1.5

Material Properties	
f'_c (psi)	f_y (psi)
4500	60000

17.2. ETABS Concrete Frame Design

ACI 318-08 Beam Section Design (Summary)



Beam Element Details

Level	Element	Unique Name	Section ID	Combo ID	Station Loc	Length (in)	LLRF	Type
3rd Floor	B18	1603	Floor Beam 15 X 10	1.246D - 1.0Ex + 1.0L	7.5	224.04	1	Sway

Section Properties

b (in)	h (in)	b _f (in)	d _s (in)	d _{et} (in)	d _{cb} (in)
10	15	10	0	1.5	1.5

Material Properties

E _c (lb/in ²)	f' _c (lb/in ²)	Lt.Wt Factor (Unitless)	f _y (lb/in ²)	f _{ys} (lb/in ²)
3823676.24	4500	1	60000	60000

Design Code Parameters

ϕ_T	ϕ_{CTied}	$\phi_{CSpiral}$	ϕ_{Vns}	ϕ_{Vs}	ϕ_{Vjoint}
0.9	0.65	0.75	0.75	0.6	0.85

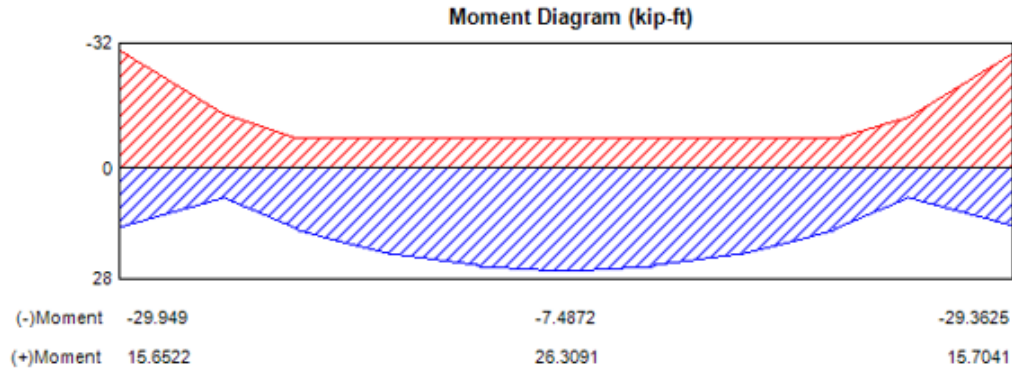
Flexural Reinforcement for Major Axis Moment, M_{u3}

	End-I Rebar Area in ²	End-I Rebar %	Middle Rebar Area in ²	Middle Rebar %	End-J Rebar Area in ²	End-J Rebar %
Top (+2 Axis)	0.508	0.34	0.1655	0.11	0.4977	0.33
Bot (-2 Axis)	0.3488	0.23	0.4528	0.3	0.35	0.23

Flexural Design Moment, M_{u3} and Axial Force, P_u

	Station Loc in	Design M _u kip-ft	Design P _u kip
Top (+2 Axis) End-I	7.5	-29.949	0
Top (+2 Axis) Middle	156.828	-7.4872	0
Top (+2 Axis) End-J	216.54	-29.3625	0
Bot (-2 Axis) End-I	44.808	15.6522	0
Bot (-2 Axis) Middle	156.828	21.59	0
Bot (-2 Axis) End-J	179.232	15.7041	0

17.3. Check for Minimum Depth Required, d_{min}



Largest moment along the beam is $-29.949 \text{ kip} - \text{ft}$

$$\therefore M_u = -29.3625 \text{ kip} - \text{ft}$$

We know,

$$\rho_b = 0.85\beta_1 \left(\frac{f_c'}{f_y} \right) \left(\frac{87000}{87000 + f_y} \right) \text{ and } \rho_{\max} = 0.75\rho_b$$

As $\beta = 0.425$ for $f_c' = 4500 \text{ psi}$,

$$\therefore \rho_{\max} = 0.75 \times 0.85 \times 0.85 \times \frac{4500}{60000} \times \frac{87000}{87000 + 60000} = 0.038$$

Now,

$$d_{req} = \sqrt{\frac{M_u}{\phi \rho f_y b \left(1 - (0.59\rho) \left(\frac{f_y}{f_c'} \right) \right)}}$$

$$= \sqrt{\frac{29.3625 \times 12 \times 10^3}{0.9 \times 0.038 \times 60000 \times 10 \left(1 - 0.59 \times 0.038 \times \frac{60000}{4500} \right)}} = 4.95 \text{ in}$$

$$d_{provided} = 15 - 1.5 = 13.5 \text{ in} > 4.95 \text{ in}$$

$\therefore$ The beam may be designed as a **singly reinforced beam**

17.4. Effective Flange Width

The effective flange width shall be the minimum of these values:

1. $16t_s + b_w = 16 \times 6 + 10 = 106 \text{ in}$
2. $\frac{\text{span}}{4} = \frac{18.67 \times 12}{4} \approx \mathbf{56.01 \text{ in}}$
3. $\text{Clear span} + b_w = 18.67 \times 12 - 2 \times \frac{15}{2} + 15 \approx 224.04 \text{ in}$

$$\therefore b_{eff} = 56.01 \text{ in}$$

17.5. Check for T-Beam

$$A_s = \frac{M_u}{\phi f_y \left(d - \frac{a}{2}\right)} \text{ where, } a = \frac{A_s f_y}{0.85 f_c' b_{eff}}, \text{ or, } A_s = \frac{0.85 f_c' a b_{eff}}{f_y}$$

$$\therefore \frac{M_u}{\phi f_y \left(d - \frac{a}{2}\right)} = \frac{0.85 f_c' a b_{eff}}{f_y}$$

For positive moment, $M_u = 26.3091 \text{ kip-ft}$

Now,

$$\frac{26.301 \times 12 \times 1000}{0.85 \times 60000 \times \left(13.5 - \frac{a}{2}\right)} = \frac{0.85 \times 4500 \times a \times 56.01}{60000}$$

For $a=6\text{in}$, $A_s = \frac{M_u}{\phi f_y \left(d - \frac{a}{2}\right)} = 0.56 \text{ in}^2$, $a=0.82$

$$\therefore c = \frac{a}{\beta_1} = \frac{0.82}{0.85} = \mathbf{0.96 \text{ in}} < t_s = 6 \text{ in}$$

So, the beam is to be designed as a **rectangular beam**.

For negative moment, the beam is always to be designed as **rectangular beam**.

17.6. Reinforcement Calculation for Flexure, A_s

17.6.1. Minimum Reinforcement Requirement

Maximum of $A_{s(\min)} = \left(\frac{3\sqrt{f_c'}}{f_y} \right) b_w d$, or, $\frac{200}{f_y} b_w d$

$$A_{s(\min)} = \left(\frac{3\sqrt{4500}}{60000} \right) \times 10 \times 13.5 = 0.453 \text{ in}^2$$

$$A_{s(\min)} = \frac{200}{60000} \times 10 \times 13.5 = \mathbf{0.45 \text{ in}^2}, 0.453 \text{ governs}$$

For Positive Moment

$$M_u = 26.3091 \text{ kip-ft}$$

Here, $b = b_{eff}$

$$\therefore A_s = \frac{M_u}{\phi f_y \left(d - \frac{A_s f_y}{0.85 f_c' b_{eff}} \right)}$$

$$A_{s(+ve)} = \frac{26.3091 \times 12 \times 1000}{0.9 \times 60000 \left(13.5 - \frac{A_s \times 60000}{0.85 \times 4500 \times 56.01} \right)}$$

for $a = 0.5 \text{ in}$, $A_s = \frac{M_u}{\phi f_y \left(d - \frac{a}{2} \right)} = 0.44 \text{ in}^2$, $a = \frac{A_s f_y}{0.85 f_c' b_{eff}} = 0.58$, NOT OKAY

for $a = 0.58$, $A_s = \frac{M_u}{\phi f_y \left(d - \frac{a}{2} \right)} = 0.443 \text{ in}^2$, $a = \frac{A_s f_y}{0.85 f_c' b_{eff}} = 0.12$, OKAY

Solving for $A_{s(+ve)}$, we find, $A_{s(+ve)} = 0.445 \text{ in}^2 < A_{s(\min)}$

$$\therefore A_{s(+ve)} = \mathbf{0.453 \text{ in}^2} \text{ (Use 2 \#5 bars)}$$

For Negative Moment

$$M_u = 29.3625 \text{ kip-ft}$$

Here, $b = b_w$

$$\therefore A_s = \frac{M_u}{\phi f_y \left(d - \frac{A_s f_y}{0.85 f_c' b_w} \right)}$$

$$A_{s(-ve)} = \frac{29.3625 \times 12 \times 1000}{0.9 \times 60000 \left(13.5 - \frac{A_s \times 60000}{0.85 \times 4500 \times 10} \right)}$$

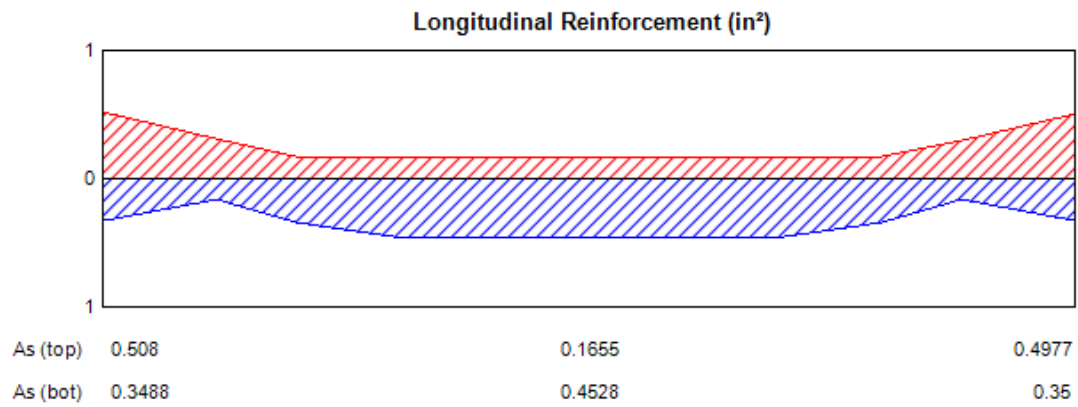
For $a = 0.5 \text{ in}$, $A_s = \frac{M_u}{\phi f_y \left(d - \frac{a}{2} \right)} = 0.492 \text{ in}^2$, $a = \frac{A_s f_y}{0.85 f_c' b_{eff}} = 0.77 \text{ in}$, NOT OKAY

For $a = 0.77 \text{ in}$, $A_s = \frac{M_u}{\phi f_y \left(d - \frac{a}{2} \right)} = 0.498 \text{ in}^2$, $a = \frac{A_s f_y}{0.85 f_c' b_{eff}} = 0.78 \text{ in}$, NOT OKAY

For $a = 0.78 \text{ in}$, $A_s = \frac{M_u}{\phi f_y \left(d - \frac{a}{2} \right)} = 0.498 \text{ in}^2$, $a = \frac{A_s f_y}{0.85 f_c' b_{eff}} = 0.78 \text{ in}$, OKAY

Solving for $A_{s(-ve)}$, we find, $A_{s(-ve)} = 0.498 \text{ in}^2 > A_{s(\min)}$

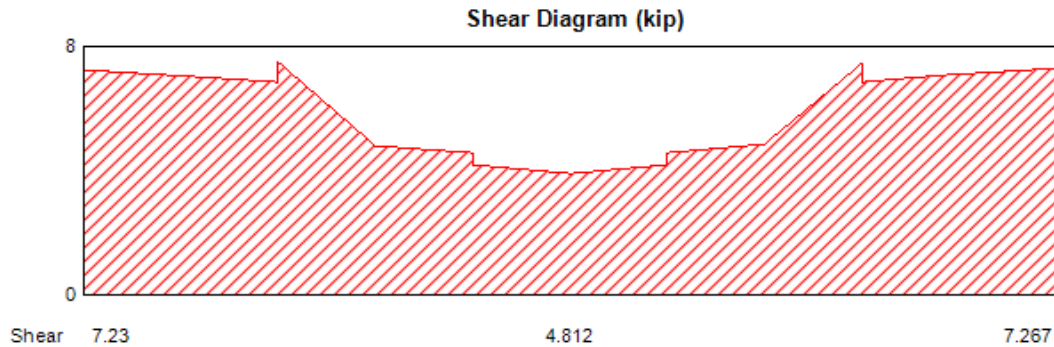
$$\therefore A_{s(-ve)} = 0.486 \text{ in}^2 \text{ (Use 2 \#5 bars)}$$



Longitudinal Reinforcement of the Beam (in ²)			
±ve	Maximum Moment (kip-ft)	A _s (Hand Calculation), in ²	A _s (ETABS), in ²
Negative	13.1684	0.443	0.508
Positive	6.5842	0.498	0.3488

There's a variation with those obtained from ETABS.

17.7. Shear Reinforcement Calculation



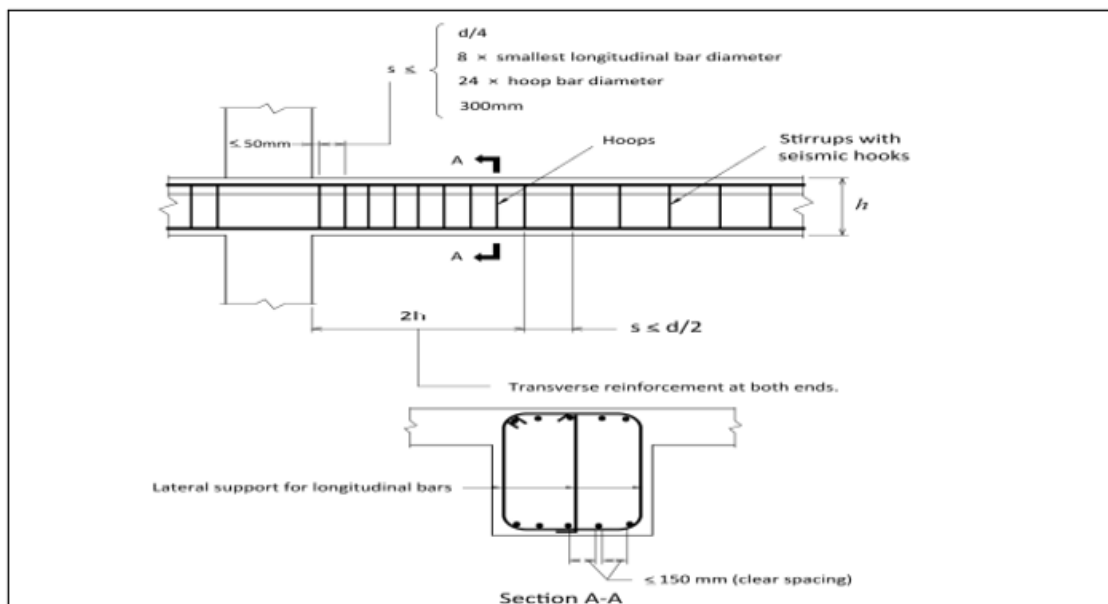
From ETABS Design Report,

$V_u = 7.23$ kip and $\phi V_c = 13.584$ kip: $V_u < \phi V_c$

$\therefore$ Shear reinforcement is not required

No stirrup is required theoretically. But as per the requirement of BNBC 8.3.4.3, it is necessary to provide shear reinforcement in the plastic hinge regions as shown in the figure below:

17.8. Provisions for Beam Reinforcement Calculation



Using #4 bars as stirrups-

$$h = 15 \text{ in} \quad \therefore 2h = 2 \times 15 = 40 \text{ in} = 30 \text{ inc}$$

$s_{\max}$ is the least of:

$$d/4 = 13.5/4 = 3.375 \text{ in governs}$$

$$8 \times d_{\text{smallest long bar}} = 8 \times \frac{5}{8} = 5 \text{ in}$$

$$24 \times d_{\text{hoop bar}} = 24 \times \frac{4}{8} = 12 \text{ in}$$

$$300 \text{ mm} = 11.8 \text{ in}$$

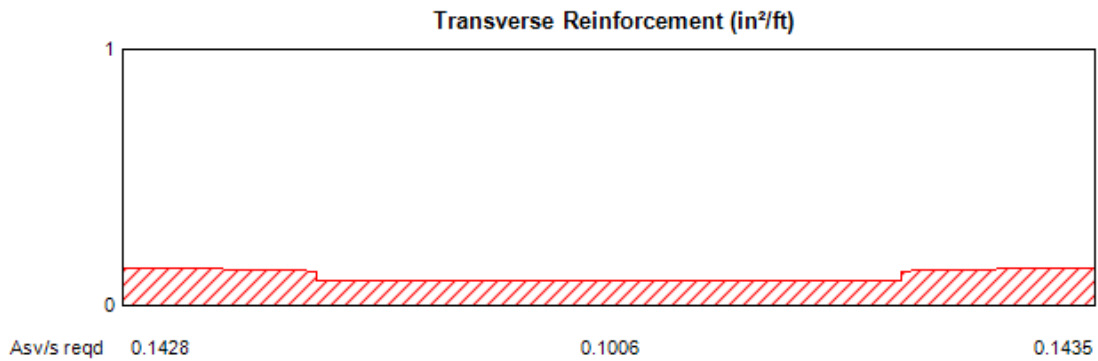
$\therefore$ Adopt $s_{\max} = 3.375 \text{ in} \sim 4 \text{ in}$

$$\text{For mid span, } s = d/2 = 13.5/2 = 6.75 \text{ in}$$

$\therefore$ Adopt $s = 6.75 \text{ in} \sim 7 \text{ in}$

Therefore, we may select **#4 bars @ 4 in c/c** spacing from near column up to **2h = 30 in** from the face of the column and **@ 7 in c/c** spacing at mid span of the column

17.9. Traverse Reinforcement



Here, for 12 in, shear reinforcement required is 0.1428 in^2
 Area of #4 bar = 0.2 in^2
 $\therefore$ For 4 in spacing, $A_{vs(\text{required})} = 0.1428 \times \frac{4}{12} = 0.0352 \text{ in}^2 < 0.2 \text{ in}^2$
 $\therefore$ For 7 in spacing, $A_{vs(\text{required})} = 0.1428 \times \frac{7}{12} = 0.0616 \text{ in}^2 < 0.2 \text{ in}^2$

17.10. Reinforcement Design

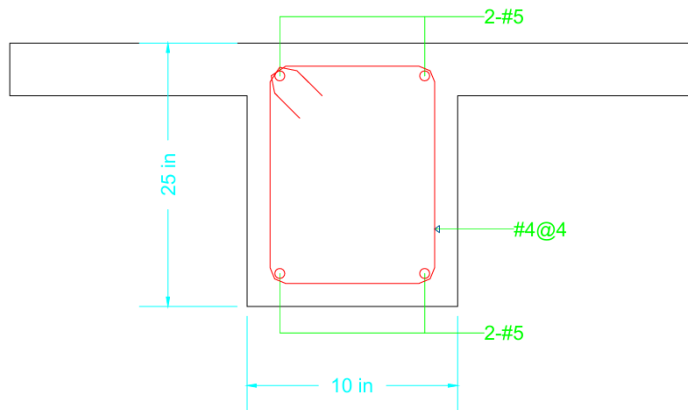


Fig: Cross Sectional Reinforcement Layout

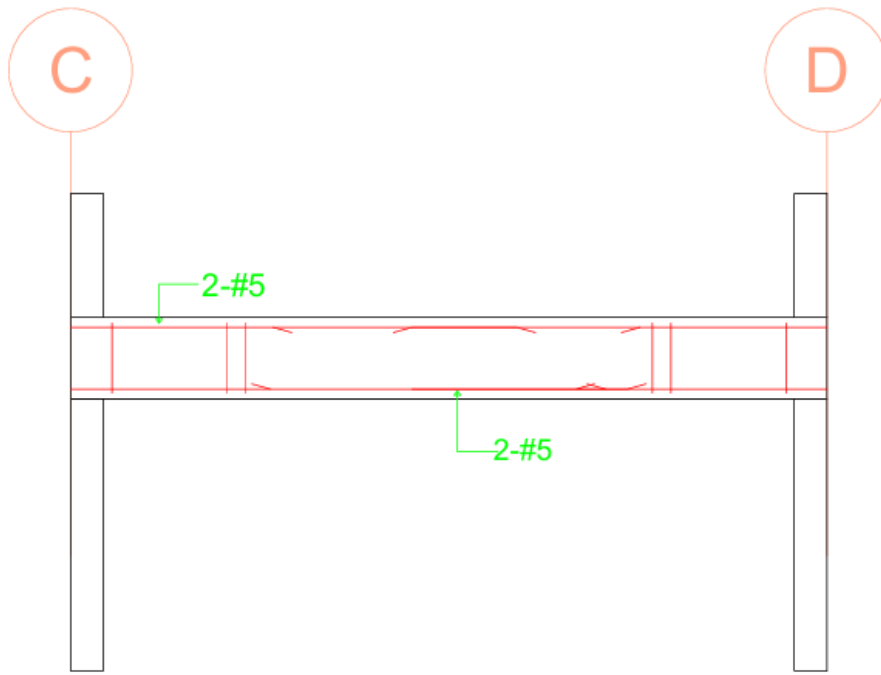


Fig: Longitudinal Reinforcement of Beam

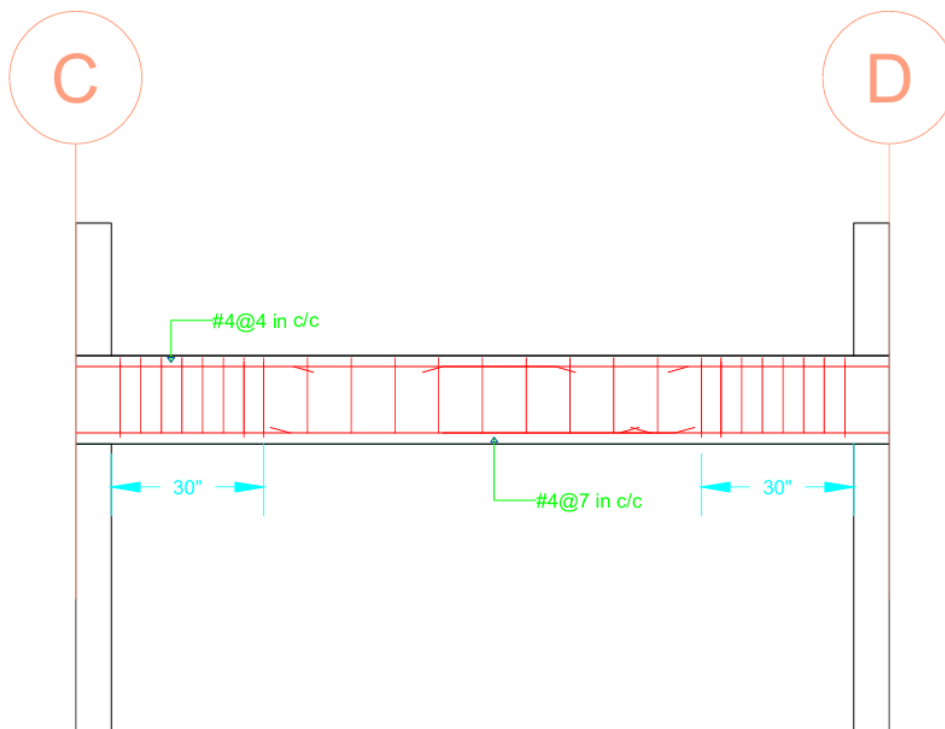


Fig: Shear Reinforcement of Beam

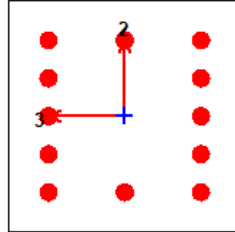
18. BEAM SCHEDULING

Concrete Strength			4500 psi
Name of Beam	Beam Size	Floor	1st Floor to Roof Floor level
Floor Beam FB	15x10	1st Floor to Roof	
Grade Beam GB	17x12	Basement & Ground Floor	

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ETABS Concrete Frame Design

ACI 318-08 Column Section Design (Summary)



Column Element Details

Level	Element	Unique Name	Section ID	Combo ID	Station Loc	Length (in)	LLRF	Type
3rd Floor	C18	313	C1 15 X 15	1.2D + 1.6L + 0.5Lr	52.5	120	0.44	Sway

Section Properties

b (in)	h (in)	dc (in)	Cover (Torsion) (in)
15	15	2.5688	1

Material Properties

E_c (lb/in ²)	f'_c (lb/in ²)	Lt.Wt Factor (Unitless)	f_y (lb/in ²)	f_{ys} (lb/in ²)
3823676.24	4500	1	60000	60000

Design Code Parameters

ϕ_T	ϕ_{CTied}	$\phi_{CSpiral}$	ϕ_{Vns}	ϕ_{Vs}	ϕ_{Vjoint}
0.9	0.65	0.75	0.75	0.6	0.85

Axial Force and Biaxial Moment Design for P_u , M_{u2} , M_{u3}

Design P_u kip	Design M_{u2} kip-ft	Design M_{u3} kip-ft	Minimum M2 kip-ft	Minimum M3 kip-ft	Rebar Area in ²	Rebar % %
426.857	-1.8643	-37.35	37.35	37.35	2.25	1

Axial Force and Biaxial Moment Factors

	C_m Factor Unitless	δ_{ns} Factor Unitless	δ_s Factor Unitless	K Factor Unitless	Length in
Major Bend(M3)	0.293896	1	1	1	105
Minor Bend(M2)	0.29475	1	1	1	105

Shear Design for V_{u2} , V_{u3}

	Shear V_u kip	Shear ϕV_c kip	Shear ϕV_s kip	Shear ϕV_p kip	Rebar A_v /s in ² /ft
Major, V_{u2}	3.865	36.561	0	5.326	0
Minor, V_{u3}	3.172	36.561	0	5.162	0

19.1. Column Properties

Column Element Details				
Element	Unique Name	Floor	Section Type	Critical Load Combination
C18	313	3 rd	Interior Column	1.2D+1.6L+0.5Lr

Section Properties			
Height (ft)	Width, b (in)	Depth, h (in)	Distance to Rebar Center, d _c (in)
10	15	15	2.5688

Material Properties	
f _c ' (psi)	f _y (psi)
4500	60000

19.2. Column Design Check for Axial Compression

Using Rebars: 8 #7.

From ETABS, Ultimate Load, $P_u = 426.857 \text{ kips}$

Now,

$$A_g = 15 \times 15 = 225 \text{ in}^2 \quad \text{and} \quad A_{st} = 8 \times 0.61 = 4.81 \text{ in}^2$$

$$\therefore \rho = \frac{A_{st}}{A_g} = \frac{4.81}{225} = 2.11 \% ; \text{ok}$$

$$\text{Nominal Strength, } P_n = 0.85f'_c(A_g - A_{st}) + A_{st}f_y$$

$$\therefore P_n = 0.85f'_c(A_g - A_{st}) + A_{st}f_y = 0.85 \times 4.5 \times (225 - 4.71) + 4.71 \times 60 = 1130.8 \text{ kips}$$

$$\therefore \phi P_n = 0.825 \times 0.65 \times 1130.8 = 606.4 \text{ kips} > P_u \text{ (OK)}$$

So, the **designed column section is adequate** for Axial Compression.

19.3. Column Design Check for Biaxial Bending:

We will evaluate the design of column under biaxial bending using **Bresler's Reciprocal Method**:

$$\frac{1}{P_n} = \frac{1}{P_{nx0}} + \frac{1}{P_{ny0}} - \frac{1}{\phi P_0}$$

In the absence of a column interaction diagram specific to $f_y = 60 \text{ ksi}$ and $f'_c = 4.5 \text{ ksi}$, we utilized the Excel-based interaction diagram tool "RECTBEAM" which is developed in accordance with the American Concrete Institute (ACI) Code 318-99. Relevant input parameters used within the tool are presented as snapshots for clarity. This tool generates interaction diagrams for axial load, P_u , and

bending moment, M_u , rather than normalized values, K_n and R_n . The values extracted from the diagram will serve as a basis for assessing the column's adequacy under specified loading conditions.

Design Axial Force & Biaxial Moment for $P_u - M_{u2} - M_{u3}$ Interaction

Design P_u kip	Design M_{u2} kip-ft	Design M_{u3} kip-ft	Controlling Combo
426.857	-1.8543	-37.35	1.2D+1.6L+0.5Lr

Modifications were made to the Excel input parameters to reflect the moment direction, and the corresponding interaction diagram was generated. The axial load, P_u value associated with each moment case was then obtained from the interaction diagram. This value was subsequently analysed using the reciprocal method and compared with the design axial load $P_u=426.857 \text{ kip}$ for verification

19.4. Spreadsheet Analysis

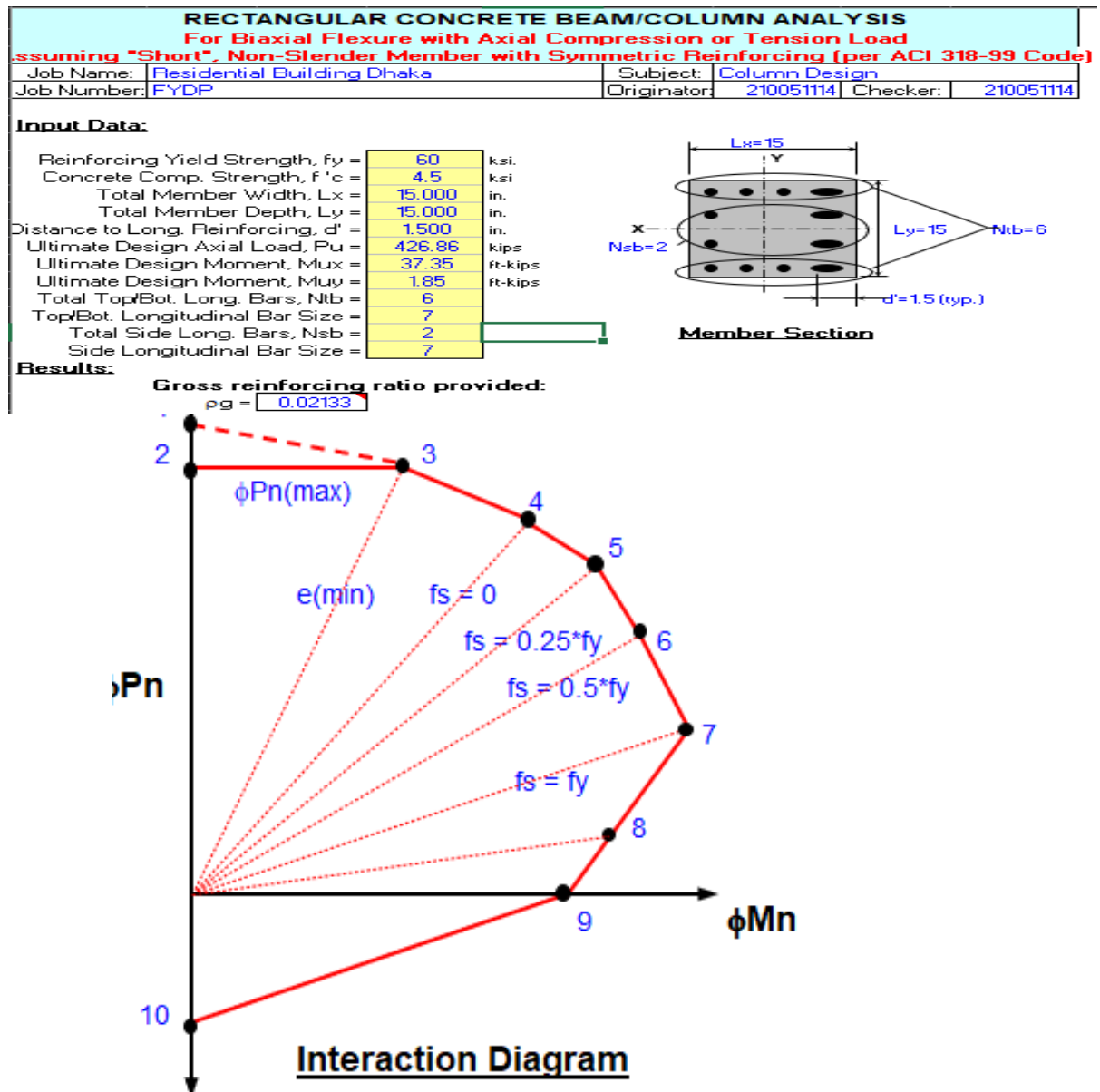
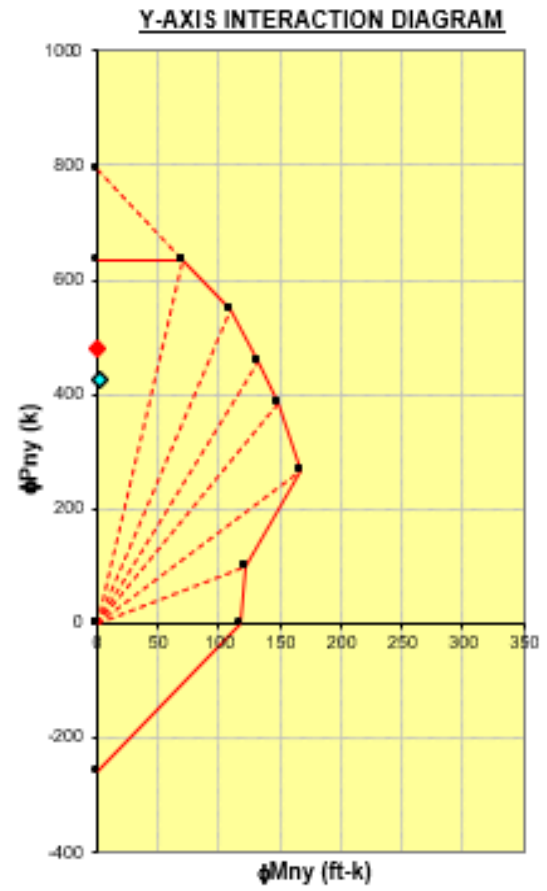
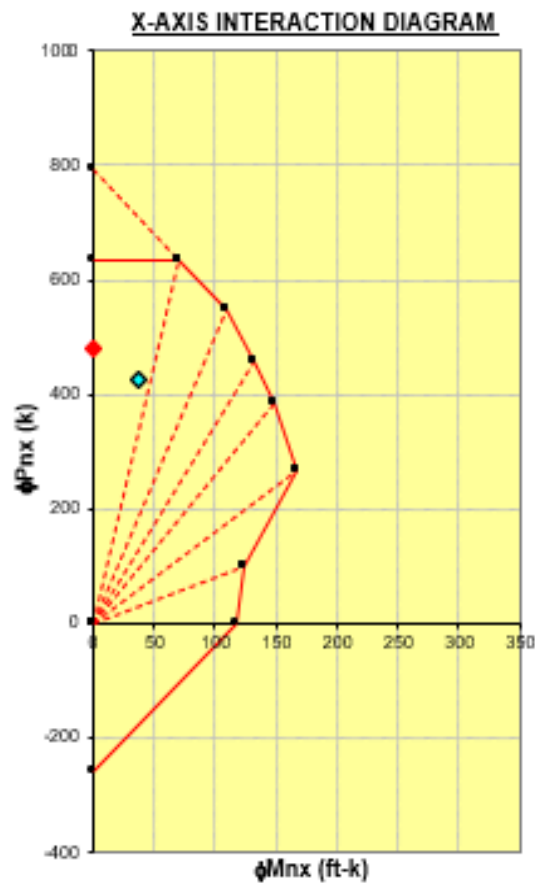


Figure 2 Spreadsheet Input and Column Interaction Diagram Key Points



19.5. Result Analysis

For governing $M_{u3} = 76.71 \text{ k} - ft$ from the graph,

$$P_{u2} = \phi P_{nx} = 631.06 \text{ kip}$$

For governing $M_{u2} = 68.09 \text{ k} - ft$ from the graph,

$$P_{u3} = \phi P_{ny} = 631.06 \text{ kip}$$

Using the reciprocal method,

$$\frac{1}{P_n} = \frac{1}{P_{nx}} + \frac{1}{P_{ny}} - \frac{1}{P_0}$$

$$\therefore P_n = 1355.8 \text{ kip}$$

As, $\phi P_n > P_u = 881.2 \text{ kips} > 426.857 \text{ kips}$, **the section is OK.**

Also, Pure Axial Compression Capacity without Reinforcements = $881.2 \text{ kips} > P_u$, this means the column is **OK** in concentric compression as well.

19.6. Verification of Interaction Diagrams with Respect to ETABS

An additional validation of the interaction diagrams was performed by comparing them with those generated in ETABS. The values of (P_{u2}) and (P_{u3}) obtained from ETABS were found to closely align with the values derived from the RECTBEAM Excel tool. This consistency confirms that the column meets the design requirements for both the applied moment and compressive load.

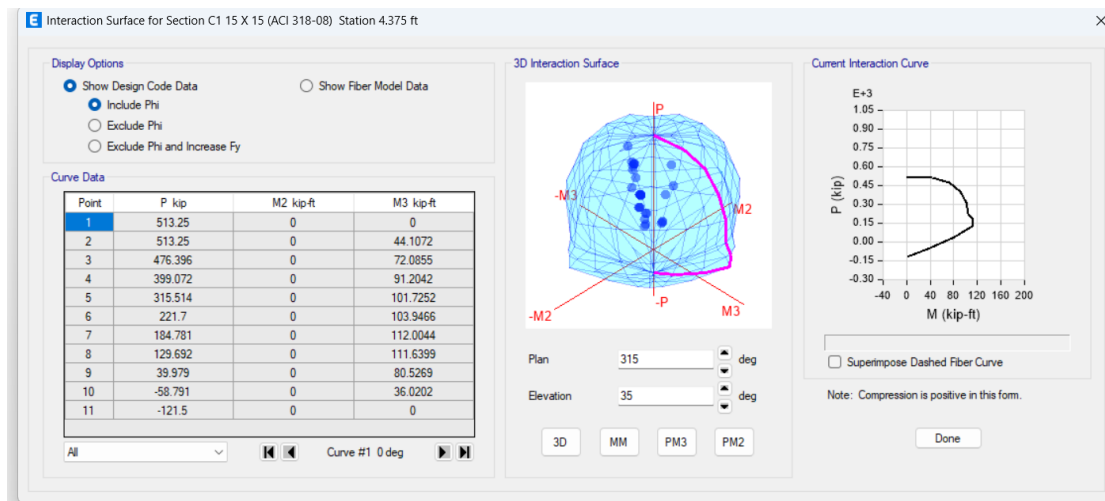


Figure : P_u - M_u Interaction

19.7. Provisions for Column Reinforcement Calculation

Tie bar spacing according to ACI guidelines

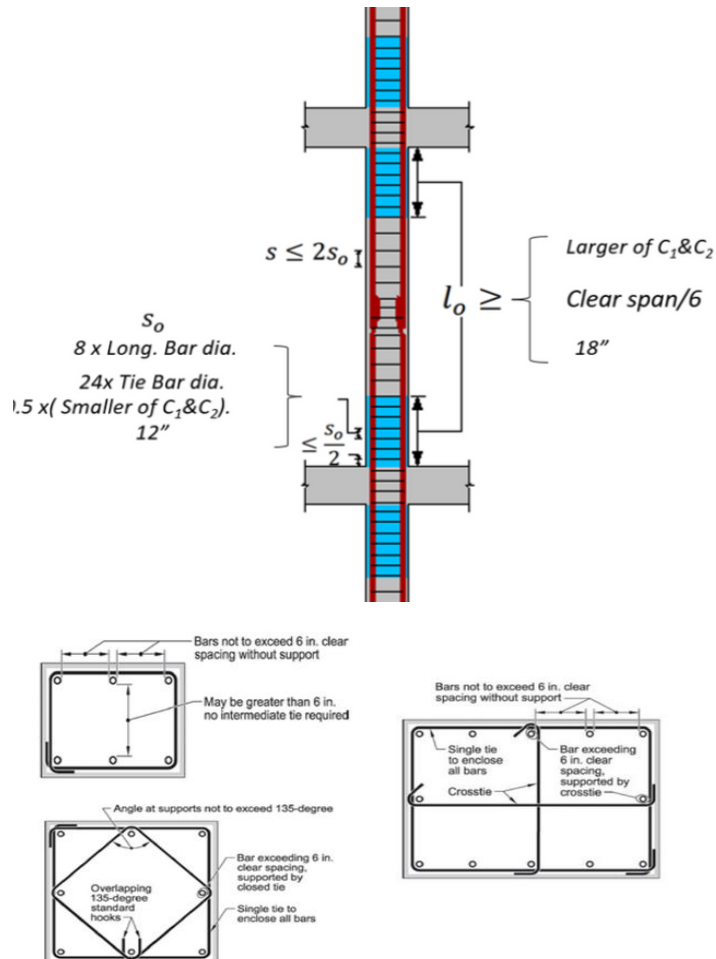


Figure 3 Transverse reinforcement requirements- rectangular hoop for members subjected to bending and axial load rectangular hoop

Determination of Tie Bar Spacing

Using #3 bars as ties.

$$\Rightarrow 0.5 \text{ smaller of } c_1 \text{ or } c_2 = 0.5 * 15 = 7.5 \text{ in}$$

$$s_o \leq \Rightarrow 8d_b = 8 * 7/8 = 7 \text{ in}$$

$$\Rightarrow 24 * \frac{3}{8} = 9 \text{ in}$$

$$\Rightarrow 12 \text{ in}$$

$$l_o \geq \Rightarrow \text{Larger dimension of column} = 15 \text{ in}$$

$$\Rightarrow \frac{\text{Clear height of column}}{6} = \frac{120 - 15 \times 2}{6} = 15 \text{ in}$$

$$\Rightarrow 18 \text{ in}$$

So, we may select **#3 bar ties @ 7 in c/c** spacing from near column-beam joint up to **18 in from the face of the beam column joint**.

Again, $s = 2s_0 = 2 \times 7 = 14 \text{ in}$

So, at **mid span #3 bar ties 14 in c/c** spacing

19.8. Reinforcement Design

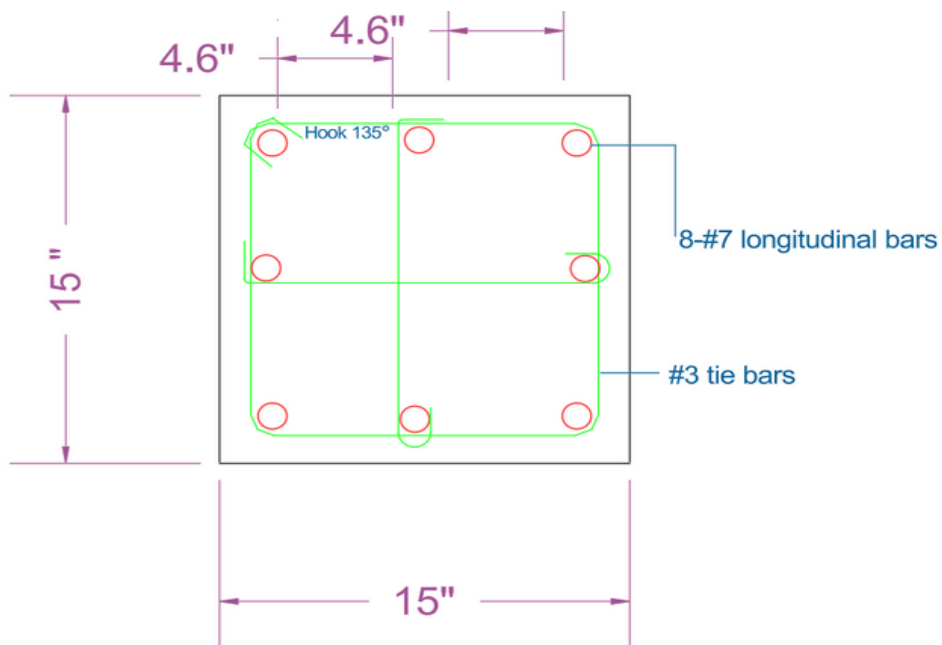


Fig: Cross Section

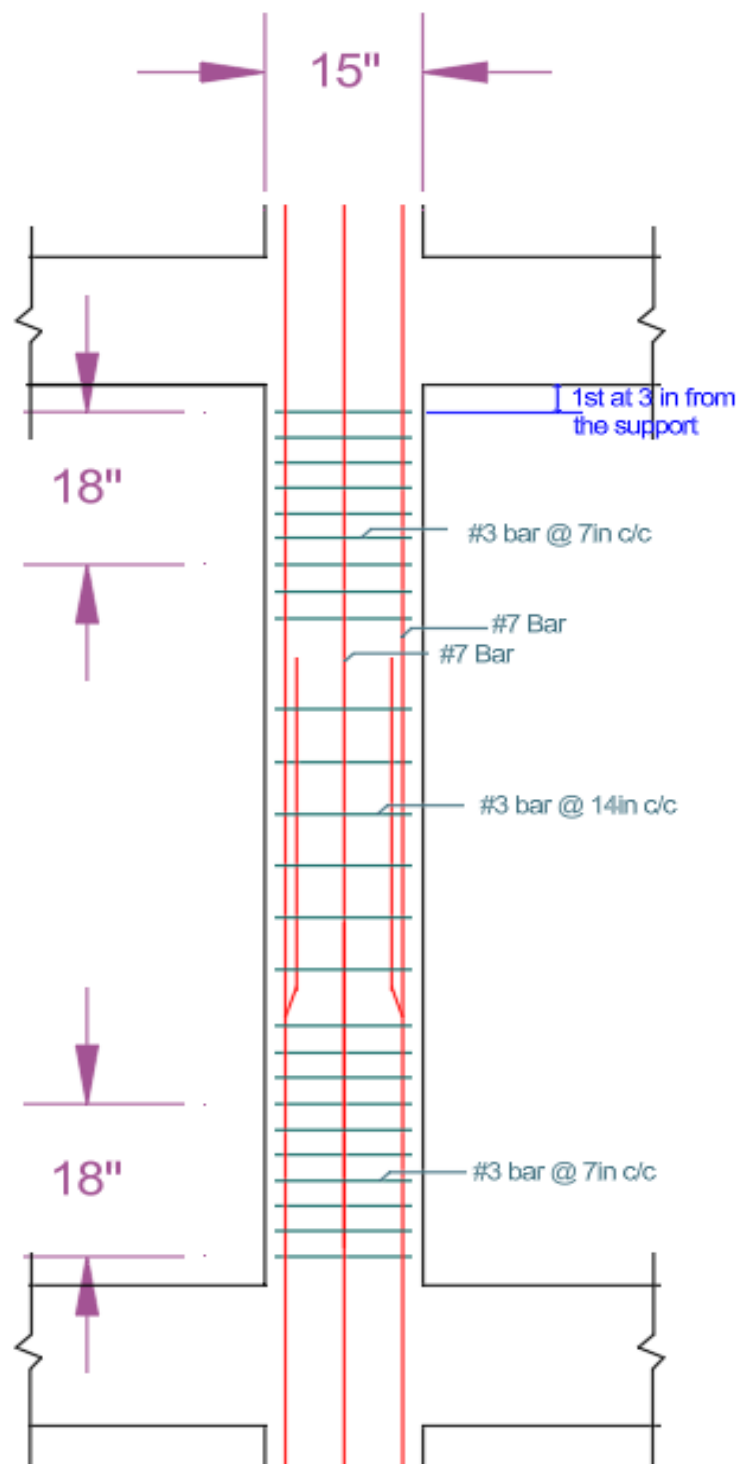
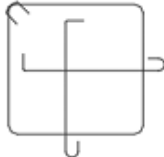
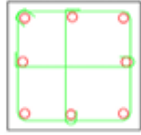
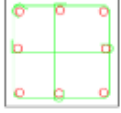
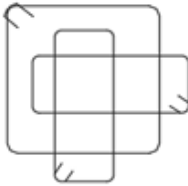
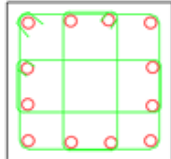
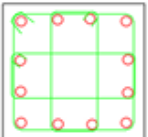
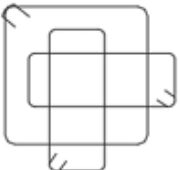
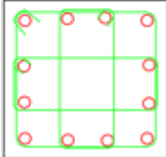
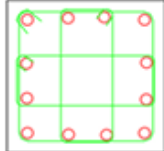
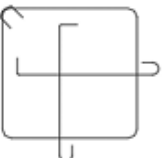
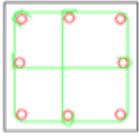
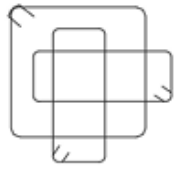
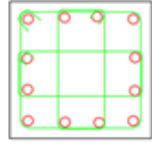
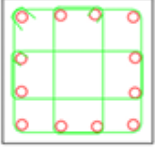
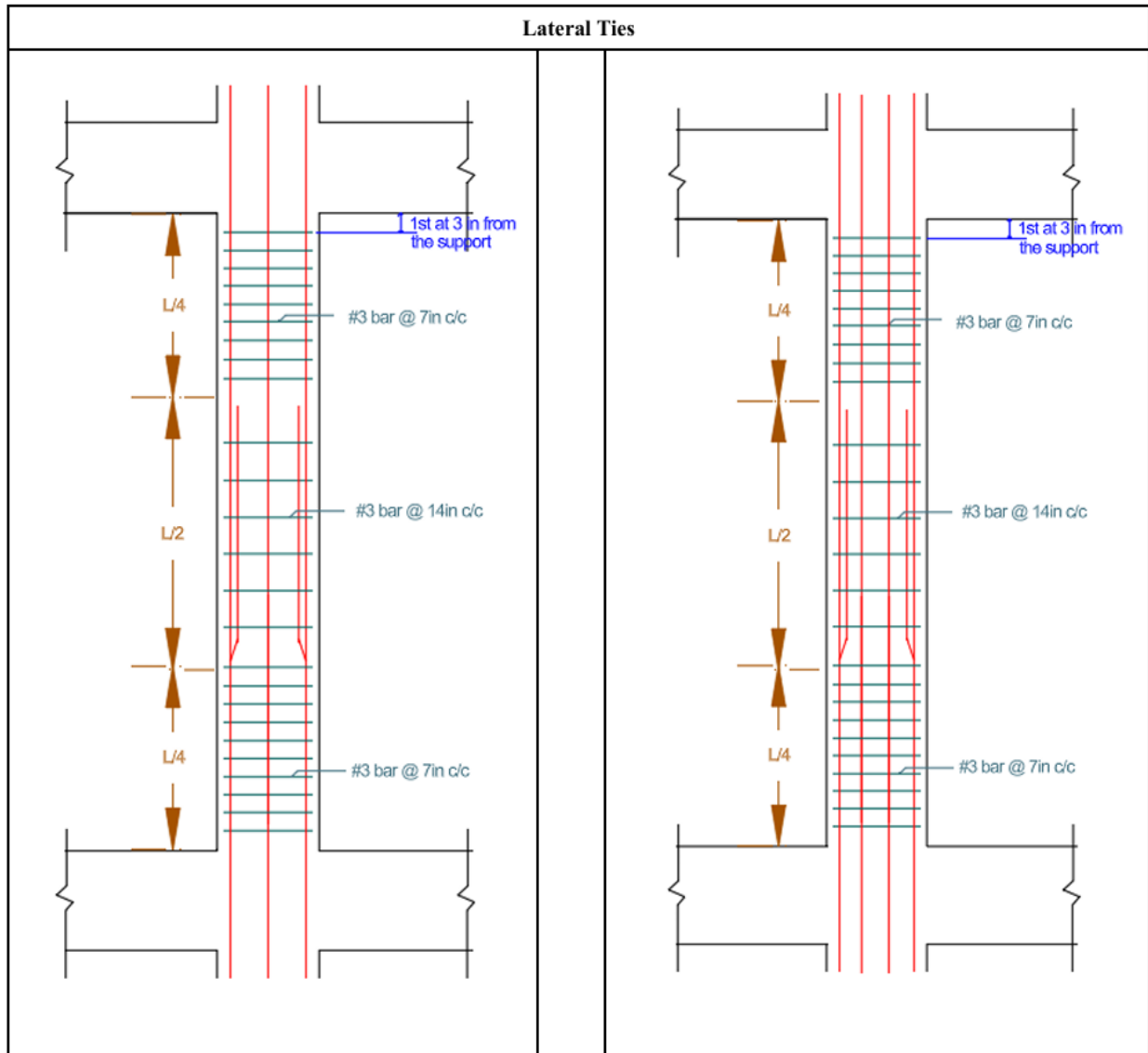


Fig: Longitudinal Section

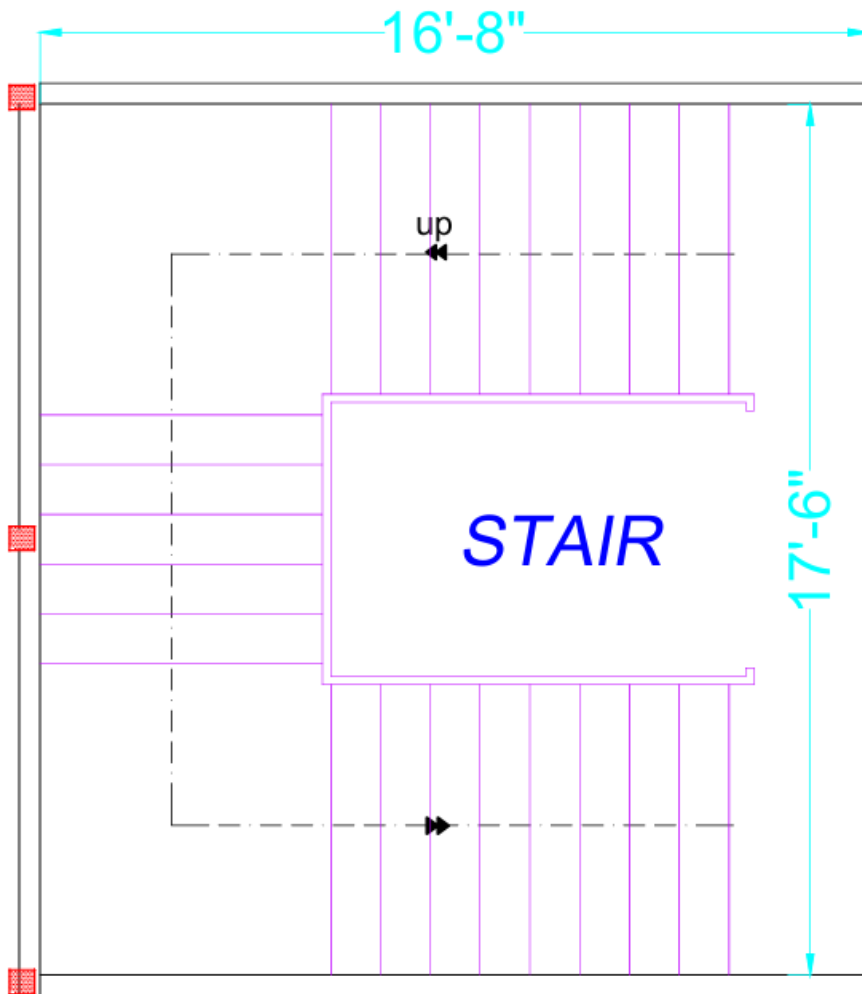
20. COLUMN SCHEDULING

Concrete Strength		Column Tie Details	4500 psi	4500 psi	4500 psi
Name of Column	Column Size		Basement and Ground Floor	1st to 4th Floor level	5th to Roof Floor level
C1	15x15		-	 8#7bar	 4#7 bars at corner + 4#6 bars at middle
C2	18x18		-	 12#6bar	 4#6 bars at corner + 8#5 bars at middle
C3	20x20		-	 12#7bar	 4#7 bars at corner + 8#6 bars at middle
GC1	17x17		 8#7bar	-	-
GC2	20x20		 12#7bar	-	-
GC3	22x22		 12#7bar	-	-

20.1. LATERAL TIES



21. STAIR DESIGN:



Given,

Floor Height= 10'

Rise Height = 6"

Total Stair Width = 17'6"

Landing Slab thickness = 8"

21.1. Load Calculation:

$$\text{Rise \& Step} = \frac{\frac{1}{2} \times \frac{8}{12} \times \frac{10}{12} \times 5.6 \times 6 \times 150}{1000} = 1.4 \text{ kip}$$

$$\text{Waist Slab} = \frac{\sqrt{\left(\frac{10 \times 6}{12}\right)^2 + 3.3^2} \times \frac{8}{12} \times 5.6 \times 150}{1000} = 3.35 \text{ kip}$$

$$\text{Dead Load, D.L} = \frac{\frac{8}{12} \times 150}{1000} + \frac{1.4 + 3.35}{5.6 \times \frac{10 \times 6}{12}} = 0.3 \text{ ksf}$$

$$\text{Total Load} = 1.4 \times DL + 1.7 LL = 1.4 \times (0.3 + 0.025) + 1.7 \times \frac{100.25}{100} = 2.16 \text{ ksf}$$

21.2. Moment Calculation:

$$\text{Max Positive Moment, } M^+ = \frac{wl^2}{14} = 43.03 \text{ k-ft}$$

$$\text{Max Negative Moment, } M^- = \frac{wl^2}{24} = 25.1 \text{ k-ft}$$

Effective Depth Check:

$$\begin{aligned}\rho_{max} &= 0.75 \times 0.85 \times \beta \times \frac{f'_c}{f_y} \times \frac{87}{87+f_y} \\ &= 0.75 \times 0.85 \times 0.85 \times \frac{3.5}{60} \times \frac{87}{87+60} \\ &= 0.018\end{aligned}$$

$$d_{req} = \sqrt{\frac{M_u}{\phi \rho f_y b (1 - 0.59 \rho \frac{f_y}{f'_c})}} = \sqrt{\frac{43.03 \times 12}{0.9 \times 60 \times 0.018 \times 12 (1 - 0.59 \times 0.018 \times \frac{60}{3.5})}} = 7.36 \text{ in}$$

$$\begin{aligned}d_{provided} &= t - 1 \\ &= 8 - 0.5 = 7.5 > d_{req}; \text{ So Design is OK.}\end{aligned}$$

21.3. Reinforcement Calculation :

$$\text{Minimum } A_s = 0.0018bt = 0.0018 \times 12 \times 8 = 0.17 \frac{\text{in}^2}{\text{ft}}$$

$A_s = \frac{M}{\phi f_y \left(d - \frac{a}{2}\right)}$	$a = \frac{A_s f_y}{0.85 f'_c b}$
$A_s + = 1.33 \text{ in}^2$	1.74
$A_s - = 0.73 \text{ in}^2$	0.05

Using 5#5 bars, $A_s = 5 \times 0.31 = 1.53 \text{ in}^2$:

Transverse Reinforcement must be provided to account for shrinkage – (using #3 bars)

$$A_s = 0.0018 \times 12 \times 8 = 0.1728 \text{ in}^2/\text{ft}$$

$$\text{Spacing} = \frac{12 \times 0.11}{0.1728} = 7.6''$$

For temp shrinkage bar

#3 bars @ 7.6" c/c

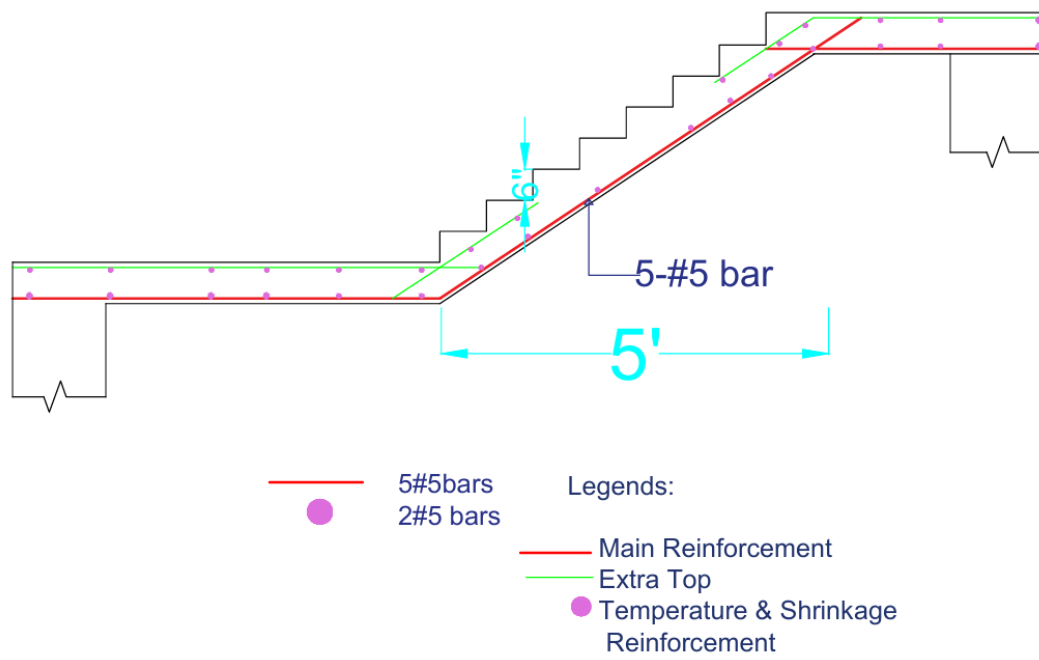


Fig: Stair Reinforcement Detailing

22. SHEAR WALL DESIGN

Shear wall thickness = $h=10''$

Shear wall length = $l_w= 20'$

From ETABS, $P_u= 838.667k$ [Ground floor]

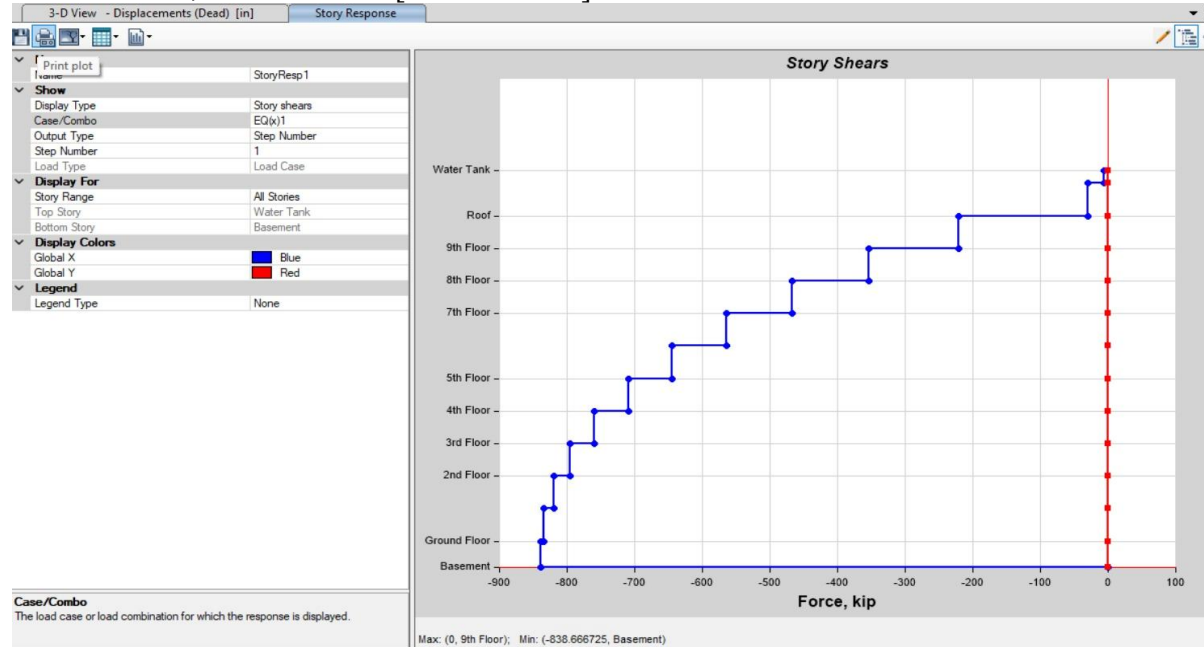


Fig: Story Response Plot from ETABS

TABLE: Story Response				
Story	Elevation	Location	X-Dir	Y-Dir
	ft		kip	kip
Water Tank	122	Top	-5.468	0
		Bottom	-5.468	0
Roof top	118	Top	-30.092	0
		Bottom	-30.092	0
Roof	108	Top	-221.25	0
		Bottom	-221.25	0
9th Floor	98	Top	-353.175	0
		Bottom	-353.175	0
8th Floor	88	Top	-467.197	0
		Bottom	-467.197	0
7th Floor	78	Top	-564.026	0
		Bottom	-564.026	0
6th Floor	68	Top	-644.427	0
		Bottom	-644.427	0
5th Floor	58	Top	-709.24	0
		Bottom	-709.24	0
4th Floor	48	Top	-759.393	0
		Bottom	-759.393	0
3rd Floor	38	Top	-795.937	0
		Bottom	-795.937	0
2nd Floor	28	Top	-820.098	0
		Bottom	-820.098	0
1st Floor	18	Top	-834.035	0
		Bottom	-834.035	0
Ground Floor	8	Top	-838.667	0
		Bottom	-838.667	0
Basement	0	Top	0	0
		Bottom	0	0

Effective Depth

$$d=0.8l_w=0.8(20') = 16'$$

Maximum Shear Strength Check

$$V_u = \Phi V_n \leq \Phi 10 \sqrt{f'_c} b d$$

$$V_u = 838.667k \leq (0.85 \times 10 \times \sqrt{3500} \times 0.8 \times 20 \times 12) / 1000 = 965.5k$$

As, $V_u \leq \Phi V_n$, **OK.**

Shear Strength Provided by Concrete,

$$V_c = 2 \sqrt{f'_c} b d = (2 \times \sqrt{3500} \times 10 \times 0.8 \times 20 \times 12) / 1000 = 227.18k$$

$$\text{As, } V_u = 838.667k > \phi V_c = 0.85 \times 227.18 = 193.1k$$

So, **shear reinforcement must be provided.**

Horizontal Shear Reinforcement

$$V_s = \frac{V_u}{\phi} - V_c = \frac{838.667}{0.85} - 227.18 = 759.48k \cong 760k$$

$$\frac{A_{vh}}{s_2} = \frac{V_s}{f_y \times d} = \frac{760}{60 \times 0.8 \times 20 \times 12} = 0.066 \text{ in}^2/\text{in}$$

Minimum Horizontal Reinforcement

$$\frac{A_{vh}}{s_2} \leq 0.0025 h = 0.0025 \times 10 = 0.025 \text{ in}^2/\text{in} < 0.066 \text{ in}^2/\text{in}$$

So, 0.066 in²/in controls.

$$\text{Spacing}(s_2) \text{ must not exceed } \leq \begin{cases} \frac{lw}{5} = \frac{20 \times 12}{5} = 48'' \\ 3h = 3 \times 10 = 30'' \\ 18'' \end{cases}$$

Using 2 layers of steel and #3 bars,

$$\frac{A_{vh}}{s_2} = 0.066$$

$$\frac{2 \times 0.11}{S_2} = 0.066$$

$$S_2 = 3.33'' \cong 3''$$

So, Using #3 bars @3'' c/c.

Vertical Shear Reinforcement

$$\frac{A_{vv}}{s_1} = \left[0.0025 + 0.5 \left(2.5 - \frac{hw}{lw} \right) \left(\frac{A_{vh}}{s_2 \times h} - 0.0025 \right) \right] \times h$$

$$\frac{A_{vv}}{s_1} = \left[0.0025 + 0.5 \left(2.5 - \frac{122}{20} \right) \left(\frac{A_{vh}}{s_2 \times h} - 0.0025 \right) \right] \times 10$$

$$\frac{A_{vv}}{s_1} = 0.025 \text{ in}^2/\text{in}$$

Using 2 layers of steel and #4 bars,

$$\frac{A_{vv}}{s_2} = 0.025$$

$$\frac{2 \times 0.20}{S_2} = 0.025$$

$$S_2 = 15''$$

So, Using #4 bars @15'' c/c.

Moment capacity with vertical steel provided shear,

$$M_u = \phi [0.5 A_{st} f_y l_w (1 - z/l_w)]$$

$$\text{Steel area, } A_{st} = 0.4 \times \frac{20 \times 12}{15} = 6.5 \text{ in}^2$$

$$\frac{z}{l_w} = \frac{1}{2 + 0.85 \beta l_w \frac{h \times f'_c}{A_{st} \times f_y}} = \frac{1}{2 + 0.85 \times 0.85 \times 20 \times 12 \times \frac{10 \times 3.5}{6.5 \times 60}} = 0.057$$

$$M_u = \phi [0.5 \times 6 \times 60 \times 20 (1 - 0.053)] = 0.9 \times [0.5 \times 6 \times 60 \times 20 (1 - 0.057)] = \mathbf{3310 \text{ k-ft}}$$

22.1. Reinforcement Design

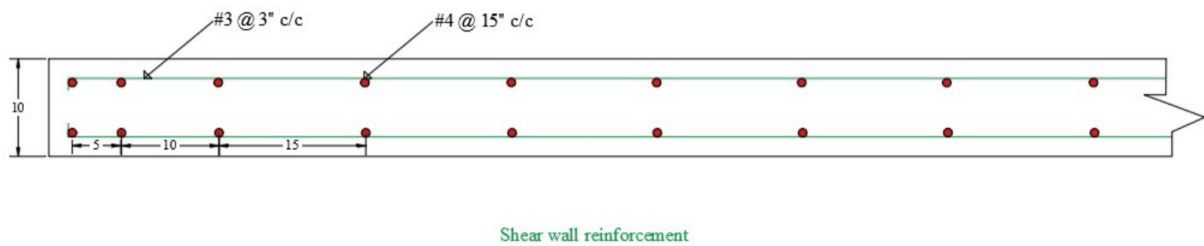


Fig: Section through wall

23. MIX DESIGN

23.1. Mix Design for Column and Beam, $f'_c = 4500 \text{ psi}$

Compressive strength, $f'_c = 4500 \text{ psi}$

Specific gravity of cement, $G_C = 2.95 \text{ (PCC)}$

Specific gravity of coarse aggregate, $G_{CA} = 2.65$

Specific gravity of fine aggregate, $G_{FA} = 2.57$

Specific gravity of water, $G_W = 1$

$s/a = 0.42$

using $\frac{3}{4}$ in 175 mm slump

Air content, $V_V = 1.14\%$

Required compressive strength, $f'_{cr} = 4500 + 1200 = 5700 \text{ psi}$

If fewer than 15 test have been made, the average strength must exceed f'_c by at least 1200 psi for f'_c between 3000 to 5000 psi; (as per [ACI Code](#)).

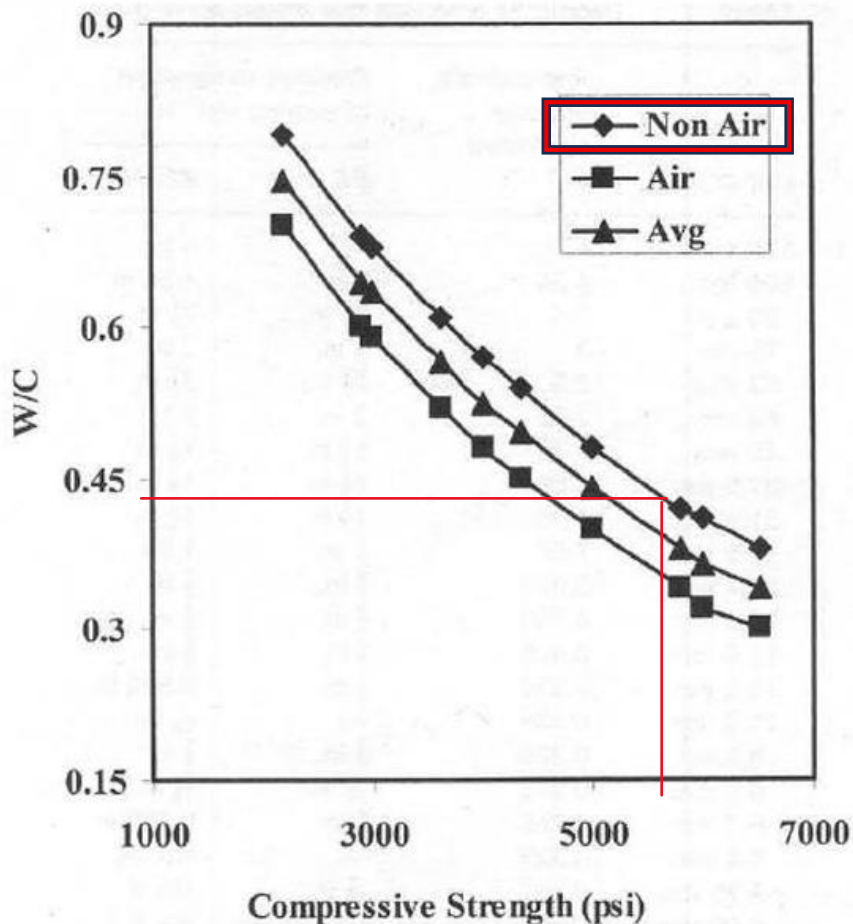


Figure : W/C vs Compressive Strength Graph

W/C ratio = 0.42

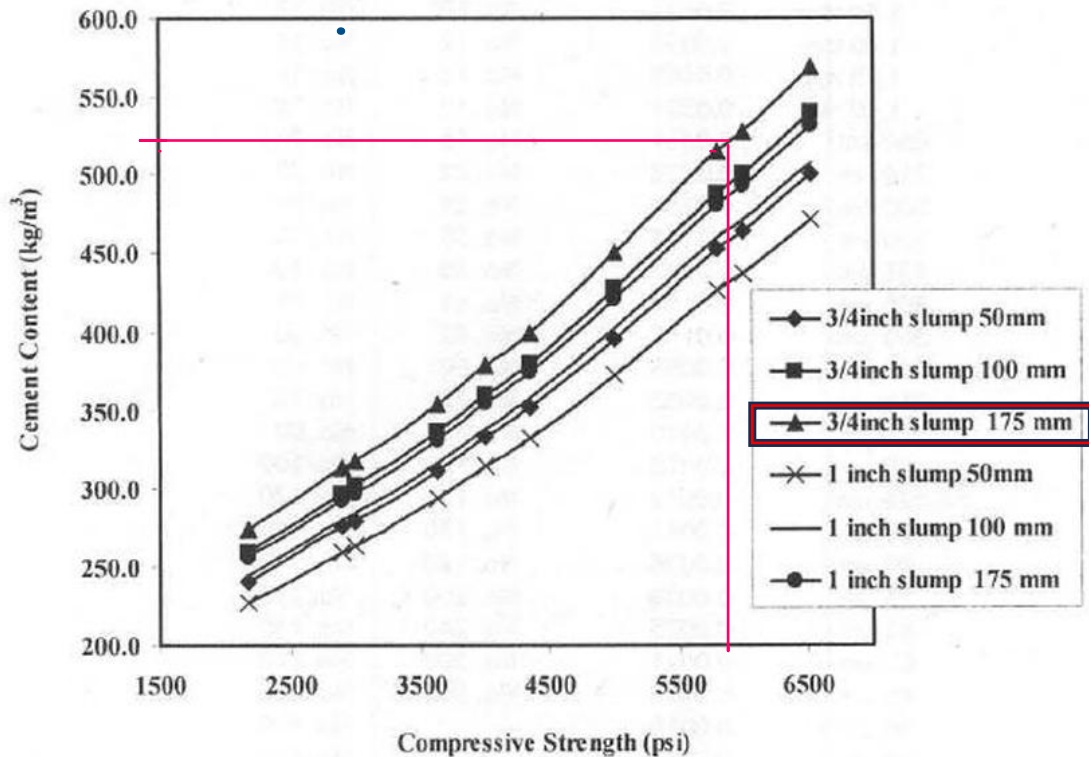


Figure: Cement Content vs Compressive Strength Graph

Cement content = 530 kg/m^3 (1 inch slump 100 mm)

Water content = $0.42 \times 530 = 222.6 \text{ kg/m}^3$

23.1.1. Calculating W_{CA} & W_{FA} :

We know,

$$\begin{aligned} \Rightarrow \frac{W_C}{G_C \times \gamma_W} + \frac{W_{CA}}{G_{CA} \times \gamma_W} + \frac{W_{FA}}{G_{FA} \times \gamma_W} + \frac{W_W}{G_W \times \gamma_W} + 0.0114 &= 1 \\ \Rightarrow \frac{530}{2.95 \times 1000} + \frac{W_{CA}}{2.65 \times 1000} + \frac{W_{FA}}{2.57 \times 1000} + \frac{222.6}{1 \times 1000} + 0.0114 &= 1 \\ \Rightarrow \frac{W_{CA}}{2650} + \frac{W_{FA}}{2570} &= 0.5863 \dots \dots \dots (i) \end{aligned}$$

Let, $\frac{V_s}{V_a} = 0.42$

$$\begin{aligned} \Rightarrow \frac{\frac{W_{FA}}{G_{FA} \times \gamma_W}}{\frac{W_{CA}}{G_{CA} \times \gamma_W} + \frac{W_{FA}}{G_{FA} \times \gamma_W}} &= 0.42 \\ \Rightarrow \frac{\frac{W_{FA}}{2570}}{\frac{W_{CA}}{2650} + \frac{W_{FA}}{2570}} &= 0.42 \\ \Rightarrow \frac{W_{CA}}{2650} \times 0.42 - \frac{W_{FA}}{2570} \times 0.58 &= 0 \dots \dots \dots (ii) \end{aligned}$$

Solving equation (i) and (ii), we find,

$$\begin{aligned} W_{CA} &= 901.13 \text{ kg/m}^3 \\ W_{FA} &= 632.88 \text{ kg/m}^3 \end{aligned}$$

23.1.2. Volume with Air Void:

We know,

$$W_C = 530 \text{ kg/m}^3$$

$$W_W = 222.6 \text{ kg/m}^3$$

$$V'_C = \frac{W_C}{\text{Unit Weight}} = \frac{530 \text{ kg}}{1294 \frac{\text{kg}}{\text{m}^3}} = 0.4096 \text{ m}^3$$

Similarly,

$$V'_W = 0.2226 \text{ m}^3$$

$$V'_{CA} = 0.6932 \text{ m}^3$$

$$V'_{FA} = 0.4887 \text{ m}^3$$

$$\therefore V'_C : V'_{FA} : V'_{CA} = 0.4096 : 0.4887 : 0.6932 = \mathbf{1 : 1.19 : 1.69}$$

23.1.3. Absolute Volume:

$$V_C = \frac{530}{2.95 \times 1000} = 0.18 \text{ m}^3$$

$$V_{CA} = \frac{901.13}{2.65 \times 1000} = 0.34 \text{ m}^3$$

$$V_{FA} = \frac{632.88}{2.57 \times 1000} = 0.2463 \text{ m}^3$$

$$V_W = \frac{222.6}{1 \times 1000} = 0.2226 \text{ m}^3$$

$$V_V = \frac{1.14}{100} = 0.0114 \text{ m}^3$$

$$\therefore \text{Total Volume} = V_C + V_{CA} + V_{FA} + V_W + V_V = 1 \text{ m}^3$$

$\therefore$ The mix design is OK.

23.2. Mix Proportion Table for Beam & Column

Material	Unit weight (kg/m ³)	Weight (kg/m ³)	Volume (m ³)
Cement	1294	530	0.4096
CA	1300	901.128	0.6932
FA	1295	632.879	0.4887
Water	1000	222.6	0.2226
Unit Weight of Concrete (kg/m ³)		2286.607	

$\therefore$ Compaction Factor = 1.81

23.3. Mix Design for Slab, $f'_c = 3500 \text{ psi}$

Compressive strength, $f'_c = 3500 \text{ psi}$

Specific gravity of cement, $G_C = 2.95 \text{ (PCC)}$

Specific gravity of coarse aggregate, $G_{CA} = 2.65$

Specific gravity of fine aggregate, $G_{FA} = 2.57$

Specific gravity of water, $G_W = 1$

$s/a = 0.42$

using $\frac{3}{4}$ in 175 mm slump

Air void, $V_v = 1.14\%$

Required compressive strength, $f'_{cr} = 3500 + 1200 = 4700 \text{ psi}$

If fewer than 15 test have been made, the average strength must exceed f'_c by at least 1200 psi for f'_c between 3000 to 5000 psi; (as per [ACI Code](#)).

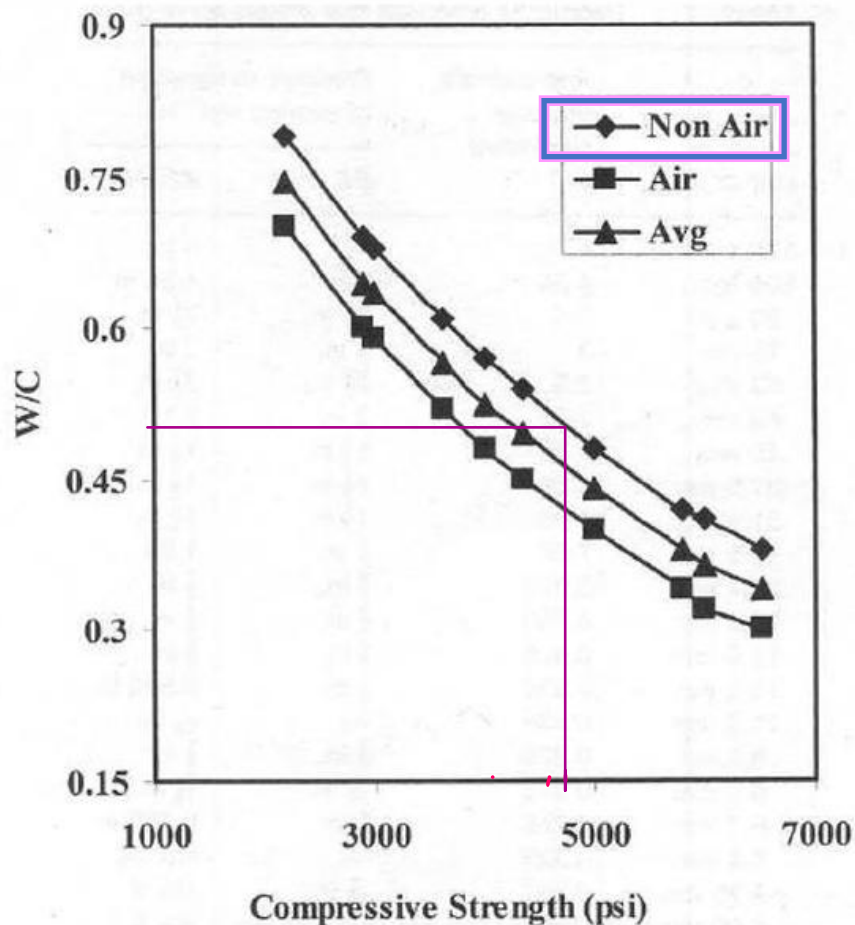


Figure : W/C vs Compressive Strength Graph

W/C ratio = 0.50

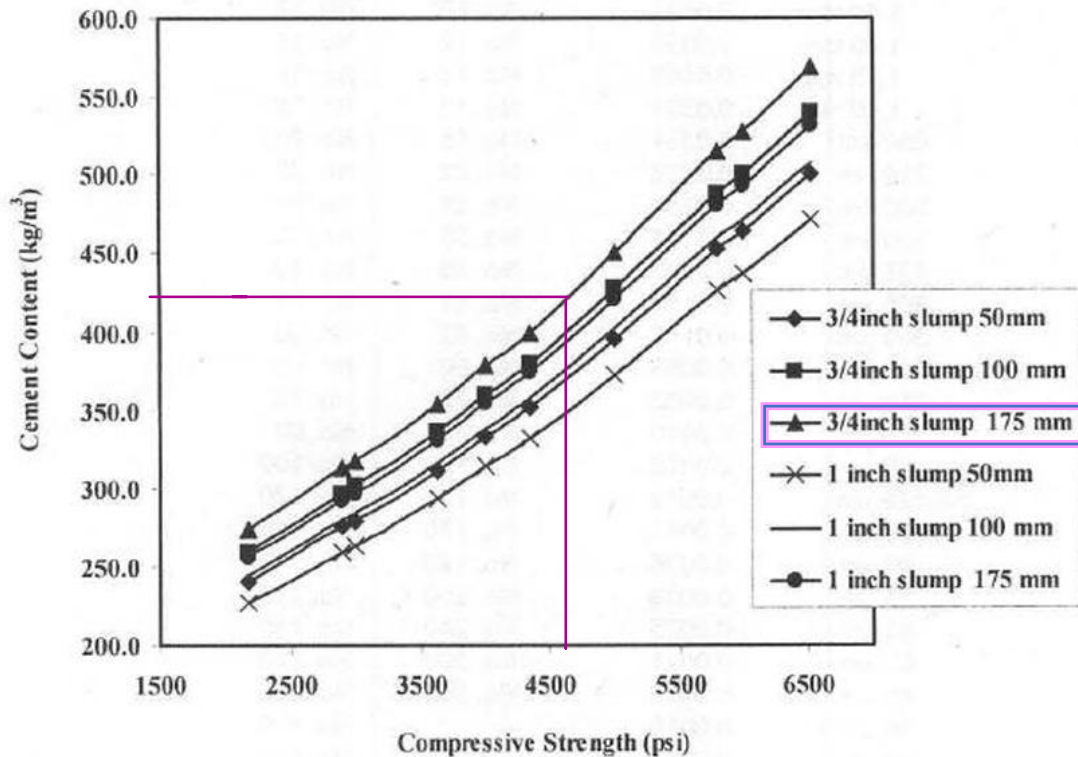


Figure : Cement Content vs Compressive Strength Graph

Cement content = 430 kg/m^3 (1 inch slump 100 mm)

Water content = $0.50 \times 430 = 215 \text{ kg/m}^3$

23.3.1. Calculating W_{CA} & W_{FA} :

We know,

$$\begin{aligned} &\Rightarrow \frac{W_C}{G_C \times \gamma_W} + \frac{W_{CA}}{G_{CA} \times \gamma_W} + \frac{W_{FA}}{G_{FA} \times \gamma_W} + \frac{W_W}{G_W \times \gamma_W} + 0.0114 = 1 \\ &\Rightarrow \frac{430}{2.95 \times 1000} + \frac{W_{CA}}{2.65 \times 1000} + \frac{W_{FA}}{2.57 \times 1000} + \frac{215}{1 \times 1000} + 0.0114 = 1 \\ &\Rightarrow \frac{W_{CA}}{2650} + \frac{W_{FA}}{2570} = 0.6278 \dots \dots \dots (i) \end{aligned}$$

Let, $\frac{V_s}{V_a} = 0.42$

$$\begin{aligned} &\Rightarrow \frac{\frac{W_{FA}}{G_{FA} \times \gamma_W}}{\frac{W_{CA}}{G_{CA} \times \gamma_W} + \frac{W_{FA}}{G_{FA} \times \gamma_W}} = 0.42 \\ &\Rightarrow \frac{\frac{W_{FA}}{2570}}{\frac{W_{CA}}{2650} + \frac{W_{FA}}{2570}} = 0.42 \\ &\Rightarrow \frac{W_{CA}}{2650} \times 0.42 - \frac{W_{FA}}{2570} \times 0.58 = 0 \dots \dots \dots (ii) \end{aligned}$$

Solving equation (i) and (ii), we find,

$$\begin{aligned} W_{FA} &= 677.67 \text{ kg/m}^3 \\ W_{CA} &= 964.905 \text{ kg/m}^3 \end{aligned}$$

23.3.2. Volume with Air Void:

We know,

$$V'_C = \frac{W_C}{\text{Unit Weight}} = \frac{430 \text{ kg}}{1294 \frac{\text{kg}}{\text{m}^3}} = 0.3323 \text{ m}^3$$

Similarly,

$$\begin{aligned} V'_{CA} &= 0.7422 \text{ m}^3 \\ V'_{FA} &= 0.5233 \text{ m}^3 \\ V'_W &= 0.215 \text{ m}^3 \\ \therefore V'_C : V'_{FA} : V'_{CA} &= 0.3323 : 0.5233 : 0.7422 = 1 : 1.57 : 2.23 \end{aligned}$$

Absolute Volume:

$$\begin{aligned} V_C &= \frac{430}{2.95 \times 1000} = 0.1458 \text{ m}^3 \\ V_{CA} &= \frac{964.905}{2.65 \times 1000} = 0.3641 \text{ m}^3 \\ V_{FA} &= \frac{677.67}{2.57 \times 1000} = 0.2637 \text{ m}^3 \\ V_W &= \frac{215}{1 \times 1000} = 0.215 \text{ m}^3 \\ V_V &= \frac{1.14}{100} = 0.0114 \text{ m}^3 \end{aligned}$$

$$\therefore \text{Total Volume} = V_C + V_{CA} + V_{FA} + V_W + V_V = 1 \text{ m}^3$$

$\therefore$ The mix design is OK.

23.4. Mix Proportion Table for Slab

Material	Unit weight (kg/m^3)	Weight (kg/m^3)	Volume (m^3)
Cement	1294	430	0.3323
CA	1300	964.905	0.7422
FA	1295	677.671	0.5233
Water	1000	215	0.215
Unit Weight of Concrete (kg/m^3)		2287.58	

$\therefore$ Compaction Factor = 1.813

23.4.1. Discussion:

The design compressive strength for column and beam was 4500 psi and for slab it was 3500 psi. For slab design, 3500psi is sufficient, economical as it mainly carries uniformly distributed loads. Beams and columns experience higher compressive stresses and bending forces. So using 4500psi is justified as they are primary load carrying members.

We used PCC (Portland Composite Cement) concrete for the mix design of slab, beam, and column members. PCC concrete is considered environmentally friendly because it incorporates supplementary cementitious materials, which reduce clinker content, lower CO₂ emissions, and conserve natural resources while maintaining adequate strength and durability for structural applications.

24. FINANCIAL ANALYSIS AND CONSTRUCTION SCHEDULE

ESTIMATIONS:

COST ESTIMATION		
Construction cost	3,500	tk/sft
Area per floor	6,804	sft
Stories	10	
construction area	68,040	sft
total construction cost	238,140,000	tk
Time of construction	2	years
Construction cost per year	123,709,091	tk/year

Benefit		
Rent	40	tk/sft
Total rentable area	6804	sft
Stories	10	
Months	12	
Rent / Revenue accumulated per month	BDT 2,721,600	
Revenue per year	BDT 32,659,200	
Increment	13	%/yr

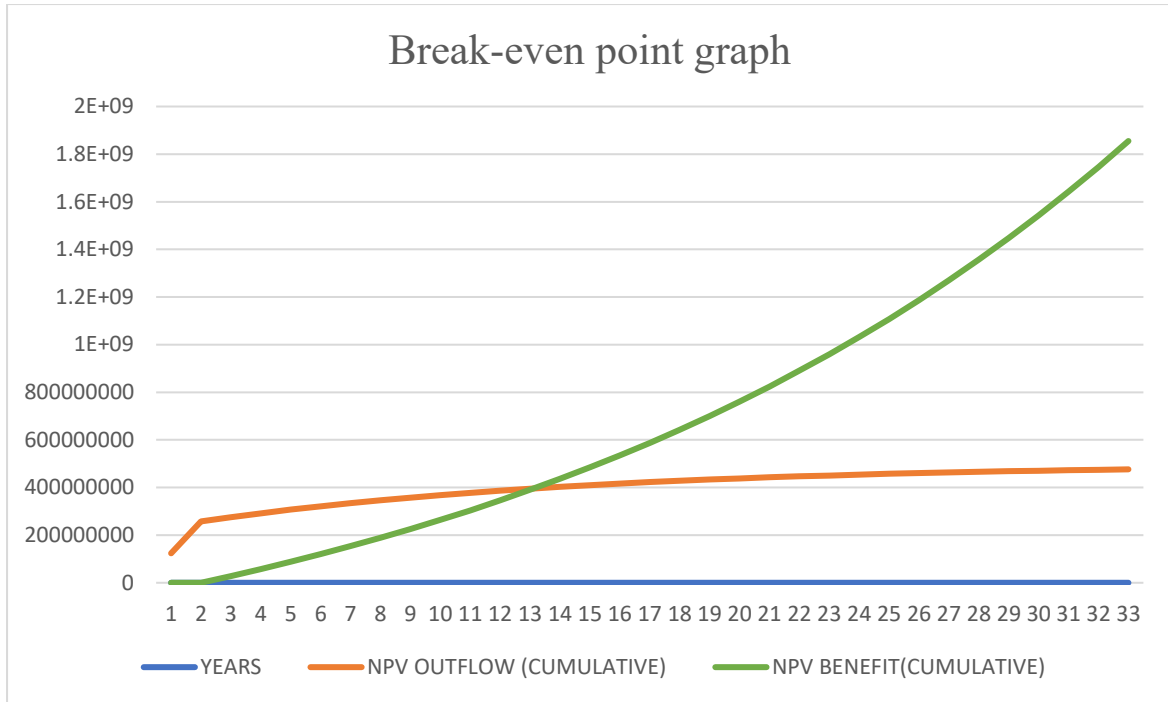
24.1. COST BENEFIT ANALYSIS:

Construction cost	YEARS	LOAN TAKEN	LOAN REPAY(INTEREST 8%)	OUTFLOW	BENEFIT (INCEMENT 13%)	NET FLOW
238140000	0	123709091	0	123709091	0	-123709091
	1	123709091	20816615.53	144525706.5	0	-144525706.5
	2		20816615.53	20816615.53	32659200	11842584.47
	3		20816615.53	20816615.53	36904896	16088280.47
	4		20816615.53	20816615.53	41702532.48	20885916.95
	5		20816615.53	20816615.53	47123861.7	26307246.17
	6		20816615.53	20816615.53	53249963.72	32433348.19
	7		20816615.53	20816615.53	60172459.01	39355843.48
	8		20816615.53	20816615.53	67994878.68	47178263.15
	9		20816615.53	20816615.53	76834212.91	56017597.38
	10		20816615.53	20816615.53	86822660.58	66006045.06
	11		20816615.53	20816615.53	98109606.46	77292990.93
	12		20816615.53	20816615.53	110863855.3	90047239.77
	13		20816615.53	20816615.53	125276156.5	104459541
	14		20816615.53	20816615.53	141562056.8	120745441.3
	15		20816615.53	20816615.53	159965124.2	139148508.7
	16		20816615.53	20816615.53	180760590.4	159943974.8
	17		20816615.53	20816615.53	204259467.1	183442851.6
	18		20816615.53	20816615.53	230813197.8	209996582.3
	19		20816615.53	20816615.53	260818913.6	240002298
	20		20816615.53	20816615.53	294725372.3	273908756.8
	21		20816615.53	20816615.53	333039670.7	312223055.2
	22		20816615.53	20816615.53	376334827.9	355518212.4
	23		20816615.53	20816615.53	425258355.6	404441740
	24		20816615.53	20816615.53	480541941.8	459725326.2
	25		20816615.53	20816615.53	543012394.2	522195778.7
	26		20816615.53	20816615.53	613604005.5	592787389.9
	27		20816615.53	20816615.53	693372526.2	672555910.6
	28		20816615.53	20816615.53	783510954.6	762694339
	29		20816615.53	20816615.53	885367378.7	864550763.1
	30		20816615.53	20816615.53	1000465138	979648522.4
	31		20816615.53	20816615.53	1130525606	1109708990
	32		20816615.53	20816615.53	1277493935	1256677319

NPV OUTFLOW	NPV BENEFIT	NPV OUTFLOW (CUMULATIVE)	NPV BENEFIT (CUMULATIVE)	NPV NET FLOW		IRR	BCR
123709091	0	123709091	0	-123709091			0
133820098.6	0	257529189.6	0	-257529189.6			0
17846892.6	28000000	275376082.2	28000000	-247376082.2			0.1017
16524900.56	29296296.3	291900982.8	57296296.3	-234604686.5			0.1963
15300833.85	30652606.31	307201816.6	87948902.61	-219252914			0.2863
14167438.75	32071708.45	321369255.4	120020611.1	-201348644.3			0.3735
13117998.84	33556509.77	334487254.2	153577120.8	-180910133.4			0.4591
12146295.22	35110051.89	346633549.5	188687172.7	-157946376.7			0.5443
11246569.65	36735517.26	357880119.1	225422690	-132457429.1			0.6299
10413490.42	38436235.65	368293609.5	263858925.6	-104434683.9			0.7164
9642120.757	40215691	377935730.3	304074616.6	-73861113.66			0.8046
8927889.59	42077528.55	386863619.9	346152145.2	-40711474.7			0.8948
8266564.435	44025562.28	395130184.3	390177707.4	-4952476.861			0.9875
7654226.329	46063782.75	402784410.6	436241490.2	33457079.56	Break-even point		1.0831
7087246.601	48196365.29	409871657.2	484437855.5	74566198.25			1.1819
6562265.371	50427678.49	416433922.6	534865534	118431611.4			1.2844
6076171.64	52762293.24	422510094.3	587627827.2	165117733			1.3908
5626084.852	55204992	428136179.1	642832819.2	214696640.1			1.5015
5209337.826	57760778.67	433345516.9	700593597.9	267248081			1.6167
4823460.95	60434888.79	438168977.9	761028486.7	322859508.8			1.7368
4466167.546	63232800.31	442635145.4	824261287	381626141.6			1.8622
4135340.321	66160244.77	446770485.7	890421531.8	443651046			1.993
3829018.815	69223219.06	450599504.6	959644750.8	509045246.3			2.1297
3545387.792	72427997.72	454144892.4	1032072749	577927856.2			2.2726
3282766.474	75781145.77	457427658.8	1107853894	650426235.5			2.4219
3039598.587	79289532.14	460467257.4	1187143426	726676169			2.5781
2814443.136	82960343.82	463281700.6	1270103770	806822069.7			2.7415
2605965.867	86801100.48	465887666.4	1356904871	891017204.3			2.9125
2412931.358	90819669.94	468300597.8	1447724541	979423942.9			3.0914
2234195.702	95024284.29	470534793.5	1542748825	1072214032			3.2787
2068699.724	99423556.71	472603493.2	1642172382	1169568888			3.4747
1915462.707	104026499.2	474518955.9	1746198881	1271679925			3.6799
1773576.581	108842540.8	476292532.5	1855041422	1378748889			3.8948

24.2. BREAK EVEN POINT GRAPH:

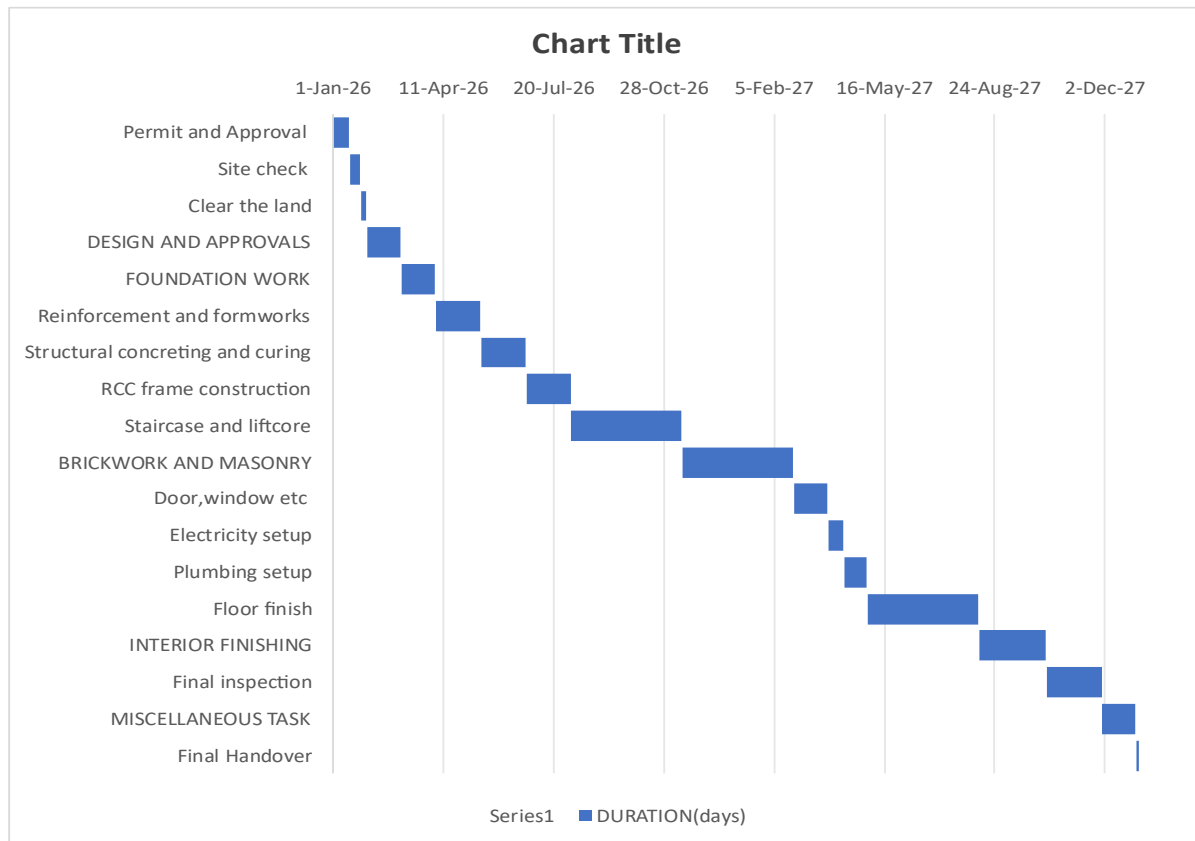
YEARS	NPV OUTFLOW (CUMULATIVE)	NPV BENEFIT(CUMULATIVE)
0	123709091	0
1	257529189.6	0
2	275376082.2	28000000
3	291900982.8	57296296.3
4	307201816.6	87948902.61
5	321369255.4	120020611.1
6	334487254.2	153577120.8
7	346633549.5	188687172.7
8	357880119.1	225422690
9	368293609.5	263858925.6
10	377935730.3	304074616.6
11	386863619.9	346152145.2
12	395130184.3	390177707.4
13	402784410.6	436241490.2
14	409871657.2	484437855.5
15	416433922.6	534865534
16	422510094.3	587627827.2
17	428136179.1	642832819.2
18	433345516.9	700593597.9
19	438168977.9	761028486.7
20	442635145.4	824261287
21	446770485.7	890421531.8
22	450599504.6	959644750.8
23	454144892.4	1032072749
24	457427658.8	1107853894
25	460467257.4	1187143426
26	463281700.6	1270103770
27	465887666.4	1356904871
28	468300597.8	1447724541
29	470534793.5	1542748825
30	472603493.2	1642172382
31	474518955.9	1746198881
32	476292532.5	1855041422



24.3. SCHEDULING:

Column1	Column2	Column3	Column4
TASK	START DATE	END DATE	DURATION(days)
SITE PREPARATION	1-Jan-26	31-Jan-26	30
DESIGN AND APPROVALS	1-Feb-26	2-May-26	90
FOUNDATION WORK	3-May-26	1-Aug-26	90
STRUCTURAL CONSTRUCTION	2-Aug-26	28-Jun-27	330
BRICKWORK AND MASONRY	29-Jun-27	27-Sep-27	90
INTERIOR FINISHING	28-Sep-27	28-Oct-27	30
UTILITIES AND INSTALLATIONS	29-Oct-27	28-Nov-27	30
MISCELLANEOUS TASK	29-Nov-27	29-Dec-27	30

24.4. GANTT CHART



DISCUSSIONS:

The total construction cost of the 10-story building is estimated at **BDT 238.14 million**, with construction completed over a **2-year period**, indicating a significant initial capital investment.

The building generates revenue through rental income at **BDT 40 per sft**, producing an initial annual revenue of **BDT 32.66 million**, which increases at a rate of **13% per year**, improving long-term cash inflow.

The **cost–benefit analysis** shows that cumulative discounted benefits gradually catch up with cumulative costs, and the **break-even point** is achieved after several operational years, confirming the project’s long-term feasibility.

Break-even point

The discounted break-even occurs between year 12 and year 13. The project will recover its investment at about year 12.

The **Benefit–Cost Ratio (BCR)** is greater than **1.0**, indicating that the present value of benefits exceeds the present value of costs, and therefore the project is economically justified.

The **Internal Rate of Return (IRR)** (20%) is higher than the discount rate(8%), suggesting that the project yields a satisfactory return on investment and is financially attractive compared to alternative investments.

A well-planned construction **schedule** ensures completion within the expected time, reducing delays and cost overruns, which impacts both IRR and BCR.

Overall, considering the positive BCR, acceptable IRR, and steady growth in rental income, the project is financially viable and suitable for long-term investment

25. LIFE CYCLE ANALYSIS



Figure 4 Plan View of the Structure

25.1. Beam Properties

Beam Element Details				
Element	Unique Name	Story	Section Type	Critical Load Combination
B143	1185	1	Interior Beam	1.2D + 1.6L

Section Properties			
Length (ft)	Width, b (in)	Depth, h (in)	Distance to Rebar Center, d_c (in)
10.5	10	15	1.5

Material Properties	
f'_c (psi)	f_y (psi)
4500	60000

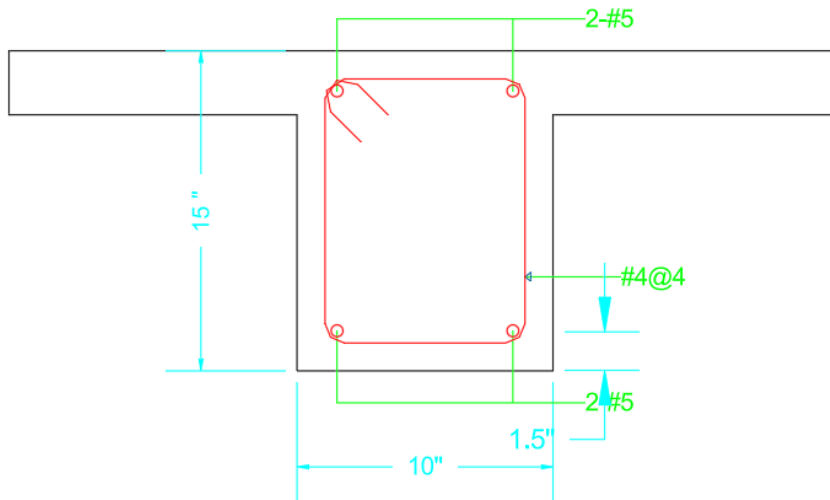


Fig: Cross Sectional Layout

25.2. Service Life Calculation

25.2.1. Time to Initiate Corrosion due to Carbonation of Concrete (T_0)

carbonation depth (mm), $CD = a\sqrt{t}$

Where, a is carbonation coefficient

t is exposure period (year)

Cover to longitudinal rebar group centroid = 1.5 inches

Provided bars = #5 bars (diameter 16 mm)

Carbonation depth, $CD = (1.5 \times 25.4 - \frac{16}{2}) = 30.1 \text{ mm}$

Carbonation coefficient, $a = 6.25 \text{ mm}/\sqrt{\text{year}}$ (assumed)

$$\therefore T_0 = \left(\frac{CD}{a}\right)^2 = \left(\frac{30.1}{6.25}\right)^2 = 23.19 \text{ years}$$

25.2.2. Time to Initiate Crack along the Main Reinforcement (T_1)

Corrosion Depth (mm), $D = 0.0116 \times I \times t \times \alpha$;

Where, 't' is the time in year

'I' is the corrosion current density in $\mu\text{A}/\text{cm}^2$

And ' α ' is the pitting factor

Here,

Depth of Corrosion, $D = 0.2 \text{ mm}$

Corrosion rate, $I = 1 \mu\text{A}/\text{cm}^2$

Pitting factor, $\alpha = 3.2$

$$\therefore D = 0.0116 \times I \times t \times \alpha$$

or,

$$\begin{aligned}
 t &= \frac{D}{\frac{0.0116 \times I \times \alpha}{0.2}} \\
 &= \frac{0.0116 \times 1 \times 3.2}{0.2} \\
 &= 5.39 \text{ years} \\
 \therefore T_1 &= 23.19 + 5.39 = 28.58 \text{ years}
 \end{aligned}$$

25.2.3. Time to spalling of concrete cover (T_2)

Depth of corrosion (mm), $D = 0.0116 \times I \times t \times \alpha$;

Where, 't' is the time in year.

'I' is the corrosion current density in $\mu A/cm^2$

And 'α' is the pitting factor.

Here,

Depth of Corrosion, $D = 0.5 \text{ mm}$

Corrosion rate, $I = 4 \mu A/cm^2$

$\alpha = 3.2$

$$\begin{aligned}
 t &= \frac{D}{\frac{0.0116 \times I \times \alpha}{0.5}} \\
 &= \frac{0.0116 \times 4 \times 3.2}{0.5} \\
 &= 3.37 \text{ years}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Time to spalling of cover concrete after construction of the building, } T_2 \\
 &= (23.19 + 5.39 + 3.37) \text{ years} \\
 &= 31.95 \text{ years} \\
 &\approx 32 \text{ years}
 \end{aligned}$$

Table : Service Life Due to Corrosion

	T_0	T_1	T_2
Service Life	23.19 years	5.39 years	3.37 years
Corrosion Depth	-	0.2 mm	0.5 mm
Corrosion Rate	-	$1 \mu A/cm^2$	$4 \mu A/cm^2$
Pitting Factor, α	-	3.2	
Carbonation Coefficient, a	$6.25 \text{ mm}/\sqrt{\text{year}}$	-	

25.3. Design Moment Capacity of the Section

Corrosion depth = $0.2 + 0.5 = 0.7 \text{ mm}$

Diameter (before corrosion) = $15.875 \text{ mm} \approx 16 \text{ mm}$ #5 Bars

Area of steel (before corrosion) = 0.31 in^2

Diameter of steel (after corrosion) = $16 - 2 \times 0.7 = 14.6 \text{ mm}$

Area of steel (after corrosion) = $\frac{\pi}{4} \times 14.6^2 = 167.42 \text{ mm}^2 = 0.2595 \text{ in}^2$

Number of steels = 2 (at bottom)

25.3.1. Moment Capacity (Before deterioration)

Compressive strength of concrete, $f'_c = 4500 \text{ psi}$

Yield strength of steel, $f_y = 60000 \text{ psi}$

Reduction Factor, $\phi = 0.9$

Depth, $d = 15 \text{ in}$

Distance to rebar center, $d_c = 1.5 \text{ in}$

Effective depth, $d_{eff} = 13.5 \text{ in}$

Width, $b = 10 \text{ in}$

$\beta_1 = 0.825$

Nominal Steel Ratio, $\rho_b = 0.85 * \beta_1 * \frac{87000}{87000 + f_y} * \frac{f'_c}{f_y} = .03113$

Maximum Steel Ratio, $= 0.75 * \rho_b = .0233452$

Area of steel, $A_s = 2 \times 0.31 \text{ in}^2 = 0.62 \text{ in}^2$

Provided Steel Ratio,

$$\begin{aligned} \rho &= \frac{0.62}{13.5 \times 10} \\ &= 0.00459259 < \rho_{\max}, \text{ tension yielding of steel governs} \\ A_s f_y &= 0.85 \times f'_c \times a \times b \end{aligned}$$

$$a = \frac{A_s f_y}{0.85 \times f'_c \times b} = \frac{0.62 \times 60000}{0.85 \times 4500 \times 10} = 0.9725 \text{ in.}$$

Design Moment Capacity,

$$\begin{aligned} M_{design} &= \phi \times A_s \times f_y \times \left(d_{eff} - \frac{a}{2} \right) \\ &= 0.9 \times 0.62 \times 60000 \times \left(13.5 - \frac{.9725}{2} \right) \\ &= 435700.35 \text{ lb-in} \\ &= 36.31 \text{ k-ft} \end{aligned}$$

$\therefore$ Moment capacity before deterioration, $M_{design1} = 36.31 \text{ k-ft}$

25.3.2. Moment Capacity (After Deterioration)

$$\text{Nominal Steel Ratio, } \rho_b = 0.85 \times \beta_1 \times \frac{87000}{87000 + f_y} \times \frac{f'_c}{f_y} = .03113$$

$$\text{Maximum Steel Ratio, } \rho_{\max} = 0.75 * \rho_b = .02334$$

$$A_s = 2 \times 0.2595 \text{ in}^2 = 0.519 \text{ in}^2$$

Provided Steel Ratio,

$$\rho = \frac{0.519}{13.5 \times 10} = .00384 < \rho_{\max}, \text{ tension yielding of steel governs}$$

$$A_s f_y = 0.85 \times f'_c \times a \times b$$

$$a = \frac{A_s f_y}{0.85 \times f'_c \times b} = \frac{0.519 \times 60000}{0.85 \times 4500 \times 10} = 0.814 \text{ in.}$$

Design Moment Capacity,

$$\begin{aligned} M_{design} &= \phi \times A_s \times f_y \times \left(d_{eff} - \frac{a}{2} \right) \\ &= 0.9 \times 0.519 \times 60000 \times \left(13.5 - \frac{0.814}{2} \right) \end{aligned}$$

$$= 366944.418 \text{ lb-in}$$

$$= 30.58 \text{ k-ft}$$

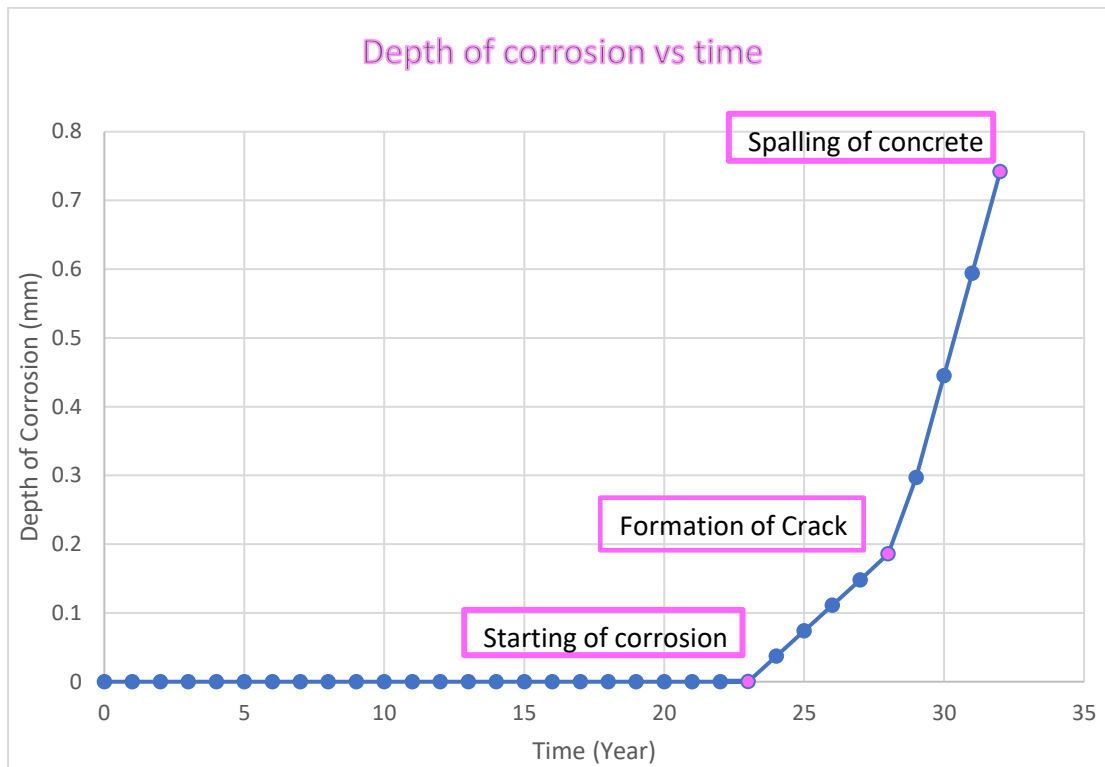
$\therefore$ Moment capacity before deterioration, $M_{design2} = 30.58 \text{ k-ft}$

25.3.3. Percent Moment Capacity Reduction

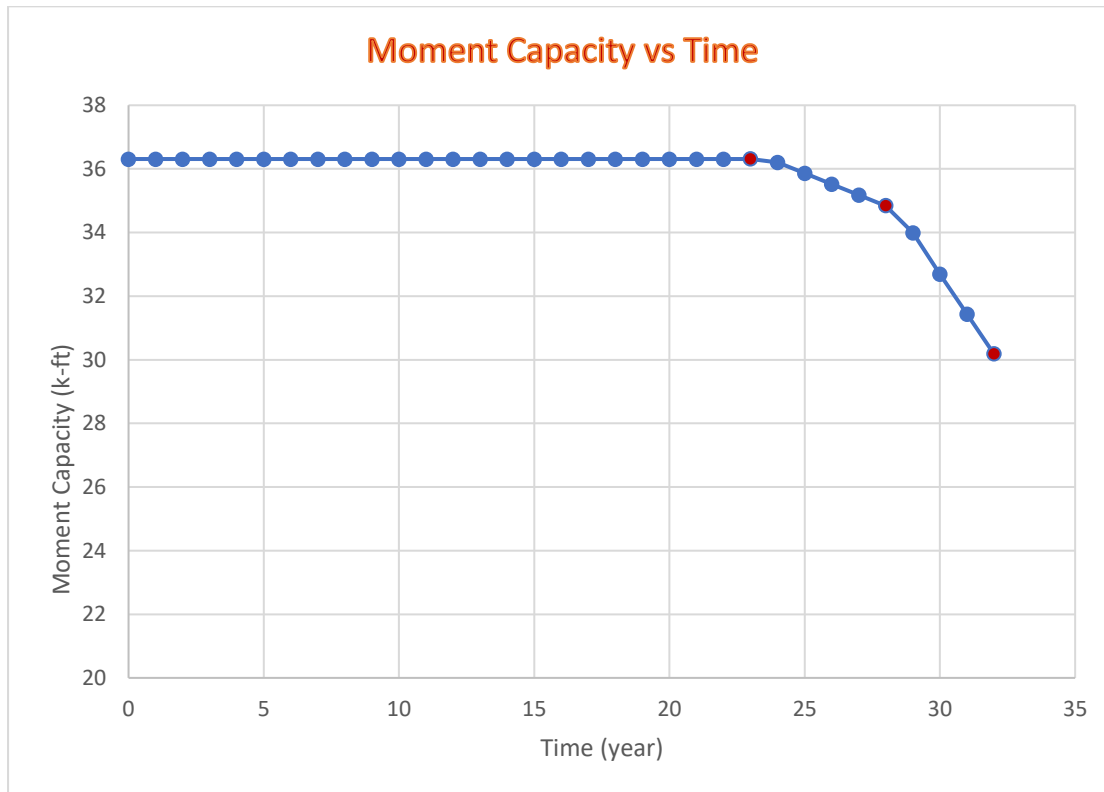
$$\text{Moment capacity reduction, } \% \Delta = \frac{36.31 - 30.58}{36.31} \times 100 = 15.78\% > 10\%$$

As the moment capacity reduction percentage is more than 10%, the structure needs to be demolished at the end of its service life.

25.4. Depth of corrosion vs Time Curve



25.5. Moment Capacity vs Time Curve



25.6. Discussion

- As per the service life calculation, the time when corrosion starts is given by $T_0 = 23.19$ years, the time when the first cracking occurs is given by $T_1 = 28.58$ years, and the time when cover spalling occurs is given by $T_2 = 31.95$ years.
- Also, the flexural moment capacity is reduced from 36.31 k-ft (before deterioration) to 30.58 k-ft (after deterioration), a reduction by 15.78%.
- Assuming a “minimum acceptable” level is a 10% moment reduction, the 10% reduction is close to the cracking stage. The 10% reduction is estimated by assuming the moment capacity reduction between the time when corrosion effects begin (T_0) and the time when spalling occurs (T_2) follows the trend shown by the calculated end values. The 10% reduction is approximately around 28.6 to 28.8 years. This is close to the time when the first cracking occurs, given by $T_1 = 28.58$ years.
- From a performance and safety perspective, it is conservative and sensible to set the “service life” at T_2 (spalling), since spalling tends to accelerate corrosion, degrade bond, and increase the risk of rapid capacity loss. Moreover, since your capacity loss at

T2 is already 15.78%, it would be unsafe to delay maintenance beyond spalling, since the structure would quickly move into the “unacceptable” region.

- To extend the service life in a sustainable way (instead of early demolition and rebuilding), the following aspects are to be considered in the design and rehabilitation of the structure:
 - **Improved durability of the concrete structure by enhancing the cover, reducing the permeability (reducing the water-cement ratio), and using good quality curing and supplementary cementitious materials.**
 - **Improved resistance to corrosion of the reinforcement by using epoxy-coated or corrosion-resistant reinforcement, corrosion inhibitors, and good quality surface protection systems.**
 - **Planned retrofitting: If the structure has to be kept in service beyond the stage of cracking, repairs and strengthening are to be done (for example, localized jacketing of the structure, fiber-reinforced polymer strengthening of the structure, and repair of the cover.**

25.7. Link to Sustainable Development Goals (SDGs)

Some of the SDGs can be linked with this work, mainly through **extending structural service life, reducing premature reconstruction, and minimizing life-cycle environmental impacts**, using the calculated deterioration timeline (T0, T1, T2) and associated capacity reduction as the basis for planned maintenance and retrofitting.

Table: Conformance to Sustainable Development Goals (SDGs) in the service-life and maintenance plan

SDG	Link with this project (service life prediction and LCA-based decisions)
SDG 9: Industry, Innovation and Infrastructure	Durability-based design decisions (cover, permeability control, corrosion protection) improve infrastructure reliability. Condition-based maintenance and retrofit planning improves resilience and reduces failure risks.
SDG 11: Sustainable Cities and Communities	Longer service life supports safer, more reliable buildings and reduces the need for disruptive demolition and reconstruction. Planned maintenance at cracking stage improves long-term habitability and safety.
SDG 12: Responsible Consumption and Production	Extending service life reduces demand for new cement, steel, and aggregates. Repair and retrofit (instead of rebuild) reduces construction waste and material extraction impacts.
SDG 13: Climate Action	Avoiding premature reconstruction reduces embodied carbon from cement and steel production. Durable concrete strategies and lifecycle maintenance reduce

SDG	Link with this project (service life prediction and LCA-based decisions)
	long-term emissions linked to replacement cycles.
SDG 6: Clean Water and Sanitation	Proper management of repair wastes (wash water, removed concrete, coatings) prevents contamination of drains and water bodies. Reduced demolition frequency lowers debris-related environmental load.
SDG 8: Decent Work and Economic Growth	Planned inspection, maintenance, and rehabilitation programs create skilled jobs and improve construction safety compared to reactive emergency repairs after severe deterioration.
SDG 17: Partnerships for the Goals	Effective service-life extension requires coordination among designers, owners, contractors, materials suppliers, and regulators for inspection schedules, repair standards, and durability compliance.

26. DISSCUSSION

The 10-storey RCC building was designed and analyzed using an Intermediate Moment Resisting Frame (IMRF) combined with a shear wall system to ensure adequate lateral stability and seismic performance. The structural configuration enhances stiffness, ductility, and overall safety under both gravity and lateral loads. After several iterations in ETABS, all structural members satisfied strength and serviceability requirements, ensuring structural reliability.

Reinforcement design was completed for slabs, beams, columns, stairs, and shear walls following code provisions. Concrete mix design for beams, slabs, and columns was carried out to achieve the required strength and durability while maintaining economic feasibility. The financial analysis further ensured that the structure is cost-effective and practically implementable.

The evaluated service life of 32 years indicates that the building meets durability considerations and reduces the likelihood of premature deterioration. By integrating structural safety, durability assessment, material optimization, and economic evaluation, the project reflects a balanced approach toward resilient and sustainable infrastructure development.

27. REFERENCES

- [1]. Housing and Building Research Institute. (2020). Bangladesh National Building Code 2020. Ministry of Housing and Public Works
<https://mccibd.org/wp-content/uploads/2021/09/Bangladesh-National-Building-Code-2020.pdf>
- [2]. Government of the People's Republic of Bangladesh. *Dhaka Imarat Nirman Bidhimala*.
https://kupdf.net/download/dhaka-imar-at-nirman-bidhimala-2008-pdf_58ee26e7dc0d60ac2eda982e_pdf
- [3]. American Concrete Institute (ACI). (2014). ACI 318-14 Building Code Requirements for Structural Concrete
https://www.concrete.org/Portals/0/Files/PDF/Previews/318-14_preview1.pdf